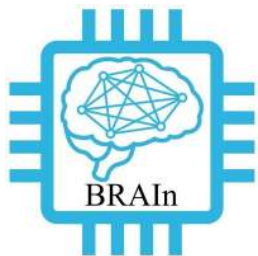




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Deep Learning for BCI applications: leveraging recent advances in computer vision

Yassine El Ouahidi

Brest, 20th March 2025

Examiners:

- Anatole Lécuyer
- Francesca Galassi
- Sylvain Baillet
- Sylvain Chevallier

Guests

- Fabien Lotte
- Karim Jerbi

Supervisors:

- Vincent Gripon
- Bastien Pasdeloup
- Giulia Lioi

Outline

- Context
- REVE: EEG Foundation Model
- A realistic Few-Shot Learning setup for BCI
- Conclusion & Perspectives



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Brain Computer Interfaces & Electroencephalography



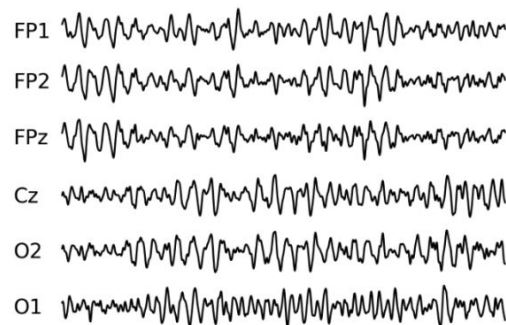
Example of a BCI system [1]

[1] <https://team.inria.fr/potioc/old-research-topics/bci-courses/>

Brain Computer Interfaces & Electroencephalography



Example of a BCI system [1]



Electroencephalography

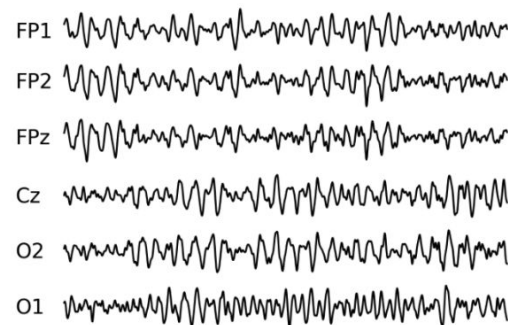
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Brain Computer Interfaces & Electroencephalography



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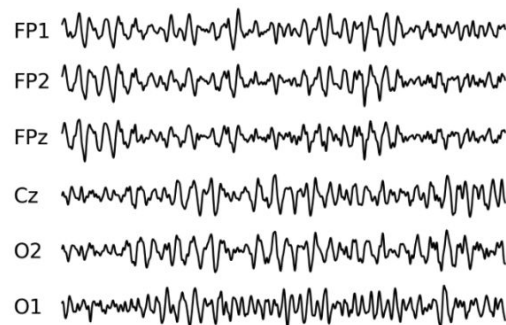
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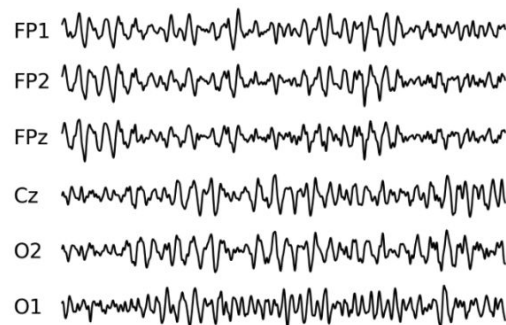
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Brain Computer Interfaces & Electroencephalography



Example of a BCI system [1]



Electroencephalography

- Recording setup / Electrode positions / Device
- BCI Task / Paradigm / Protocol
- Subjects / Sessions / Runs / Trials
- High variability / Very Noisy

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Deep Learning methods for BCI

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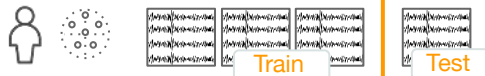
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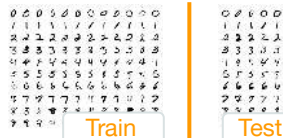
- + Rich Literature
- No clear consensus
- Lack of clearly established benchmark

Evolution of Deep Learning

EEG



Vision

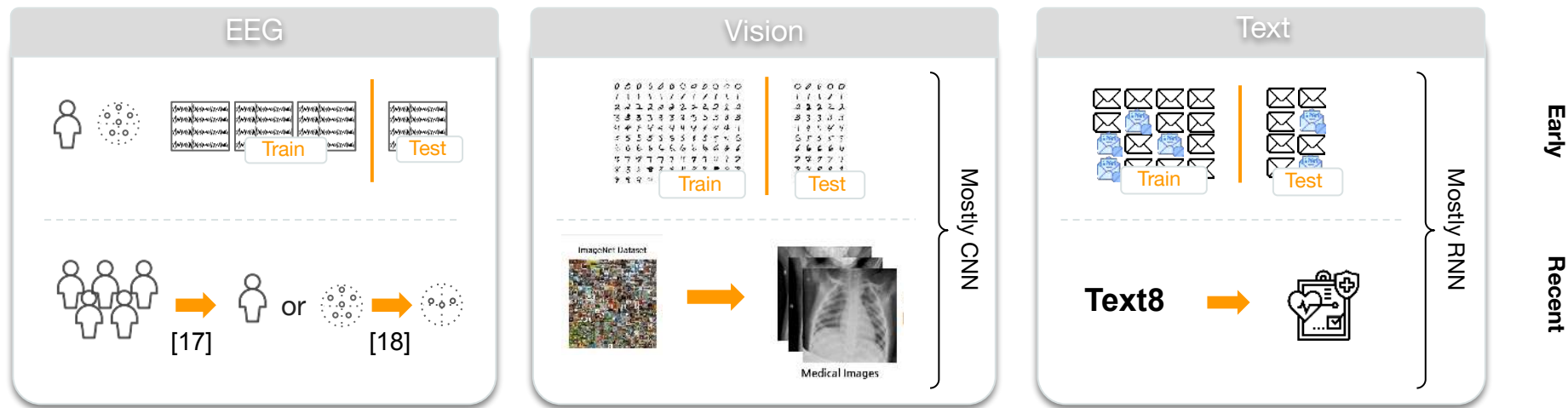


Text



Early

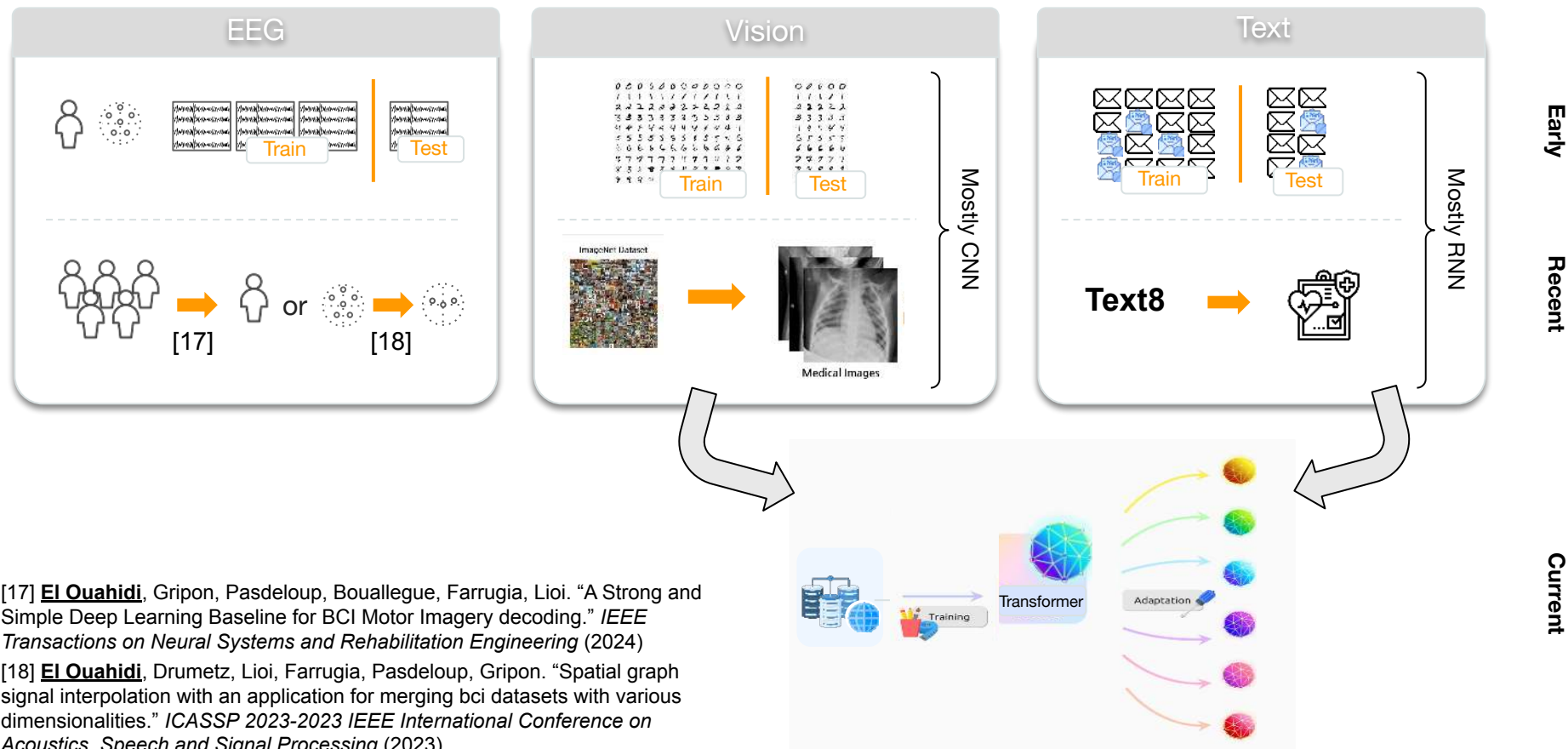
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[17] **El Ouahidi**, Gripon, Padeloup, Bouallegue, Farrugia, Lioi. “A Strong and Simple Deep Learning Baseline for BCI Motor Imagery decoding.” *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (2024)

[18] **El Ouahidi**, Drumetz, Lioi, Farrugia, Padeloup, Gripon. “Spatial graph signal interpolation with an application for merging bci datasets with various dimensionalities.” *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing* (2023)

Evolution of Deep Learning



Problem Statement

The Shift in Model Training

- **Early paradigm:** Given domain D , and task T , train a specialised model $f_{T,D}$

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- **Early paradigm:** Given domain D , and task T , train a specialised model $f_{T,D}$
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Problem Statement

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- **Early paradigm:** Given domain D , and task T , train a specialised model $f_{T,D}$
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Does applying deep learning to BCI demand specialized methods?

Or, can we leverage recent advances and established techniques from vision and language and adapt them for EEG?

Outline

- Context
- **REVE: EEG Foundation Model**
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What's happening in vision and language

└ Foundation Models

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What's happening in vision and language

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Many Models and Actors

- **Vision:** MAE, Dino, Sam, Stable Diffusion



- **Text:** Llama, GPT, Gemini, Bert, Mistral

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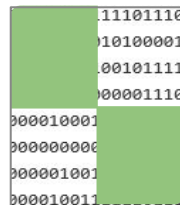
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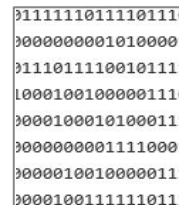


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Self Supervised Learning (SSL)



Reconstruct →



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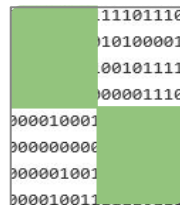


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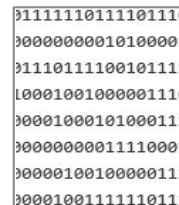
Large Scale Datasets



Self Supervised Learning (SSL)



Reconstruct



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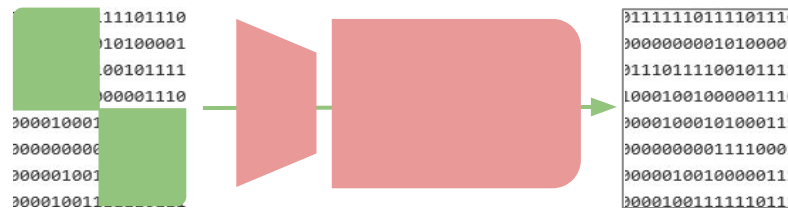


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Large Scale Datasets



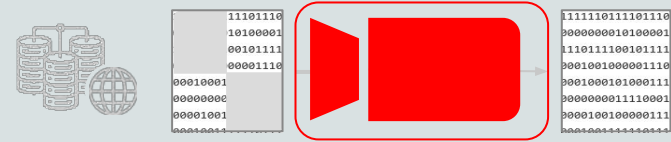
Self Supervised Learning (SSL)



Transformers

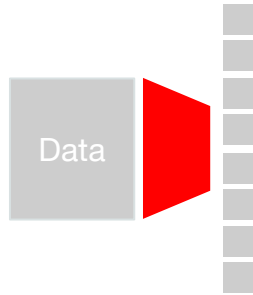
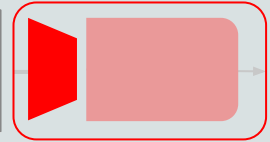
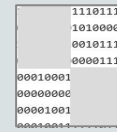
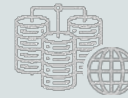
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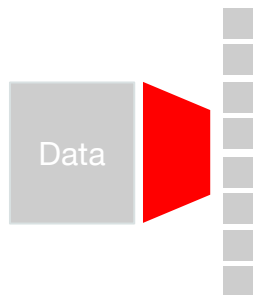
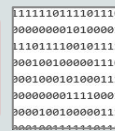
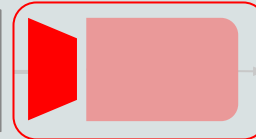
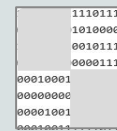
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What's happening in vision and language

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Modality dependent

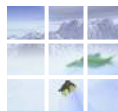
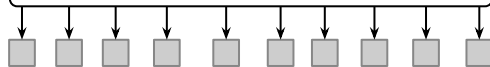
Today, I love Brain Computer Interfaces

Tokenization

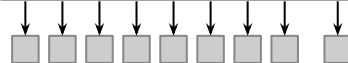
['To', 'day', 'I', 'love', 'Brain', 'Com', 'put', 'er', 'Inter', 'faces']

[98, 1452, 205, 1082, 3211, 7, 9823, 2048, 21, 200]

Learned Codebook

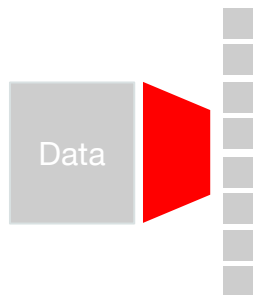
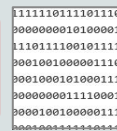
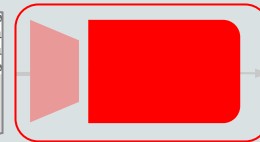
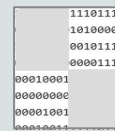


Linear projection of flattened patches

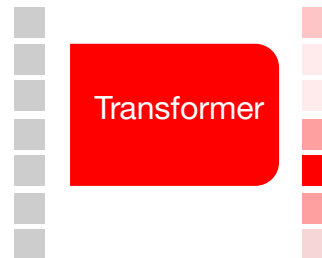


What's happening in vision and language

└ Transformer



Modality dependent



Modality agnostic

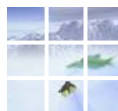
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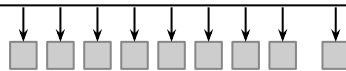
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Learned Codebook



Linear projection of flattened patches



$$Z = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

with:

$$X = [x_1, x_2, \dots, x_n] \in \mathbb{R}^{n \times d}$$

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

n : sequence length,

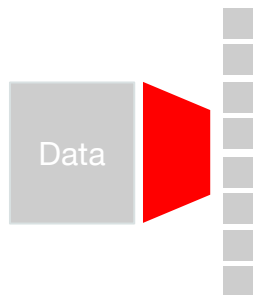
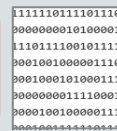
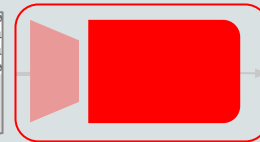
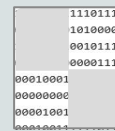
d : embedding dimension,

$W_Q, W_K, W_V \in \mathbb{R}^{d \times d_k}$: learned projection matrices,

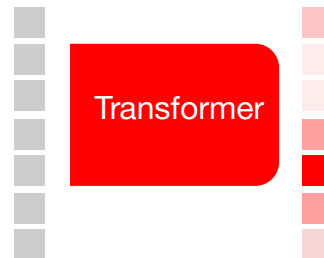
d_k : dimension of each key/query vector.

What's happening in vision and language

└ Transformer



Modality dependent
(~1-8% weights)



Modality agnostic
(~90-99% weights)

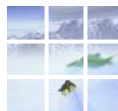
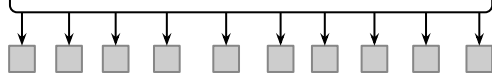
Today, I love Brain Computer Interfaces

Tokenization

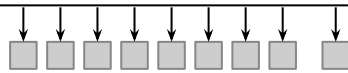
['To', 'day', 'I', 'love', 'Brain', 'Com', 'put', 'er', 'Inter', 'faces']

[98, 1452, 205, 1082, 3211, 7, 9823, 2048, 21, 200]

Learned Codebook



Linear projection of flattened patches



$$Z = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

with:

$$X = [x_1, x_2, \dots, x_n] \in \mathbb{R}^{n \times d}$$

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

n : sequence length,

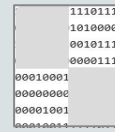
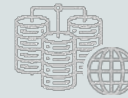
d : embedding dimension,

$W_Q, W_K, W_V \in \mathbb{R}^{d \times d_k}$: learned projection matrices,

d_k : dimension of each key/query vector.

What's happening in vision and language

└ Positional Encoding

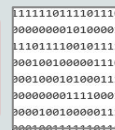
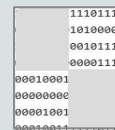


'To', 'day', 'I', 'love', 'Brain', 'Com', 'put', 'er', 'Inter', 'faces'

Position 0 1 2 3 4 5 6 7 8 9

What's happening in vision and language

└ Positional Encoding



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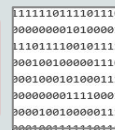
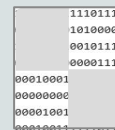
Integer encoding

-0.02	5
0.04	5
0.01	5
0.22	5
-0.09	5
0.12	5
-0.02	5
-0.11	5

+

What's happening in vision and language

└ Positional Encoding



'To', 'day', 'I', 'love', 'Brain', 'Com', 'put', 'er', 'Inter', 'faces'

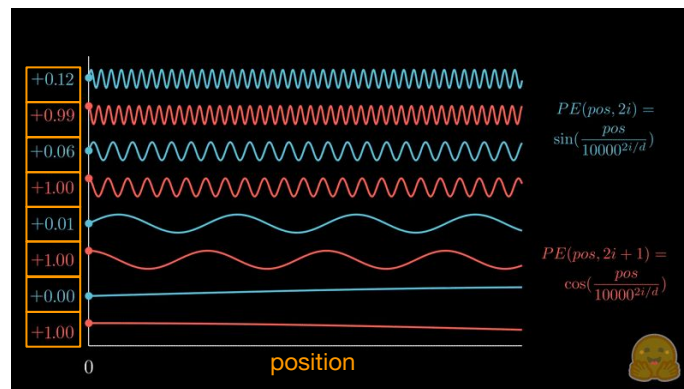
Position 0 1 2 3 4 5 6 7 8 9

Integer encoding

-0.02	5
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+

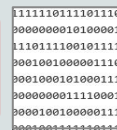
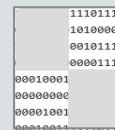
Fourier encoding [19]



[19] Blogpost: You could have designed state of the art positional encoding, Christopher Fleetwood <https://huggingface.co/blog/designing-positional-encoding>

What's happening in vision and language

└ Positional Encoding



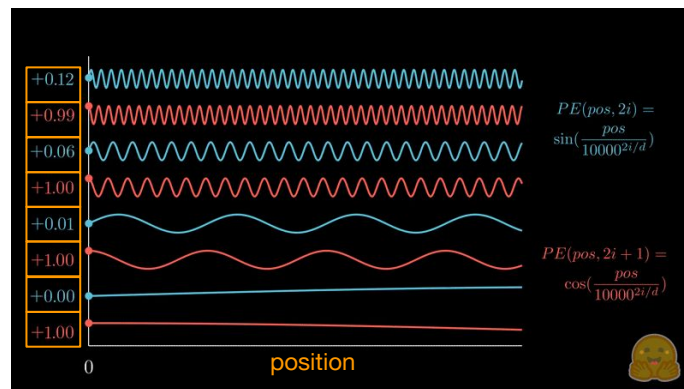
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Fourier encoding [19]



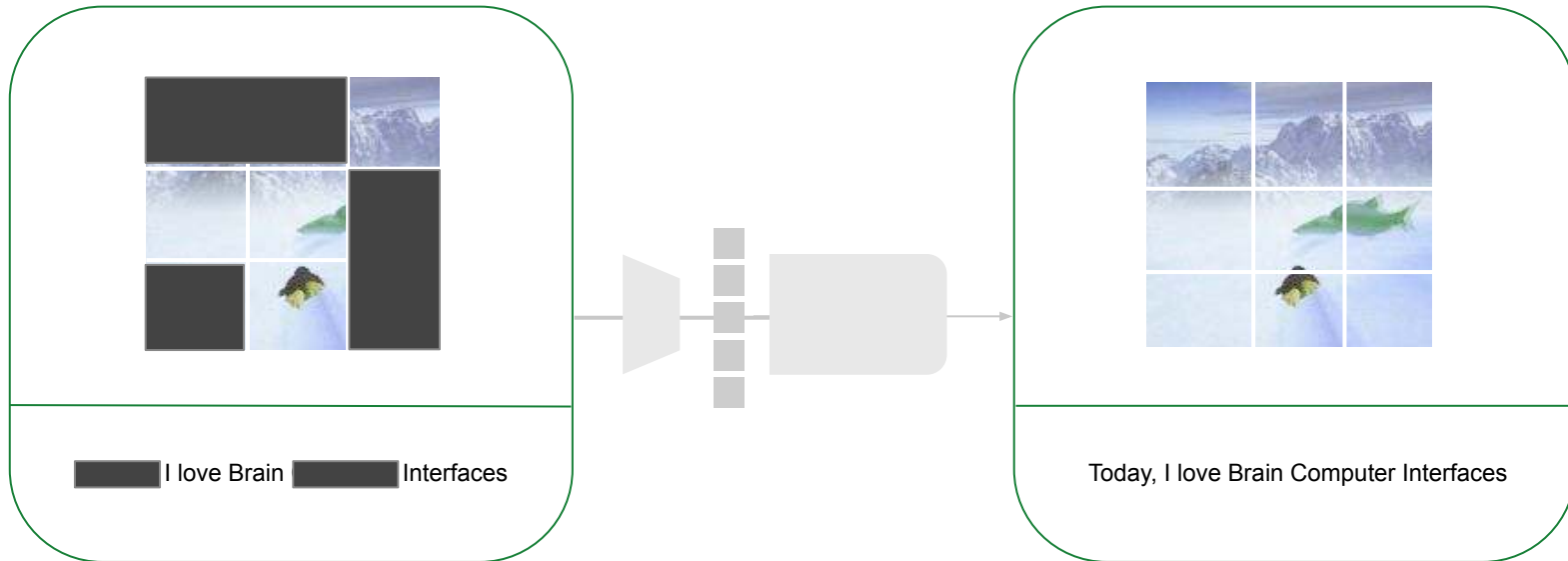
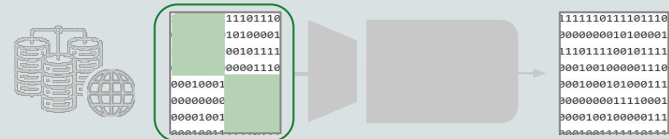
Learnable embeddings

$e_{i,0}$
$e_{i,1}$
$e_{i,2}$
$e_{i,3}$
$e_{i,4}$
$e_{i,5}$
$e_{i,6}$
$e_{i,7}$

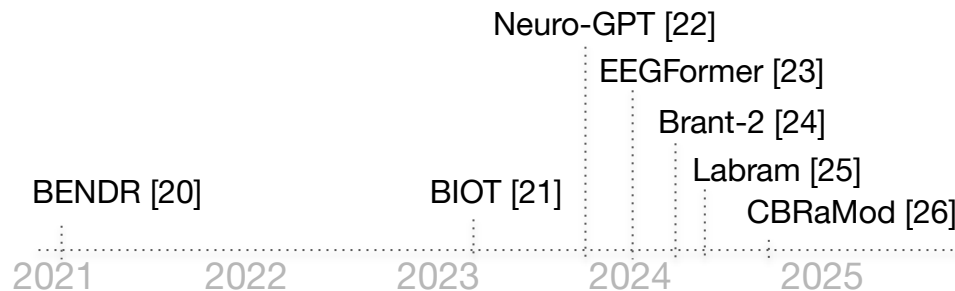
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What's happening in vision and language

└ Self-Supervised Learning



Foundation Models for EEG



[20] Kostas et al. "BENDR: Using transformers and a contrastive self-supervised learning task to learn from massive amounts of EEG data." *Frontiers in Human Neuroscience* 15 (2021)

[21] Yang et al. "Biot: Biosignal transformer for cross-data learning in the wild." *Advances in Neural Information Processing Systems* 36 (2023)

[22] Cui et al. "Neuro-gpt: Towards a foundation model for eeg." *IEEE International Symposium on Biomedical Imaging* (2024)

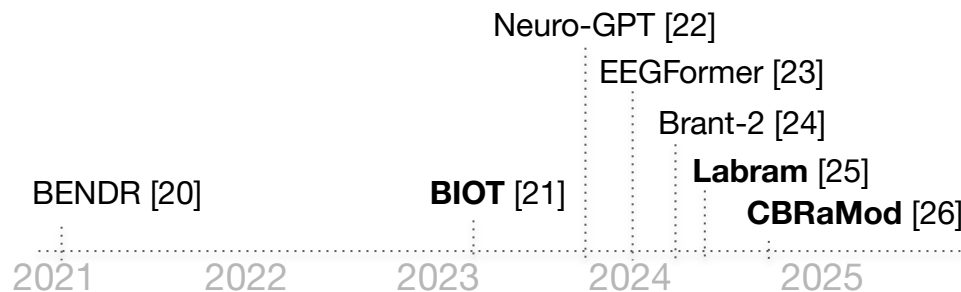
[23] Chen et al. "EEGFormer: Towards transferable and interpretable large-scale EEG foundation model." *arXiv preprint* (2024)

[24] Yuan et al. "Brant-2: Foundation model for brain signals." *arXiv preprint* (2024)

[25] Jiang et al. "Large brain model for learning generic representations with tremendous EEG data in BCI." *International Conference on Learning Representations* (2024)

[26] Wang et al. "CBRaMod: A Criss-Cross Brain Foundation Model for EEG Decoding." *International Conference on Learning Representations* (2025)

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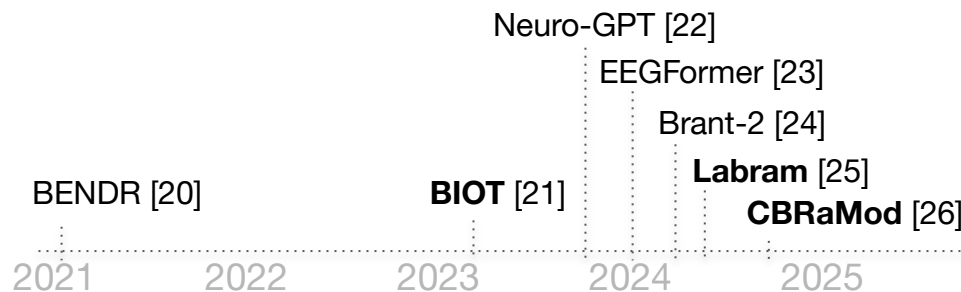
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	Infer and generalize to any EEG Configuration
BIOT	~
Labram	~
Cbramod	~
REVE (ours)	✓

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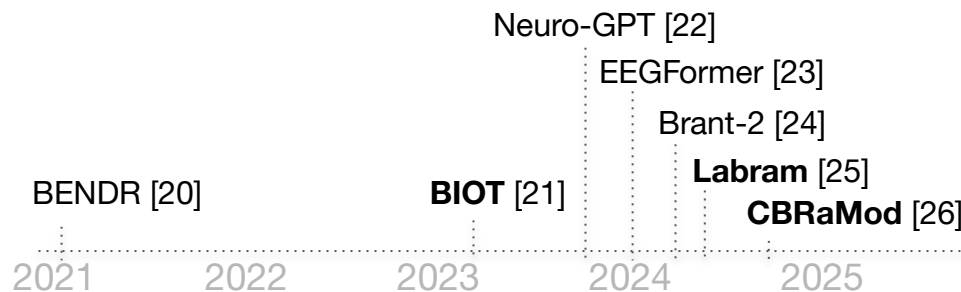
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	Infer and generalize to any EEG Configuration	Trained on Multiple EEG Configuration
BIOT	~	✗
Labram	~	✓
Cbramod	~	✗
REVE (ours)	✓	✓

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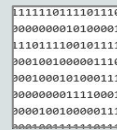
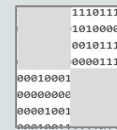
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	Infer and generalize to any EEG Configuration	Trained on Multiple EEG Configuration	Can be used without tuning the weights
BIOT	~	✗	~
Labram	~	✓	~
Cbramod	~	✗	~
REVE (ours)	✓	✓	✓

REVE: A foundation model for EEG

└ Dataset



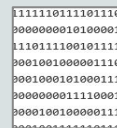
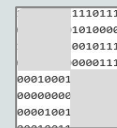
Data Collection



- 19 TB of raw data
- 24,274 subjects
- 150,833 sessions
- 61,415 hours (80% clinic)
- 92 datasets (28 BCI, 8 clinic)
- 396 unique electrodes

REVE: A foundation model for EEG

└ Dataset



Data Collection



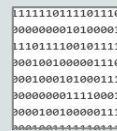
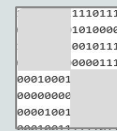
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Data Processing

- 1) Resampling 200 Hz, Band-pass [0.5,99.5] Hz

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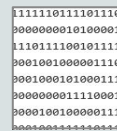
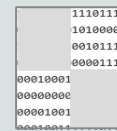
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- 2) Z-Normalization (per channel, per session)

Problems addressed

- A) Different amplitudes

REVE: A foundation model for EEG

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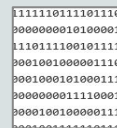
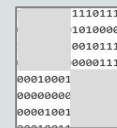
- 1) Resampling 200 Hz, Band-pass [0.5,99.5] Hz
- 2) Z-Normalization (per channel, per session)
- 3) Clipping

Problems addressed

- A) Different amplitudes
- B) Outliers and big spikes

REVE: A foundation model for EEG

└ Dataset



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- 19 TB of raw data
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Data Processing

- 1) Resampling 200 Hz, Band-pass [0.5,99.5] Hz
- 2) Z-Normalization (per channel, per session)
- 3) Clipping
- 4) 3D electrode positions alignment
- 5) 10 second windowing

Problems addressed

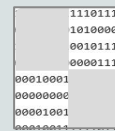
- A) Different amplitudes
- B) Outliers and big spikes
- C) Different montage with different coordinate systems

REVE: A foundation model for EEG

└ 4D Positional Encoding Strategy

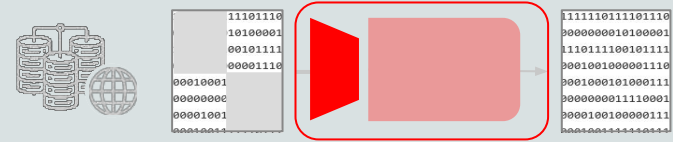
Problems addressed

- A) Electrode configurations (number and placement)
- B) Signal Duration
- C) Generalize to unseen configurations



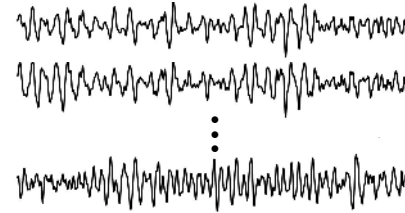
REVE: A foundation model for EEG

└ 4D Positional Encoding Strategy

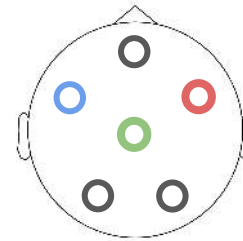


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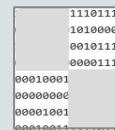


- Base the strategy on the 4D coordinates



REVE: A foundation model for EEG

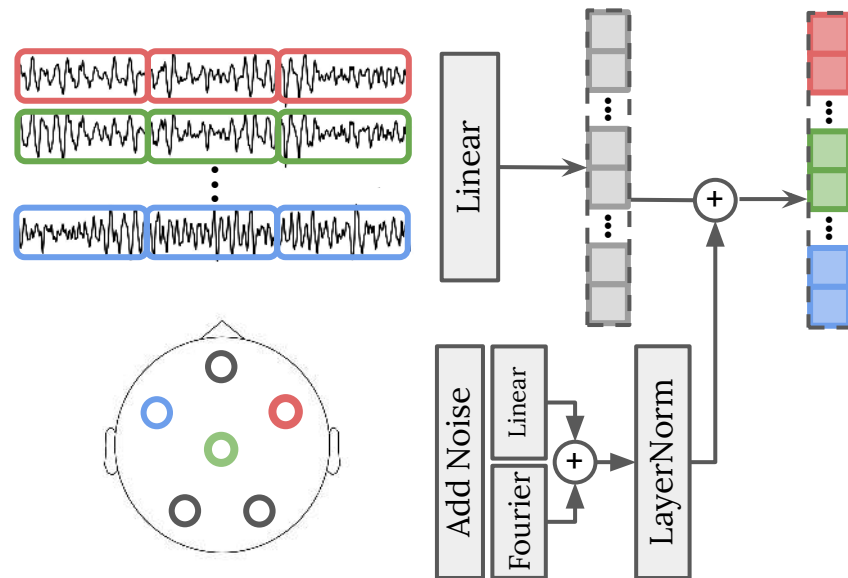
└ 4D Positional Encoding Strategy



Problems addressed

- A) Electrode configurations (number and placement)
- B) Signal Duration
- C) Generalize to unseen configurations

- Base the strategy on the 4D coordinates
- **Spatial Noise:** Introduce more diversity
- **Fourier4D:** Extension of the 2D proposed in [27]
- **MLP4D:** Allows more adaptation



[27] Défossez et al. "Decoding speech perception from non-invasive brain recordings." *Nature Machine Intelligence* 5.10 (2023)

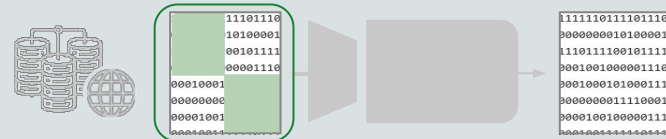
What's happening in vision and language

└ Spatio-Temporal Block Masking Strategy

Problems addressed

A) Choice of the SSL method

- Same as MAE [28]



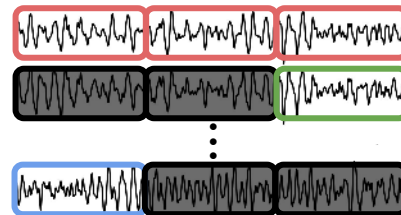
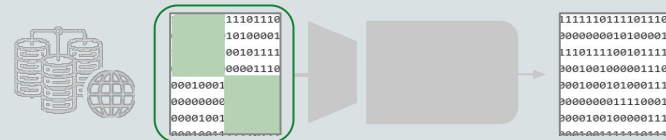
[28] He et al. "Masked autoencoders are scalable vision learners." *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (2022)

What's happening in vision and language

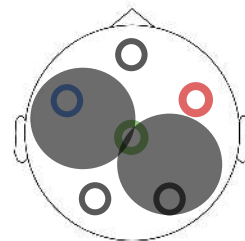
└ Spatio-Temporal Block Masking Strategy

Problems addressed

- A) Choice of the SSL method
- B) Balancing temporal and spatial characteristics

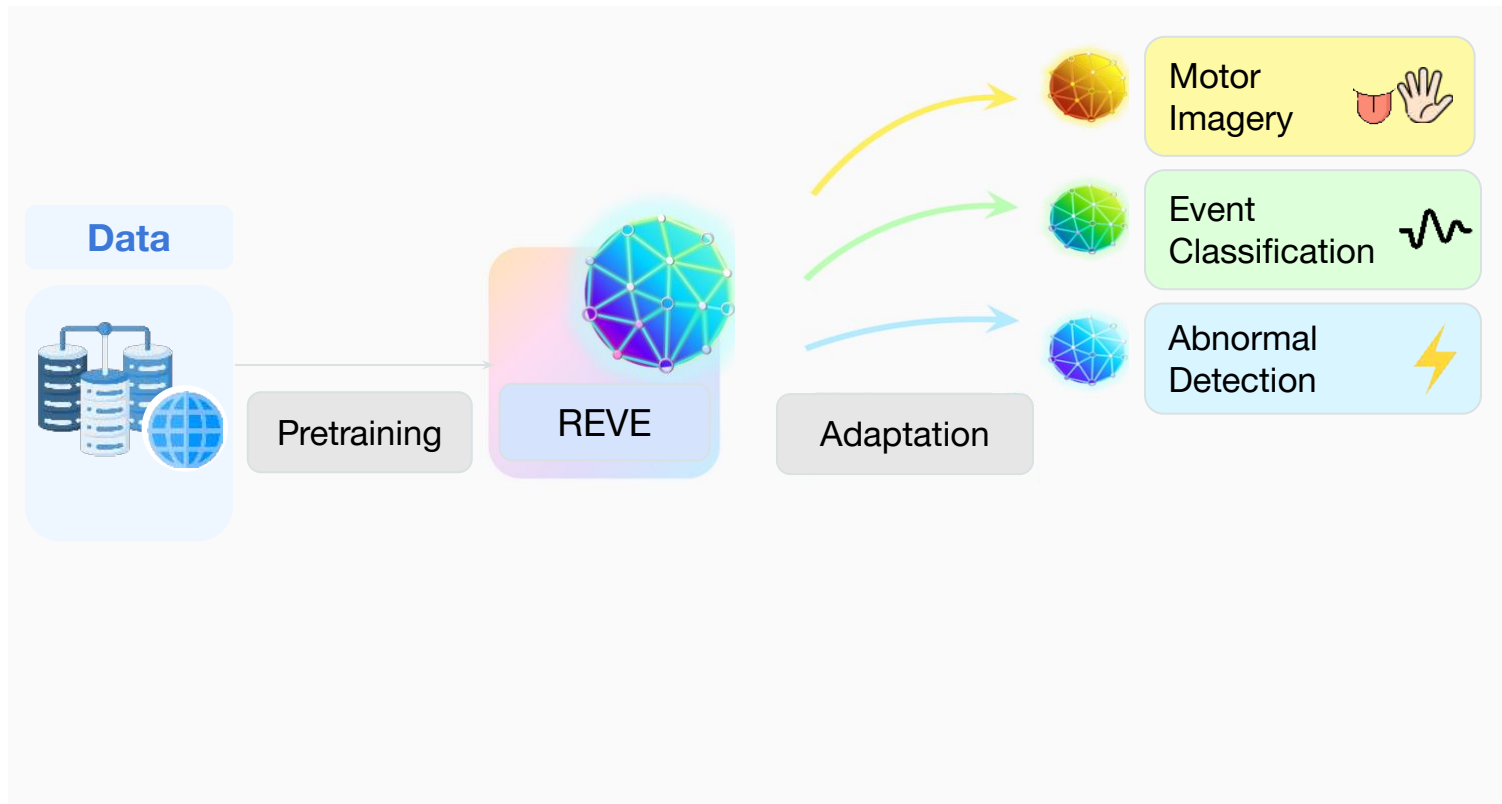


- Same as MAE [28]
- Novel masking strategy with spatio-temporal blocks
- Overcomes limit of random masking

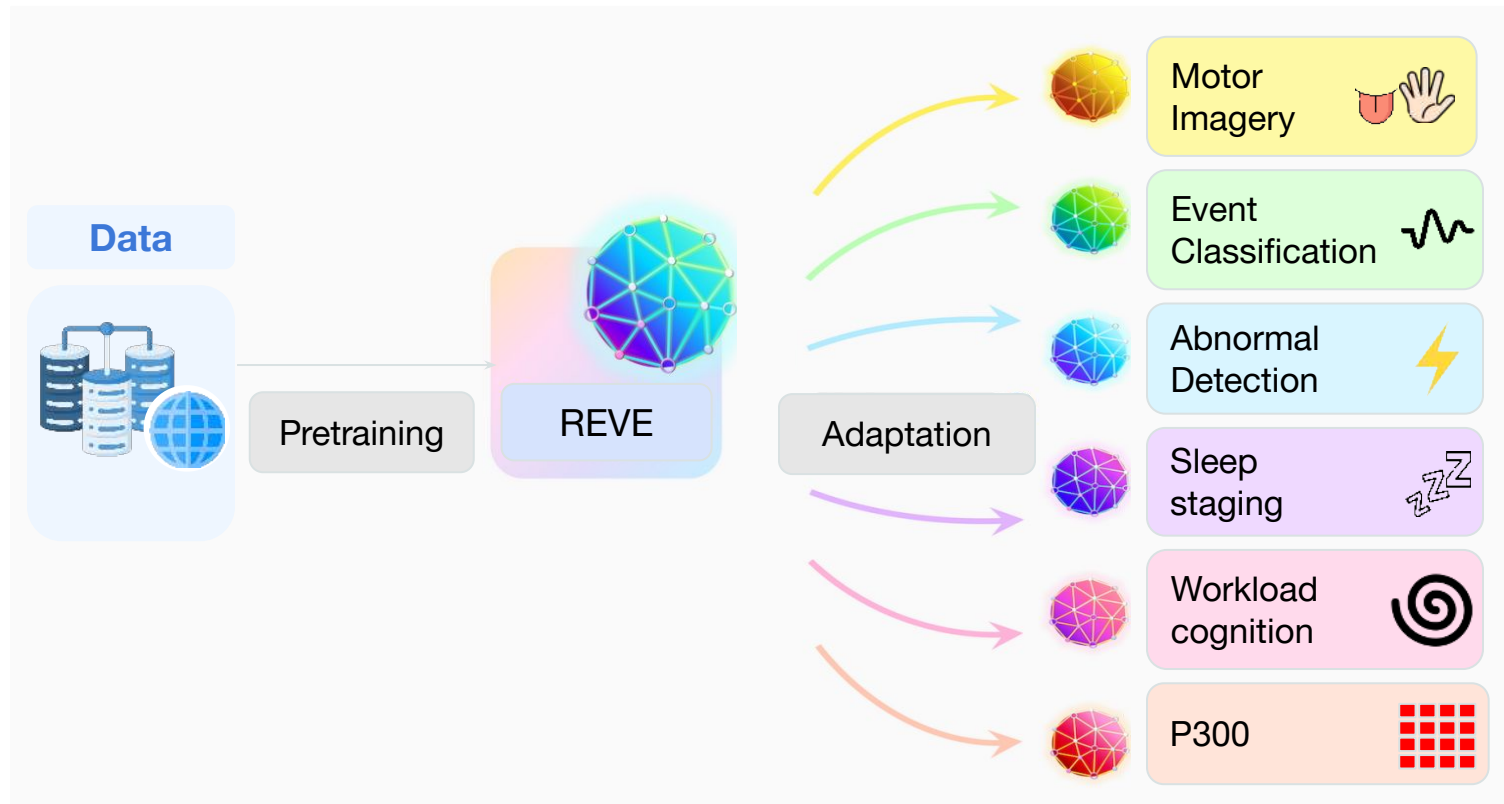


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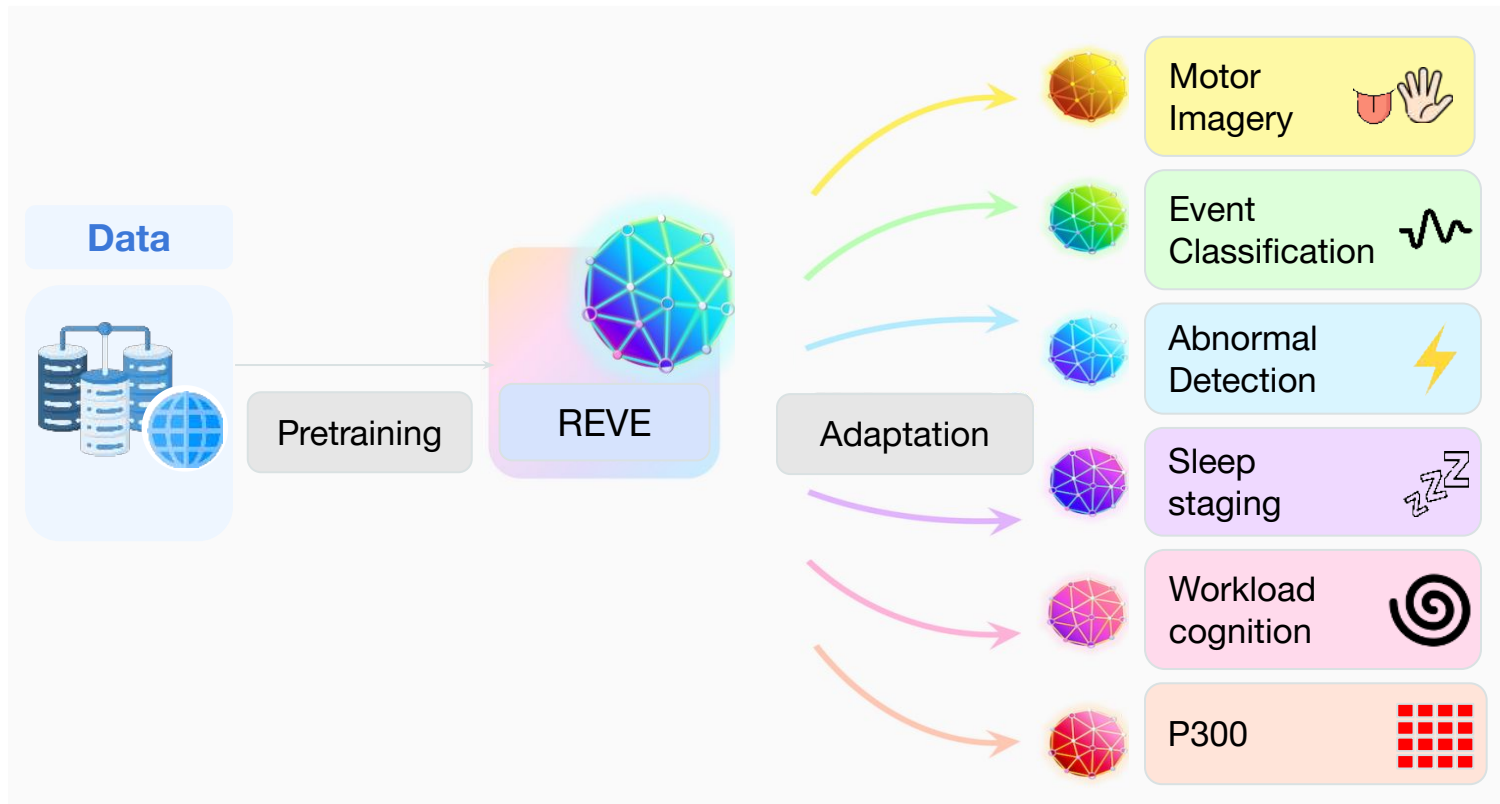
Results



Results



Results



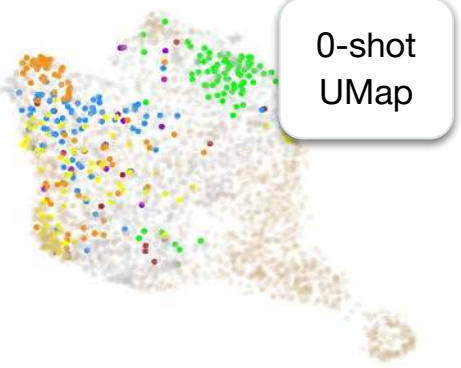
Average gain
across all tasks:

+1.5%

Results

- Provides quality embeddings, with strong generalization even when frozen **+16.7%** (55.1% - 38.5%)

Embeddings visualization

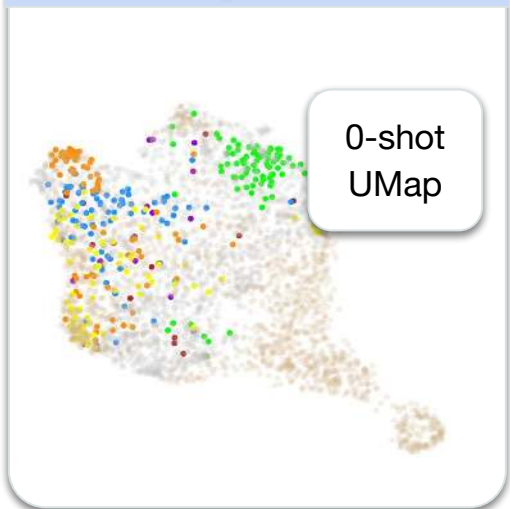


0-shot
UMap

Results

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- Pretraining leads to important performance gains **+8.9%** (11.1% - 2.21%)

Embeddings visualization



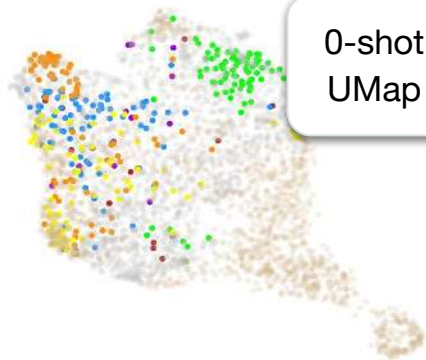
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- Provides interpretable attention maps

Embeddings visualization

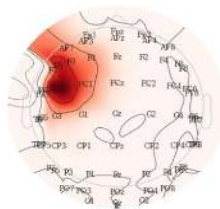
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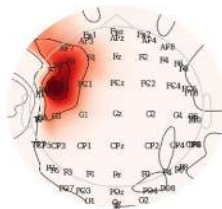
Task-Specific Attention Maps

Motor Imagery (Left / Right)

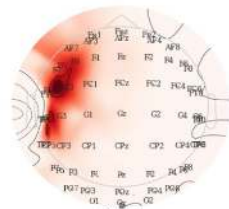
P300



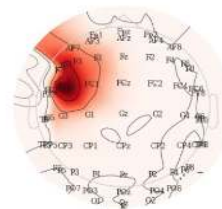
[0-1] s.



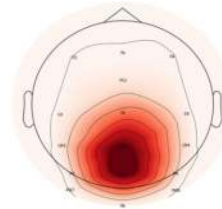
[1-2] s.



[2-3] s.



[0-3] s.



[0,1] s.

Outline

- Context
- REVE: EEG Foundation Model
- **A realistic Few-Shot Learning setup for BCI**
- Conclusion & Perspectives



IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom

Calibration in BCI systems

Why Is Calibration Essential?

- **User Variability:** Unique brain signal
- **Session-to-Session Variability:** Changes in brain activity
- **Influencing Factors:** Experimental Conditions, Mental State, Electrode Placement..

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Why is Calibration a Challenge?

- Requires New Data & Retraining
- Time-Consuming & Impractical

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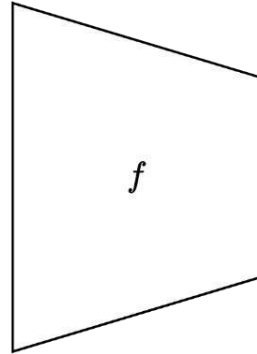
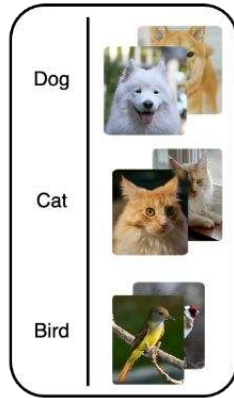
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Impact on BCI Systems?

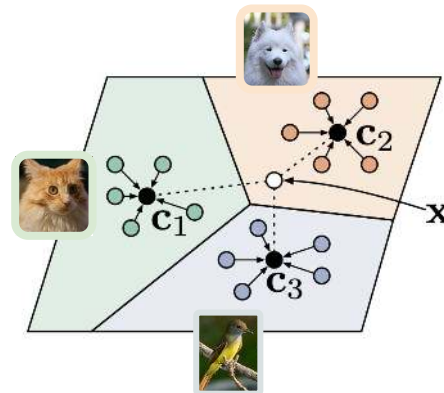
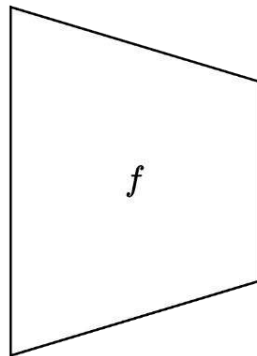
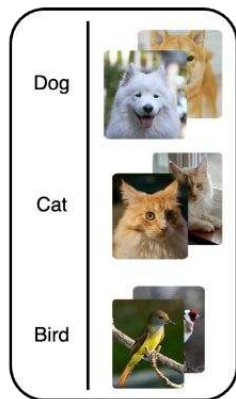
- Limits Practical Applications
- Restrict Widespread Adoption

Link between reducing calibration and FSL?

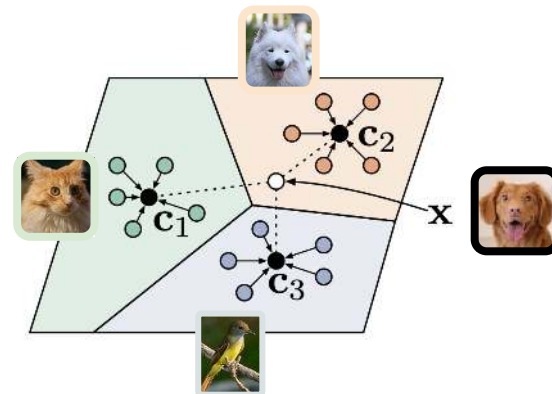
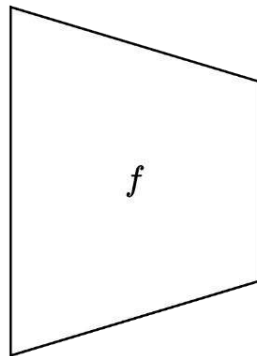
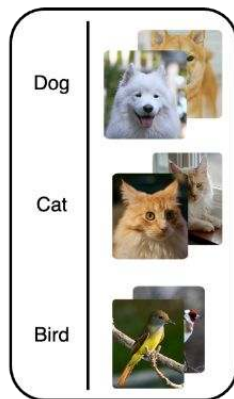
Few-Shot Learning



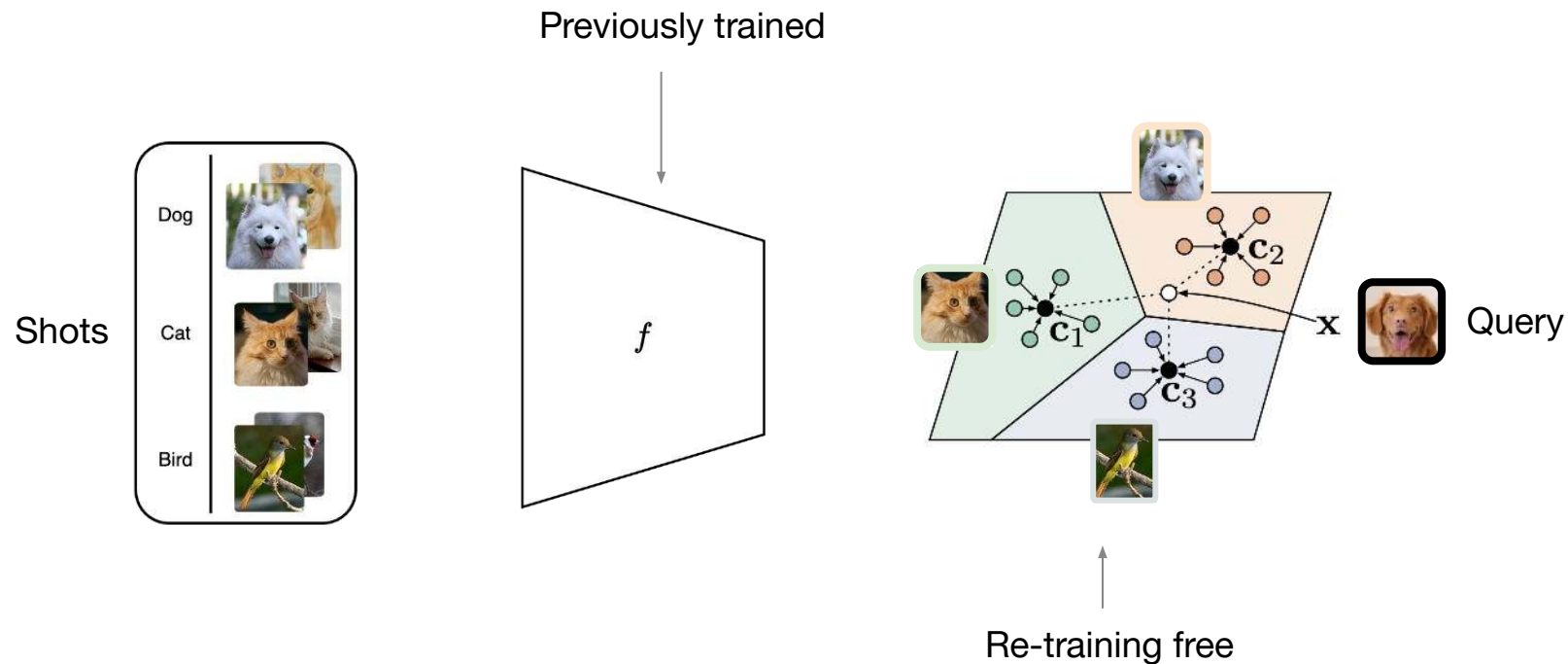
Few-Shot Learning



Few-Shot Learning



Few-Shot Learning



Reducing calibration

└ Setups



Eval. Subject



Single-Subject

Reducing calibration

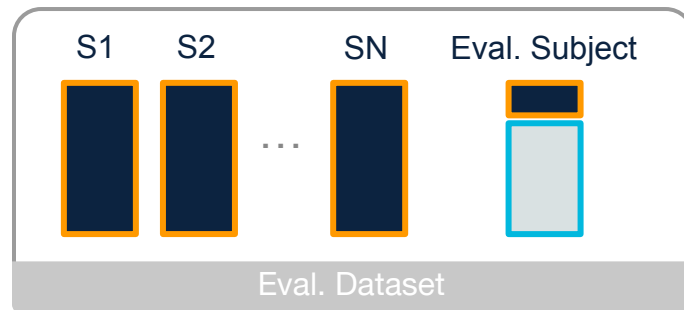
└─ Setups



Eval. Subject



Single-Subject



Cross-Subject

In-Dataset

Reducing calibration

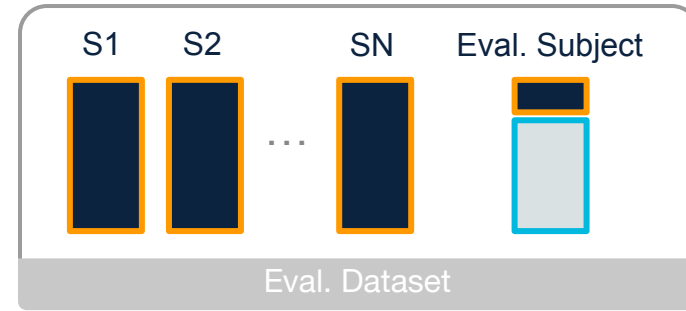
└─ Setups



Eval. Subject

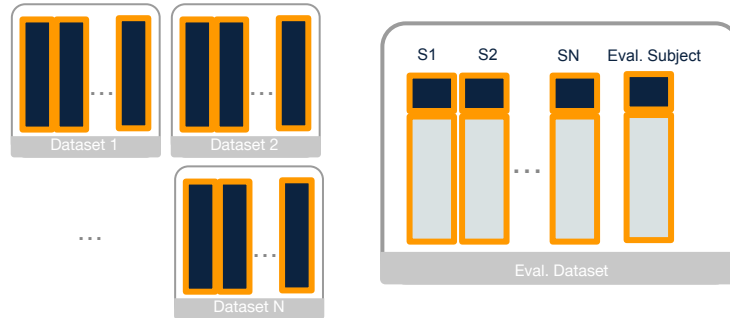


Single-Subject



Cross-Subject

In-Dataset



BEETL Challenge [29]

Cross-Dataset

[29] Wei et al. "BEETL competition: Advancing transfer learning for subject independence and heterogenous EEG data sets." *NeurIPS*. PMLR (2022)

Reducing calibration

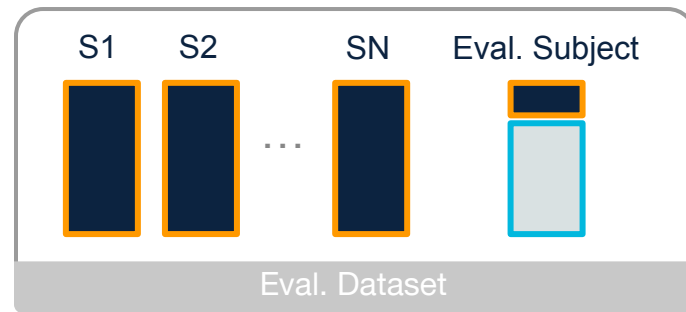
└─ Setups



Eval. Subject

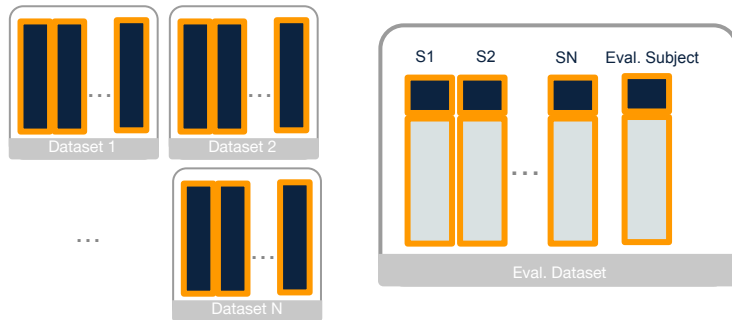


Single-Subject

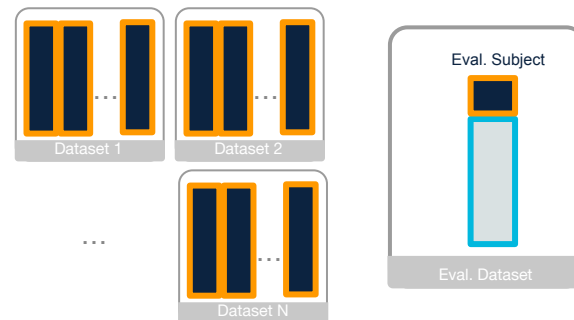


Cross-Subject

In-Dataset



BEETL Challenge [29]



Our realistic setup

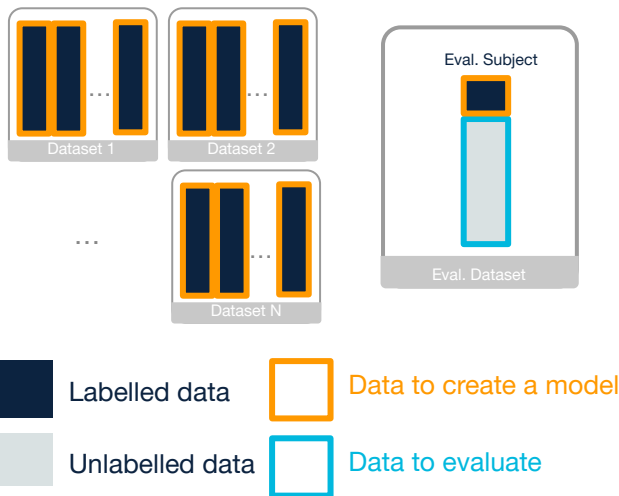
Cross-Dataset

[29] Wei et al. "BEETL competition: Advancing transfer learning for subject independence and heterogenous EEG data sets." *NeurIPS*. PMLR (2022)

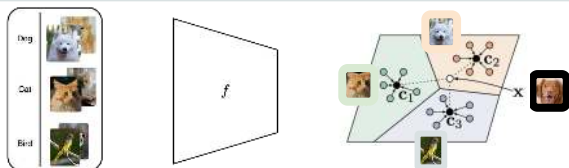
Reducing calibration

└ Our Setup

Data splits in our setup



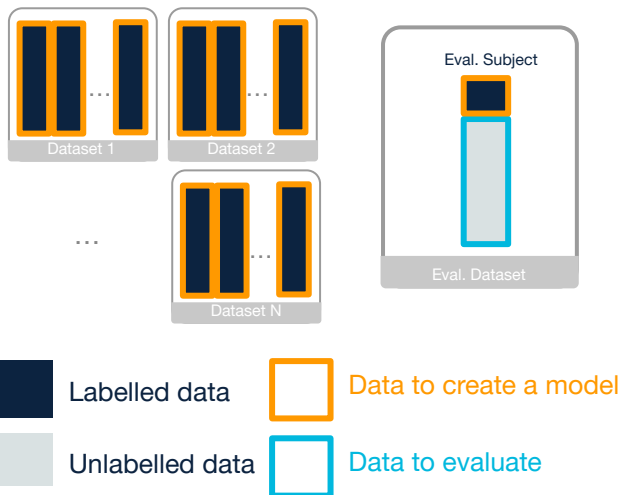
FSL method



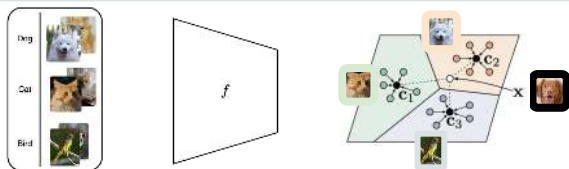
Reducing calibration

└ Our Setup

Data splits in our setup



FSL method



1) How to train our feature extractor?



- 12 Motor Imagery datasets
- ~ 400 Subjects
- ~ 104,528 trials

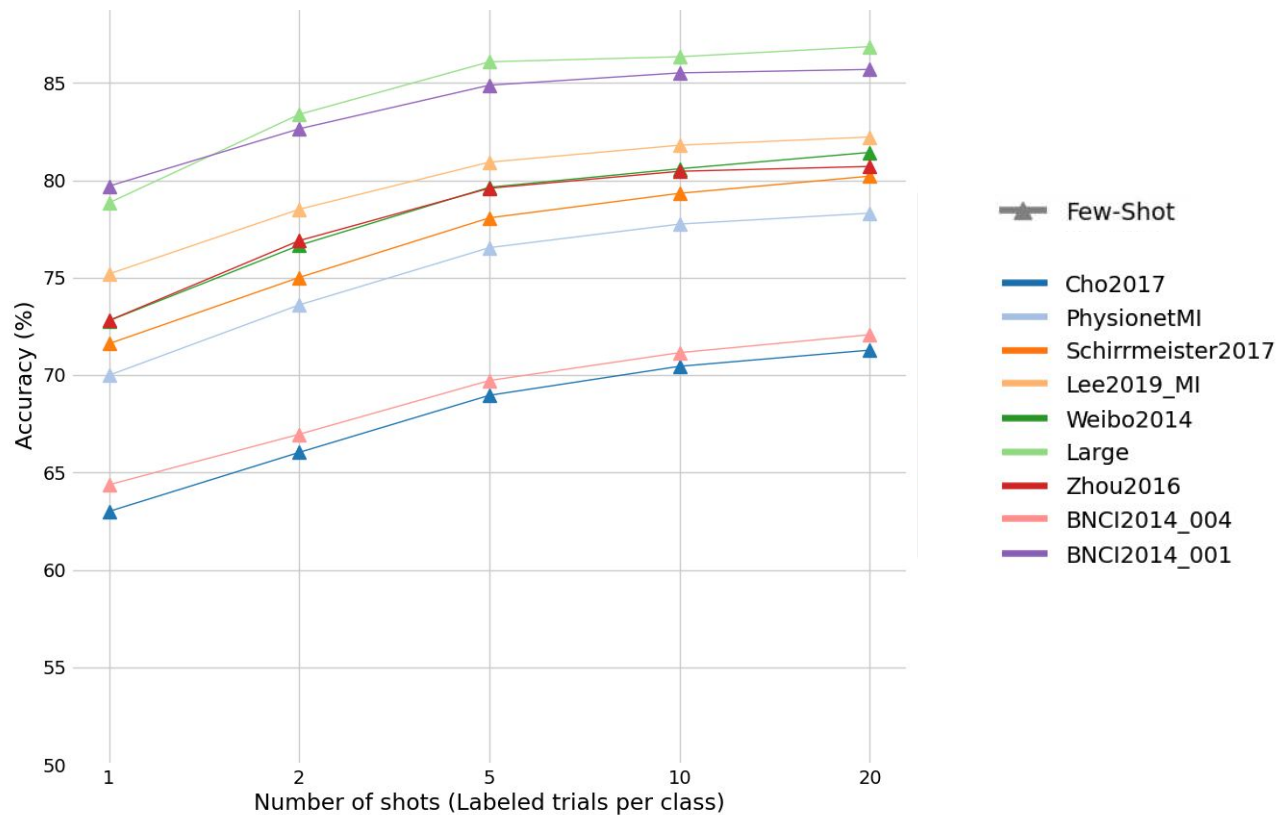
2) How to evaluate ?

- Inductive (NCM)
- Adaptive ([30] + Soft K-means)
- Intra / inter session comparisons

[30] **El Ouahidi**, Lioi, Farrugia Padeloup, Gripon. "Unsupervised Adaptive Deep Learning Method For BCI Motor Imagery Decoding". In *2024 32nd European Signal Processing Conference* (2024)

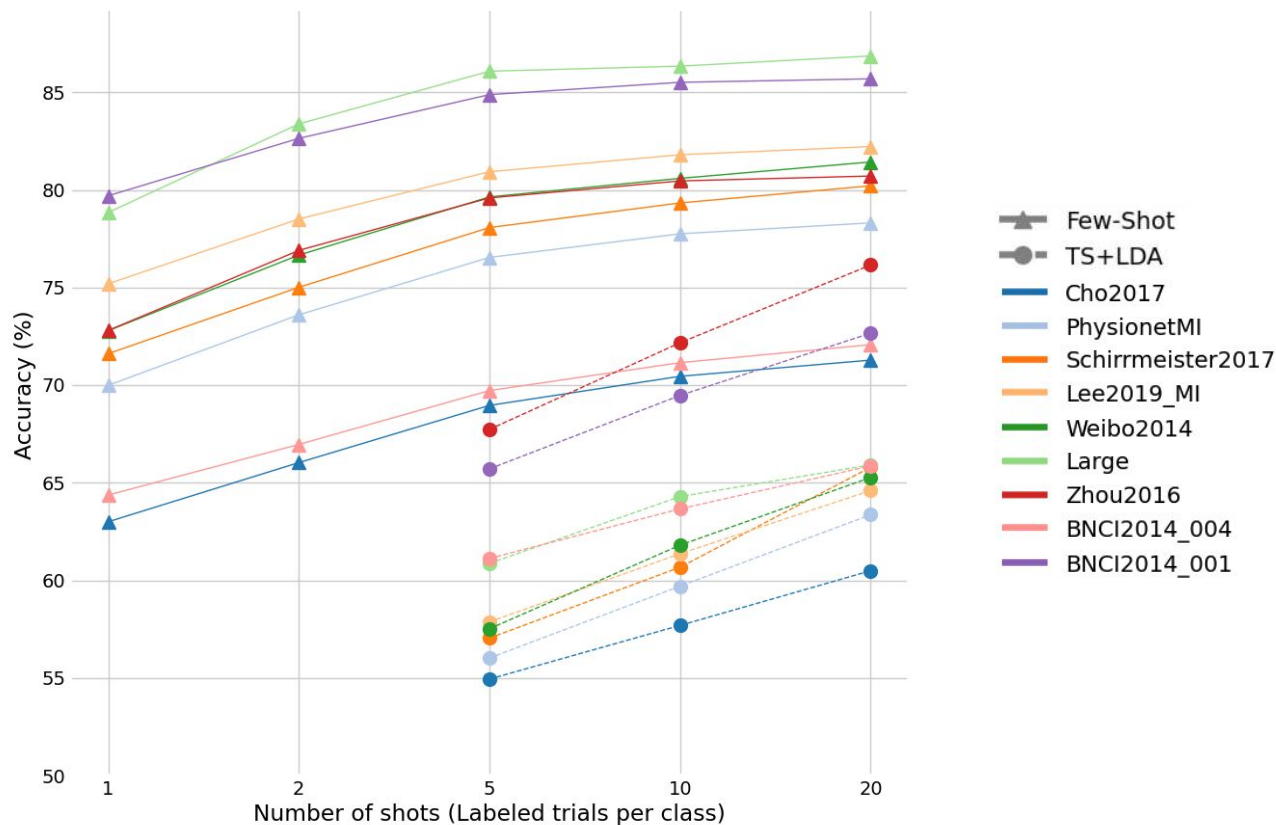
Results

└ Left vs. Right Motor Imagery



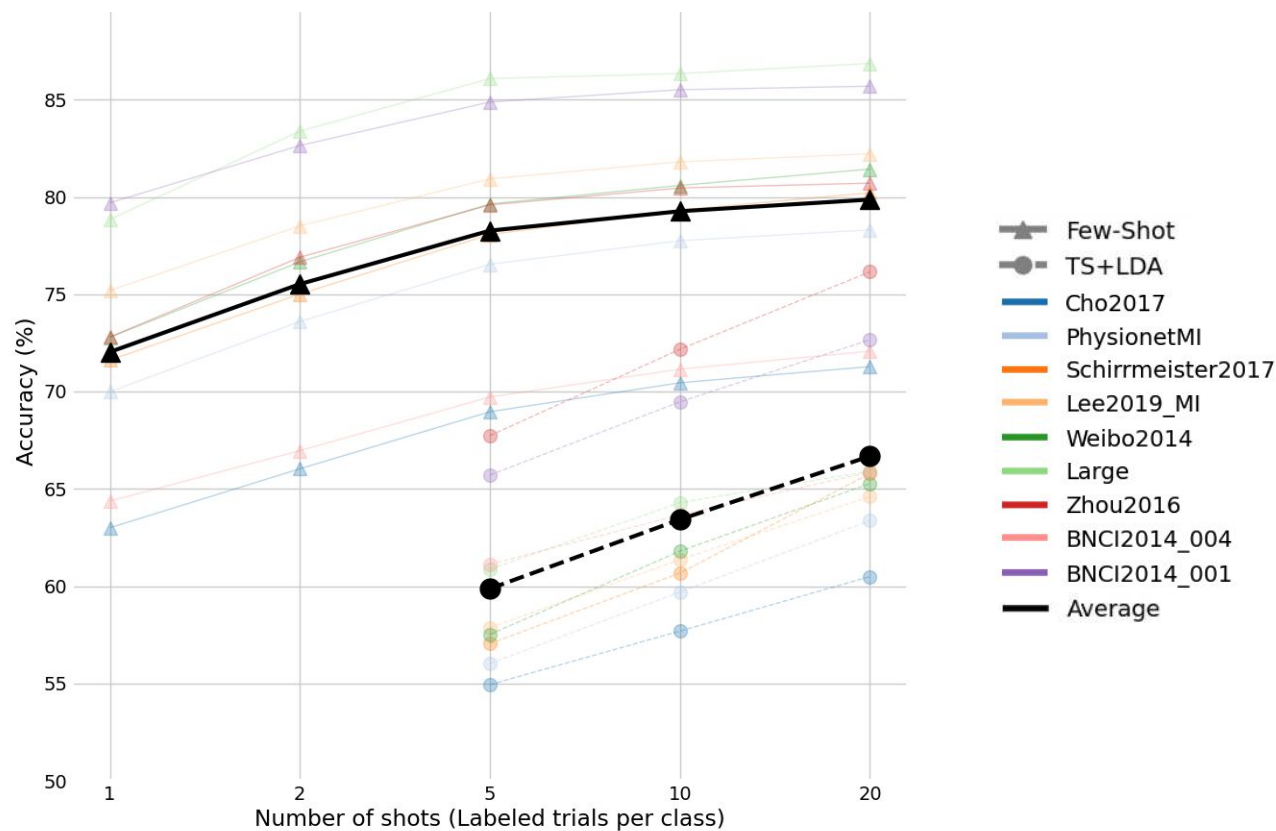
Results

└ Left vs. Right Motor Imagery



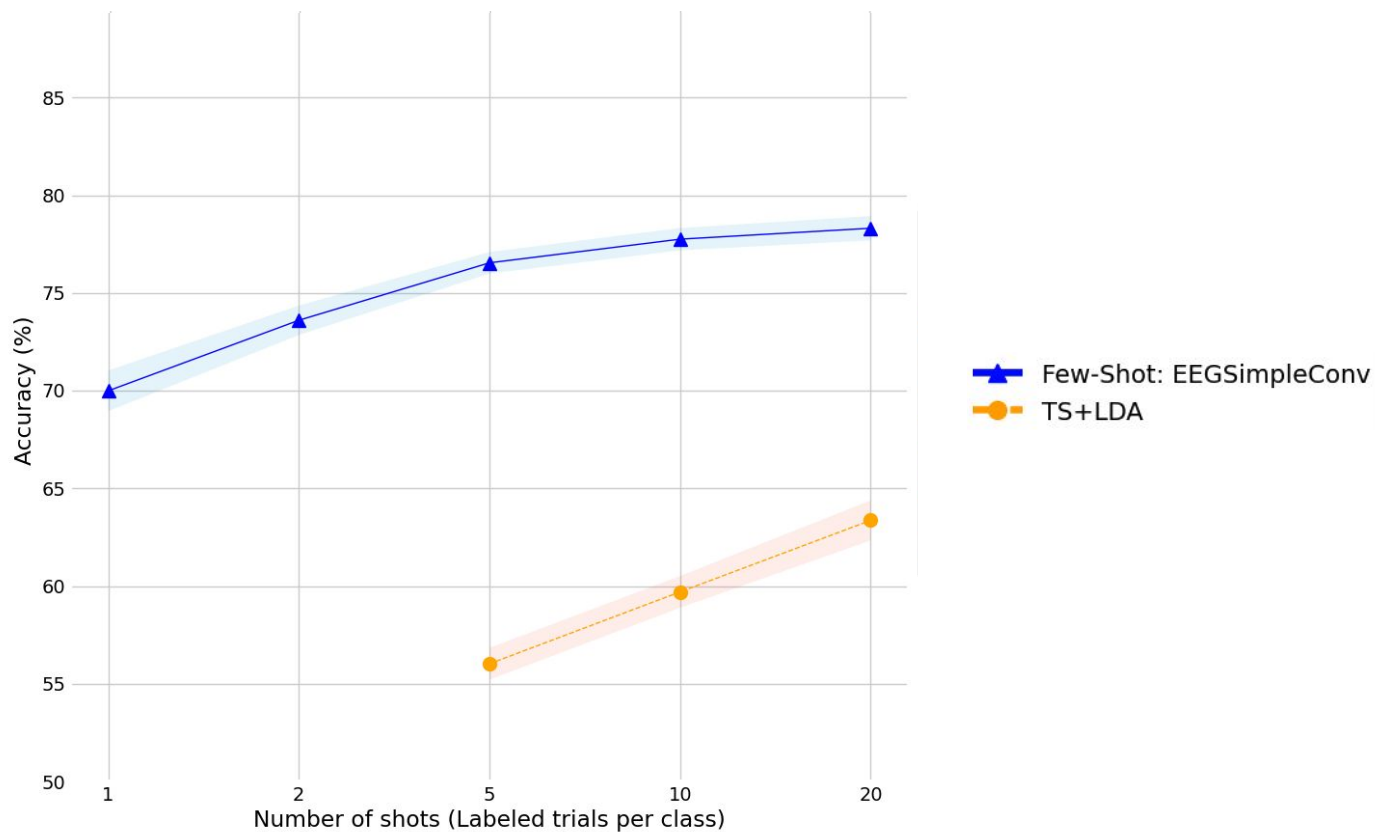
Results

└ Left vs. Right Motor Imagery



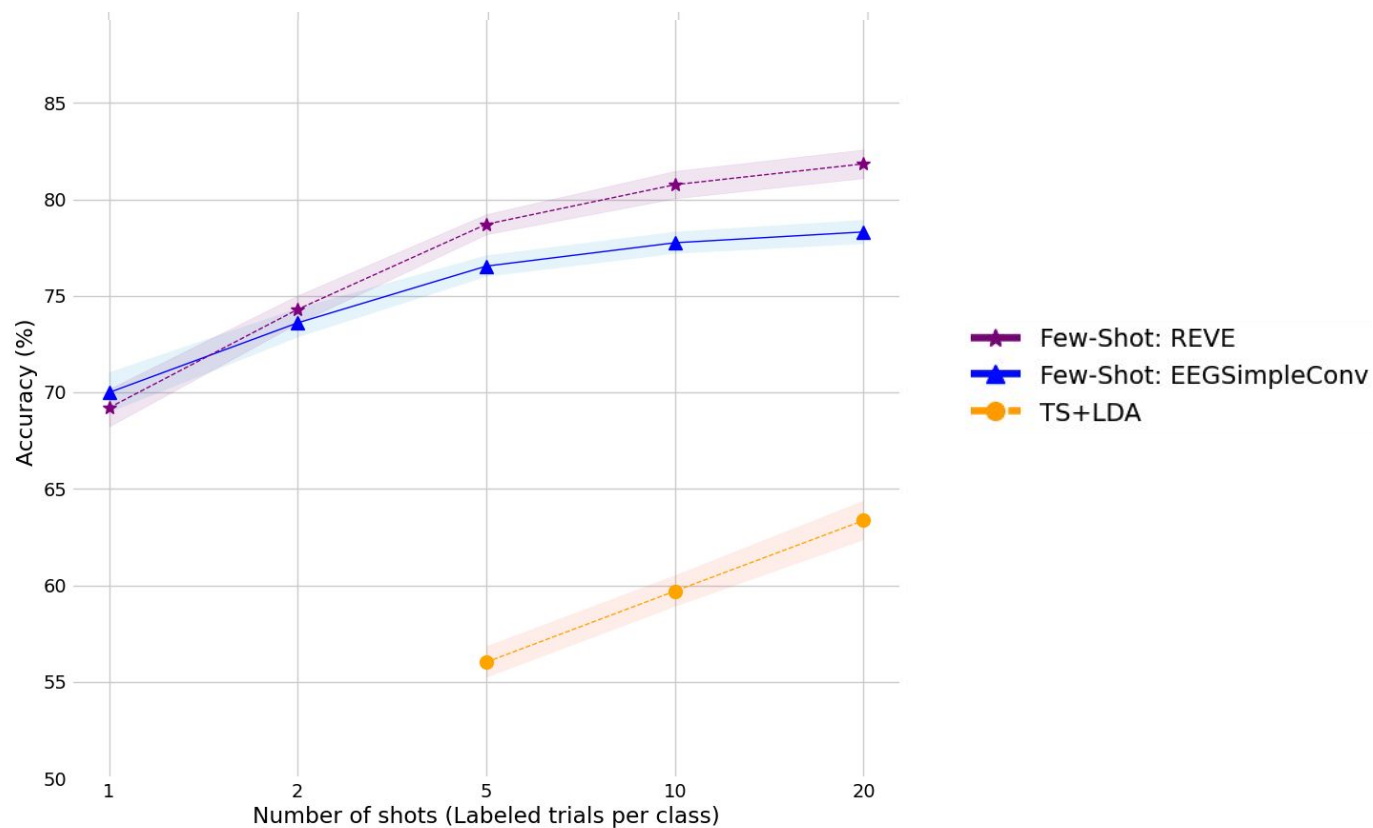
Results

└ Left vs. Right Motor Imagery (PhysionetMI)



Results

└ Left vs. Right Motor Imagery (PhysionetMI)



Comparison with Fine-Tuning approach

- **Below 5 shots:** Few-Shot > Fine-Tuning (2–3 minutes of calibration)

Comparison with Fine-Tuning approach

- **Below 5 shots:** Few-Shot > Fine-Tuning (2–3 minutes of calibration)

Impact of Adaptive methods

- Bring a consistent boost
- Gain more pronounced in inter-session

Comparison with Fine-Tuning approach

- **Below 5 shots:** Few-Shot > Fine-Tuning (2–3 minutes of calibration)

Impact of Adaptive methods

- Bring a consistent boost
- Gain more pronounced in inter-session

Comparison with other setups

- Our method in other setups
- Analysis of performance difference

Other experiments

- **More ML methods:** very competitive, notably in multiclass
- More DL backbones

Outline

- Context
- REVE: EEG Foundation Model
- A realistic Few-Shot Learning setup for BCI
- **Conclusion & Perspectives**



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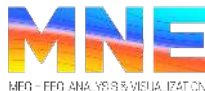
Conclusion and Perspectives

Key contribution

- Foundation model adaptable to any setup
- 4D Encoding strategy and Large Scale Dataset
- Efficient in many DT
- Realistic FSL BCI setup to reduce calibration in BCI

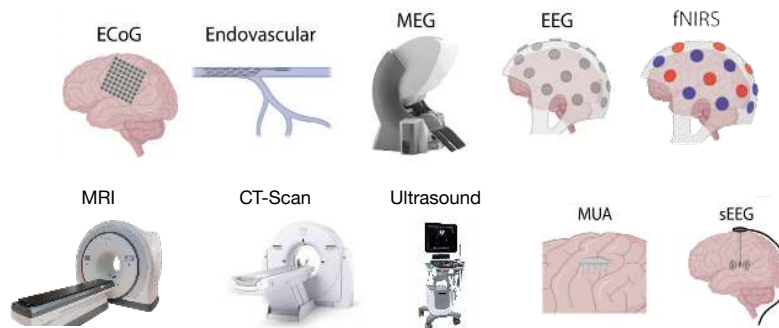
My view on the field

- Lack of clearly established benchmarks
- Rich open-source ecosystem



Perspectives

- Improve REVE (Data curation / More advanced techniques)
- REVE and FSL in real-life BCI system
- **Multimodality:**
 - Brain Foundation model
 - Electrophysiology
 - BCI modalities



- Other health recording techniques
- Other modalities, Text and Vision

Contributions

An efficient 1D-CNN for EEG Decoding

- **El Ouahidi**, Gripon, Pasdeloup, Bouallegue, Farrugia, Lioi. "A Strong and Simple Deep Learning Baseline for BCI Motor Imagery decoding." *IEEE Transactions on Neural Systems and Rehabilitation Engineering* (2024)
- **El Ouahidi**, Drumetz, Lioi, Farrugia, Pasdeloup, Gripon. "Spatial graph signal interpolation with an application for merging BCI datasets with various dimensionalities." *In ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing. IEEE* (2023)

A Foundation Model for EEG

- **El Ouahidi**, Thoeleke, Farrugia, Pasdeloup, Gripon, Lioi, Jerbi. "A review on EEG Foundation Models." *Article in preparation*.
- **El Ouahidi**, Lys, Thoeleke, Farrugia, Pasdeloup, Gripon, Jerbi, Lioi. "REVE: A Foundation Model for EEG - Adapting to Any Setup with Large-Scale Pretraining on 25,000 Subjects." *Under Review. ICML* (2025)

Reducing calibration for BCI systems

- **El Ouahidi**, Lioi, Farrugia, Pasdeloup, Gripon. "Unsupervised Adaptive Deep Learning Method For BCI Motor Imagery Decoding." *In 2024 32nd European Signal Processing Conference. IEEE* (2024)
- **El Ouahidi**, Lotte, Farrugia, Pasdeloup, Gripon, Lioi. "Toward a realistic Few shot Learning setup for BCI Motor Imagery decoding: methods, setups and algorithms." *Article in preparation*.

Other Contributions

- **El Ouahidi**, Tessier, Lioi, Farrugia, Pasdeloup, Gripon. "Pruning Graph Convolutional Networks to select meaningful graph frequencies for fMRI decoding." *In 30th European Signal Processing Conference. IEEE* (2022)
- El Ouahidi, **El Ouahidi** et al. "Machine learning for pacemaker implantation prediction after TAVI using multimodal imaging data." *Scientific Reports* 14 (2024)
- Benoit, Lioi, **El Ouahidi**, Farrugia, Mattout, Maby, Bougard, Dauguet. "Adjusting classical BCI paradigms parameters to optimize performance with a wearable dry-electrode EEG device: towards operational conditions." *Article in preparation*.

Outline

- Context
- REVE: EEG Foundation Model
- A realistic Few-Shot Learning setup for BCI
- Conclusion & Perspectives
- **Additional Slides**



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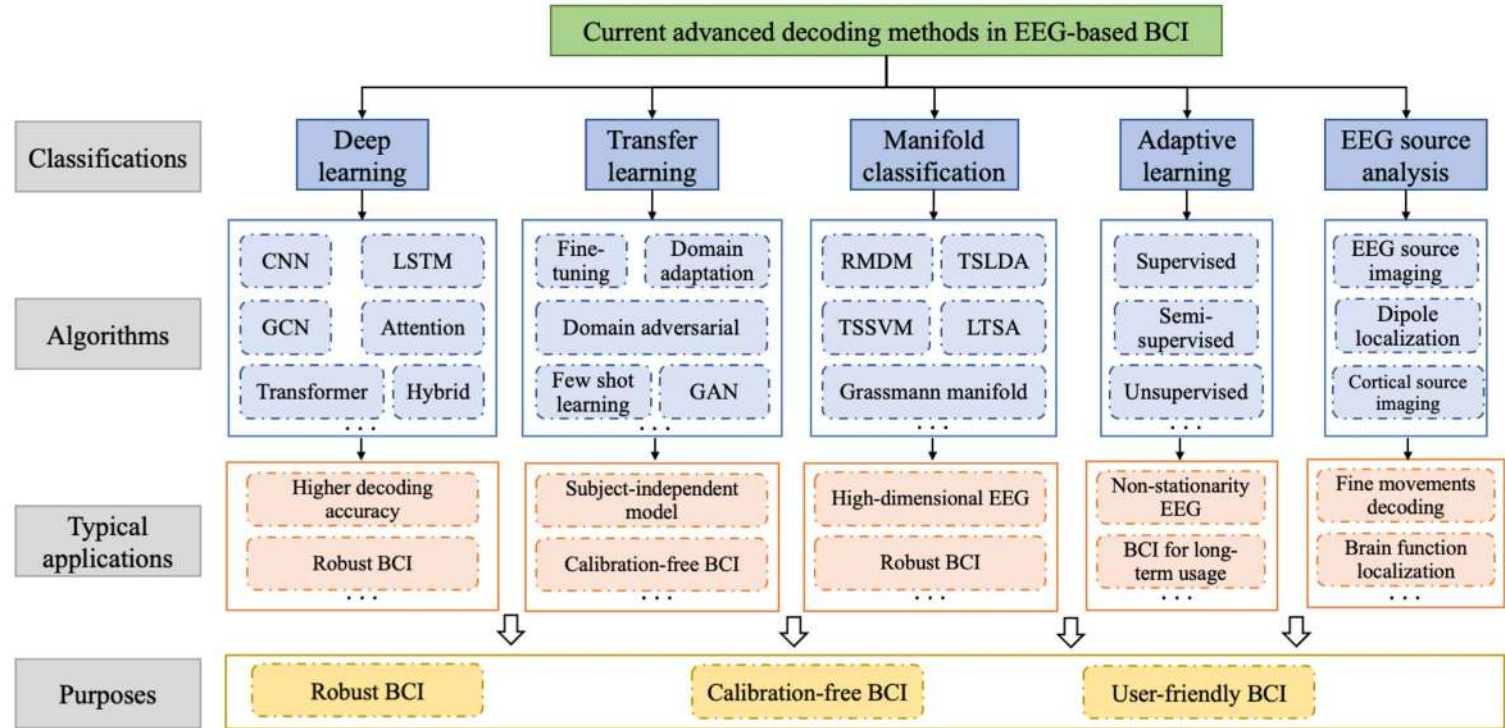


Fig. 6. Overview of recent decoding methods used in EEG-based BCI. These methods can be broadly separated into five main categories: Deep learning, transfer learning, manifold classification, adaptive learning, and EEG source analysis. Each of these categories consists of detailed algorithms that aim to improve various aspects of EEG decoding and BCI performance. In the end, all of these approaches aim to develop robust, calibration-free and user-friendly BCIs.

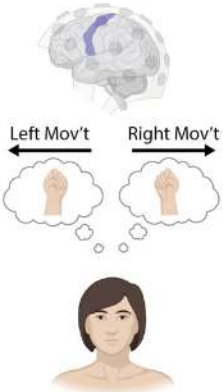
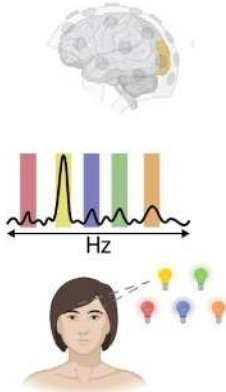
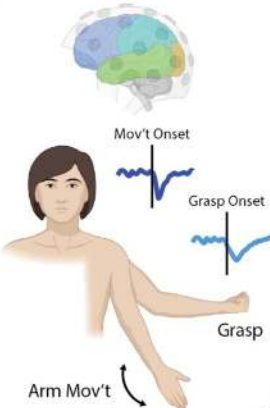
Endogenous		Exogenous		
Frequency Domain		Time Domain		
Motor	Parietal	Visual	Parietal	Whole-Brain
Motor Imagery	Overt Spatial Attention	SSVEPs	P300	Slow Cortical Potentials
				

Fig. 2. Overview of neural signals used for noninvasive brain-computer interface control. These signals can be broadly categorized according to being endogenous or exogenous in origin, frequency or time domain, and by the brain region/electrode coverage from which they are acquired. Motor imagery tasks generate an event-related (de)synchronization that can be detected from motor electrodes and which generate a velocity-based signal for end effector control direction. Overt spatial attention refers to a gaze-based action that is detected in electrodes covering the parietal cortex and which drives an end effector in a particular direction. Steady-state visual evoked potentials (SSVEPs) refer to increases in narrow band power in electrodes covering the visual cortices upon attending to a stimulus flickering at the corresponding frequency. The P300 response is elicited during an oddball-type context when a user's choice is selected. Finally, slow cortical potentials refer to electrical potential deflections that are time-locked to movement events or that correspond to limb kinematics.

EEGSimpleConv: a strong, simple deep learning baseline for BCI Motor Imagery decoding

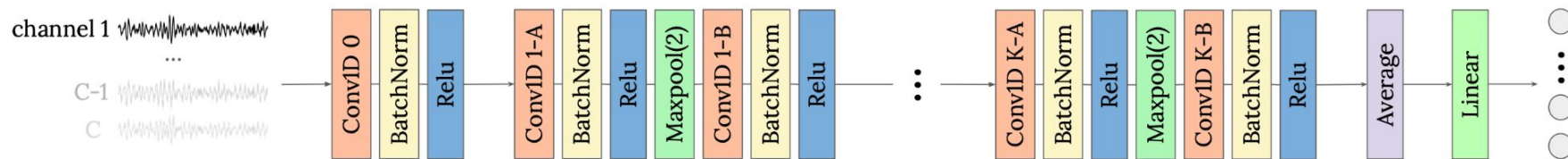


Figure 2.1 – EEGSimpleConv architecture

EEGSimpleConv: a strong, simple deep learning baseline for BCI Motor Imagery decoding

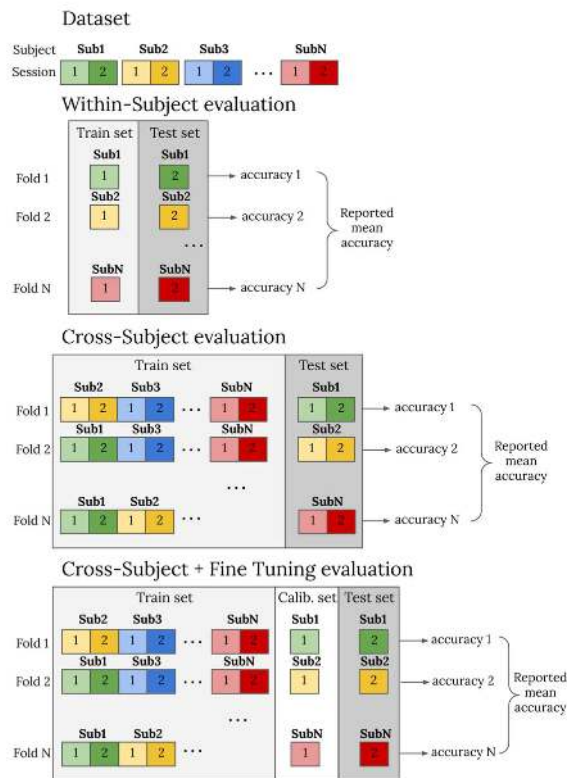


Figure 1.4 – Evaluation strategies for EEG datasets: Within-Subject, Cross-Subject, and Cross-Subject with Fine-Tuning, showing train, test, and calibration set splits for performance assessment, under a Leave-One-Out protocol.

EEGSimpleConv: a strong, simple deep learning baseline for BCI Motor Imagery decoding

		W-S	C-S	C-S F-T
DL	EEGSimpleConv	78.4 ± 10.6	72.1 ± 7.3	86.2 ± 6.3
	EEG Conformer [23]	78.7 ± 14.4	—	—
	EEG-ITNet [86]	76.7 ± 11.1	69.4 ± 8.9	78.7 ± 9.4
	EEG-TCNet [82] ¹	74.5 ± 10.1	65.1 ± 10.9	75.8 ± 10.2
	CNN-SPDNet [97]	74.2	—	—
	Shallow ConvNet [14]	73.7	—	—
	EEGNet [15] ¹	73.7 ± 11.1	64.0 ± 11.6	73.9 ± 12.2
	EEGNet [15] ²	—	66.2 ± 9.9	—
	EEG-Inception [85] ¹	73.5 ± 9.1	66.3 ± 8.7	75.0 ± 9.8
	Tensor-SPDNet [94]	73.0	—	—
	Multi-view CNN [101]	72.5 ± 14.1	—	—
	GNN-SPDNet [95]	72.0	—	—
	Hybrid ConvNet [14]	71.6	—	—
	Deep ConvNet [14]	70.9	—	—
	Residual ConvNet [14]	67.7	—	—
	DFNN [112]	—	64.4	—
	CCNN [100]	—	55.3	—
ML	CMO-CNN [77]	—	63.3	—
	Multi-branch 3D [99]	—	52.2	—
	TIDNet [83]	—	65.4 ³	77.4 ³
	CSP+LDA ²	57.7 ± 14.9	—	—
	FBCSP+LDA ²	63.7 ± 10.4	—	—
	TS+LDA ²	65.4 ± 12.9	—	—

¹ Results reproduced by [86] ² by us ³ 1 subject removed

Table 2.2 – Performance (accuracy expressed in %) comparison to literature methods on BNCI using **offline** evaluation setups. Within-Subject (W-S), Cross-Subject (C-S), and Cross-Subject with Fine-Tuning (C-S F-T). Methods are grouped based on whether they use deep learning (DL) or machine learning (ML) methods. Standard deviations (stds or ±) represent the variation across subjects.

Offline evaluation					Online evaluation			
W-S		C-S		C-S F-T	W-S	C-S		C-S F-T
Test	Session 2	Sess. 1&2	Sess. 2	Sess. 2	Sess. 2	Sess. 1&2	Sess. 2	Sess. 2
S0	86.3 ± 1.3	78.9 ± 1.7	79.7 ± 1.5	89.0 ± 1.5	81.0 ± 2.0	62.5 ± 1.5	64.7 ± 2.0	86.0 ± 1.1
S1	59.1 ± 1.1	57.7 ± 1.4	57.3 ± 1.9	72.8 ± 2.3	54.7 ± 2.4	50.2 ± 1.0	49.0 ± 2.0	65.8 ± 4.0
S2	91.3 ± 0.9	83.0 ± 1.1	85.4 ± 2.4	94.1 ± 1.3	87.4 ± 1.3	67.4 ± 2.3	66.8 ± 3.2	87.6 ± 1.6
S3	77.3 ± 1.7	66.7 ± 1.6	71.5 ± 2.2	89.3 ± 0.8	71.9 ± 1.9	54.5 ± 3.0	57.6 ± 3.2	85.7 ± 1.8
S4	68.3 ± 2.3	70.9 ± 1.8	71.0 ± 2.0	82.2 ± 2.0	44.3 ± 5.0	56.0 ± 2.7	54.9 ± 4.1	71.5 ± 3.0
S5	68.3 ± 1.3	66.6 ± 0.7	65.6 ± 1.5	80.3 ± 2.3	54.5 ± 4.0	47.2 ± 2.7	44.8 ± 2.4	68.8 ± 4.5
S6	89.9 ± 1.0	75.7 ± 1.3	74.0 ± 1.9	92.9 ± 0.6	86.5 ± 2.9	69.1 ± 1.5	70.2 ± 2.0	74.8 ± 1.7
S7	87.2 ± 1.0	77.5 ± 1.5	78.1 ± 1.0	88.7 ± 1.4	84.0 ± 1.7	60.9 ± 2.6	62.2 ± 2.2	86.7 ± 1.5
S8	77.8 ± 0.7	71.6 ± 1.6	71.7 ± 1.9	86.1 ± 1.2	64.4 ± 1.2	58.0 ± 3.8	54.3 ± 4.1	81.9 ± 2.8
Average	78.4 ± 10.6	72.1 ± 7.3	72.7 ± 7.7	86.2 ± 6.3	70.0 ± 15.1	58.4 ± 6.9	58.3 ± 7.9	78.8 ± 8.1
+EOG	82.2 ± 9.2	79.6 ± 6.1	80.1 ± 7.2	90.2 ± 5.5				

Table 2.3 – EEGSimpleConv performance (accuracy expressed in %) on BNCI on various evaluation setups. Stds (±) on each subject lines represent the variation across runs, while stds on the Average line represent the variation across subjects.

	BNCI	Zhou	Physionet	Cho	Avg. Gain
Full pipeline	72.1 ± 7.3	81.8 ± 1.2	64.6 ± 4.4	75.4 ± 4.6	—
- BN	67.1 ± 8.4	77.3 ± 2.9	62.8 ± 4.2	74.5 ± 5.5	-3.0
- EA	66.7 ± 6.5	80.5 ± 1.3	63.0 ± 4.6	74.6 ± 4.6	-2.3
- Session	70.4 ± 7.2	79.9 ± 1.4	—	—	-1.8
Online	58.4 ± 6.9	73.1 ± 6.6	61.0 ± 4.9	73.5 ± 5.3	-6.8
- Mixup	69.1 ± 9.2	80.1 ± 0.7	60.0 ± 4.2	73.8 ± 5.2	-2.7
- Reg S	71.4 ± 7.9	81.8 ± 1.0	63.6 ± 4.1	74.2 ± 4.4	-0.72
- Everything	56.4 ± 9.0	74.5 ± 3.6	57.4 ± 4.6	72.6 ± 4.9	-8.3
+ EOG	79.8 ± 5.8	81.9 ± 1.6	—	—	+3.9
- LMSO	—	—	65.0 ± 14.3	76.2 ± 10.3	+0.6

Table 2.4 – Ablation study of our training pipeline ingredients (accuracy expressed in %). Stds ($\pm$) represent the variation across subjects. The “Average Gain” column is computed by subtracting the average accuracy of the full pipeline over each dataset (the first line) to the average accuracy of a given ablation line over each dataset.

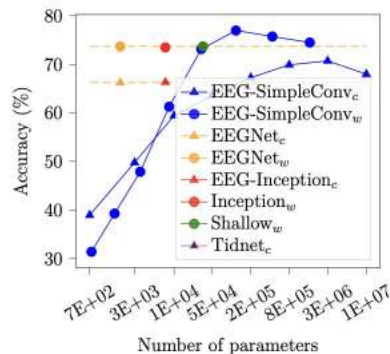
EEGSimpleConv: a strong, simple deep learning baseline for BCI Motor Imagery decoding

	Inference time (s)		Size (p)
	GPU	CPU	
Shallow ConvNet	$2.65e^{-04}$	$7.68e^{-04}$	38404
EEGSimpleConv _w	$3.60e^{-04}$	$5.38e^{-04}$	245654
EEGNet	$4.82e^{-04}$	$1.17e^{-03}$	1972
EEGSimpleConv _c	$1.05e^{-03}$	$2.56e^{-03}$	2462246
TIDNet	$1.06e^{-03}$	$2.38e^{-03}$	187028
EEG-Inception	$1.67e^{-03}$	$2.54e^{-03}$	9952

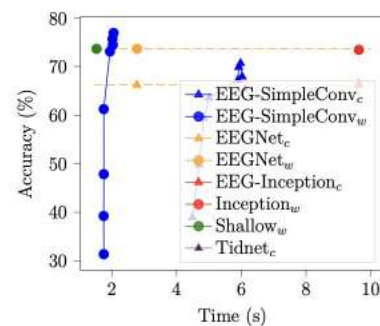
Table 2.5 – Latency and size comparison to literature methods

Model	A100	3090	CPU
REVE-tiny	3.3×10^{-3}	3.6×10^{-3}	1.6×10^{-2}
REVE-small	3.1×10^{-3}	3.6×10^{-3}	8.1×10^{-2}
REVE-base	1.4×10^{-2}	1.8×10^{-2}	2.5×10^{-1}
REVE-Large	1.4×10^{-2}	2.1×10^{-2}	1.1
EEGSimpleConv	1.4×10^{-3}	3.2×10^{-3}	7.7×10^{-3}

Table 1: Computation time in seconds comparison of different models across hardware configurations.



(a) Impact of the size



(b) Impact of the latency

Figure A.5 – Impact of the size and the latency on the classification performance of EEGSimpleConv. x-axis of (a) is in log-scale.

EEGSimpleConv: a strong, simple deep learning baseline for BCI Motor Imagery decoding

	TS + LDA	CSP+LDA	FBCSP+LDA
S0	77.4	68.1	75.4
S1	52.8	52.8	56.3
S2	84.0	73.6	78.8
S3	60.1	53.1	64.6
S4	47.9	27.8	50.7
S5	49.3	42.4	47.2
S6	64.2	55.6	74.3
S7	74.0	71.5	63.2
S8	78.8	74.3	63.2
mean	65.4 ± 12.9	57.7 ± 14.9	63.7 ± 10.4

Table A.1 – Subjects performances (accuracy expressed in %) using machine learning baselines performances on BNCI. Stds ($\pm$) represent the variation across subject

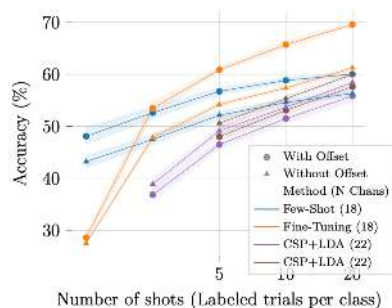
Session	BN	EA	Z0	Accuracy (%)
✓	✓	✓	✓	72.1
✓	✓	✓	✗	71.9
✓	✓	✗	✓	67.7
✓	✓	✗	✗	67.5
✓	✗	✓	✓	67.1
✓	✗	✓	✗	64.0
✓	✗	✗	✓	58.4
✓	✗	✗	✗	58.3
✗	✓	✓	✓	71.4
✗	✓	✓	✗	70.4
✗	✓	✗	✓	67.1
✗	✓	✗	✗	68.1
✗	✗	✓	✓	63.6
✗	✗	✓	✗	63.5
✗	✗	✗	✓	58.2
✗	✗	✗	✗	58.6

Table A.3 – Normalisation pipeline full ablation

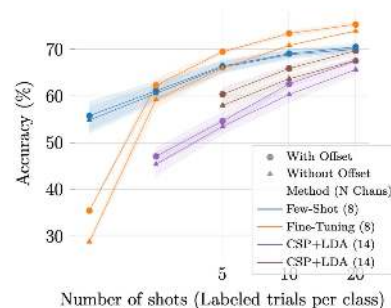
EEGSimpleConv: a strong, simple deep learning baseline for BCI Motor Imagery decoding

Remove offset (0.5 s.)	ICA EOG artifact removal	Acc. (%)	
		BNCI	Zhou
x	x	72.1	80.2
x	✓	67.0	79.1
✓	x	63.0	77.6
✓	✓	61.5	76.9

Table A.4 – Ablation study (accuracy in %) to evaluate the impact of non MI brain activity, and ocular artifacts



(a) Analysis on BNCI2014001 (4 classes)



(b) Analysis on Zhou2016 (3 classes)

Figure B.1 – Offset removal analysis

EEGSimpleConv: a strong, simple deep learning baseline for BCI Motor Imagery decoding

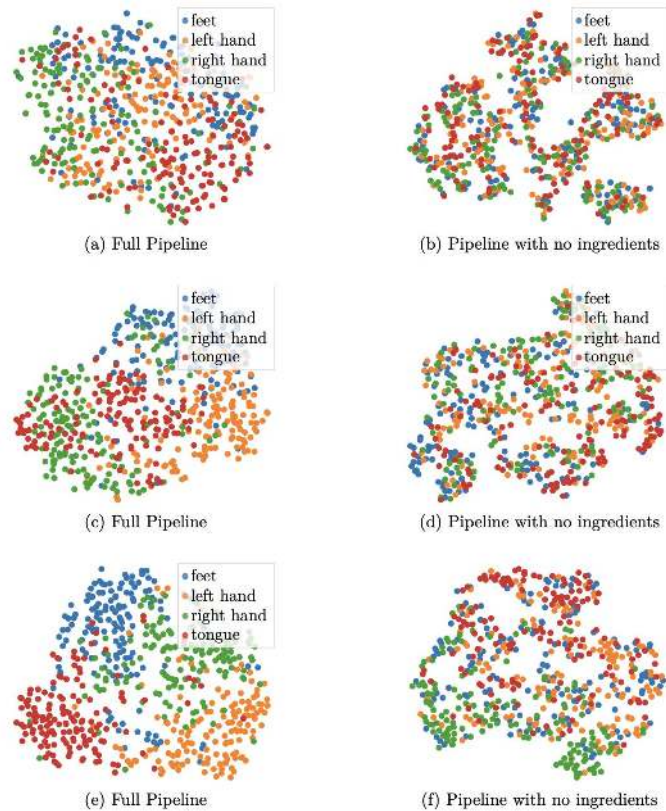


Figure A.2 – t-SNE Embeddings. From top to bottom: (a,b) the “hardest”, (c,d) the “average” and (e,f) the “easiest” BNCI subject to classify

EEGSimpleConv: a strong, simple deep learning baseline for BCI Motor Imagery decoding

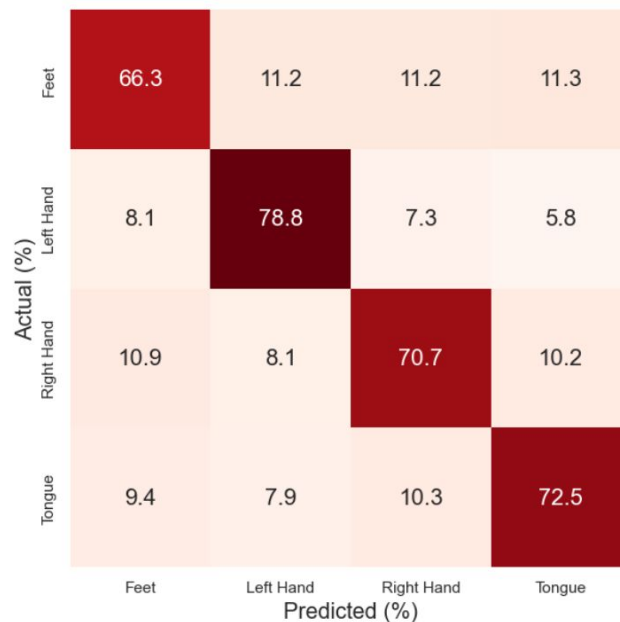


Figure A.3 – Confusion Matrix: average of the confusion matrices obtained across runs and subjects on BNCI (C-S, offline)

	Within	Cross
Kernel Size	[12 – 17]	[5 – 8]
Length (W)	[64 – 144]	[64 – 144]
Depth (K)	1	[2 – 4]
Sampling Freq. (Hz.)	[70 – 100]	[50 – 80]
High-Pass Freq. (Hz.)	0.5	0.5

Table A.5 – Hyperparameter Search space

EEGSimpleConv: a strong, simple deep learning baseline for BCI Motor Imagery decoding

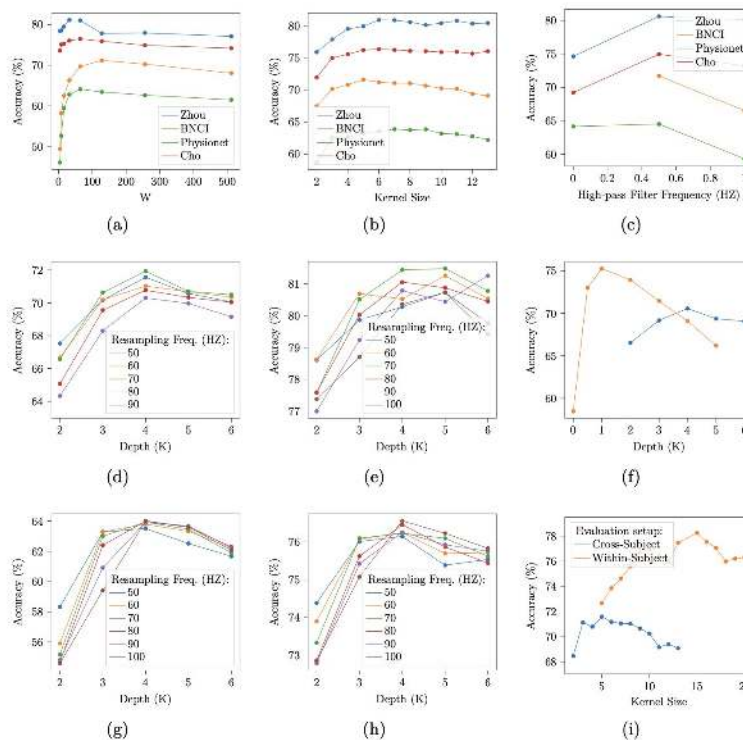
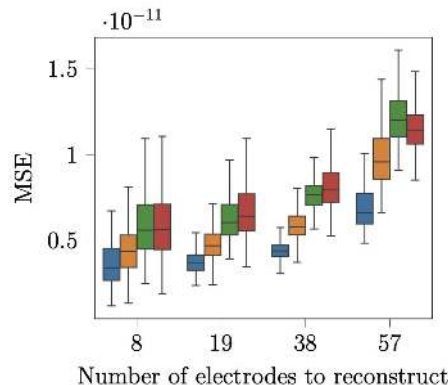
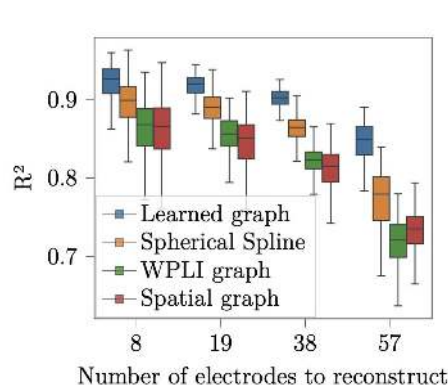


Figure A.4 Impact of the variations of EEGSimpleConv's hyperparameters of EEGSimpleConv on the Cross-Subject classification accuracy on four datasets. (a) variation of the width W , (b) variation of the size of the convolutions kernels, (c) variation of the High-pass filter frequency, variation of the Depth K on (d) BNCL, (e) Zhou, (g) Physionet, (h) Cho. Impact of the variation of (f) the Depth K and (i) the kernel size on Cross-Subject performances compared to Within-Subject on BNCL.

Remove offset (0.5 s.)	ICA EOG artifact removal	Acc. (%)	
		BNCI	Zhou
x	x	72.1	80.2
x	✓	67.0	79.1
✓	x	63.0	77.6
✓	✓	61.5	76.9

Table A.4 – Ablation study (accuracy in %) to evaluate the impact of non MI brain activity, and ocular artifacts

Spatial Graph Signal Interpolation with an application for Merging BCI Datasets with various Dimensionalities

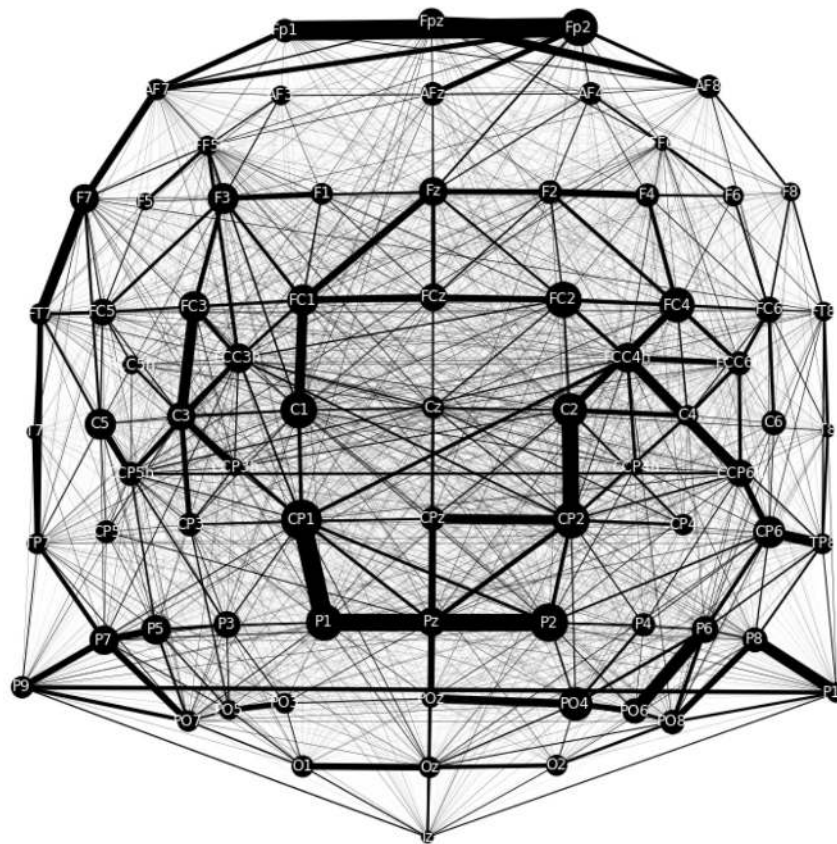


Method	Physionet [110]	BNCI [107]	Cho [109]	Shin [129]	Zhou [108]
Transfer	52.5±3.6	56.3±30.4	18.8±11.8	33.8±17.3	8.3±18.1
Transfer + FT	86.1±1.2	93.2±2.8	77.1±2.3	90.4±1.7	70.0±9.2
Spatial	59.3±2.6	79.3±4.8	35.0±6.8	73.4±4.2	32.9±18.4
Spherical	80.6±4.5	94.9±1.5	75.6±4.4	85.9±4.2	54.1±21.8

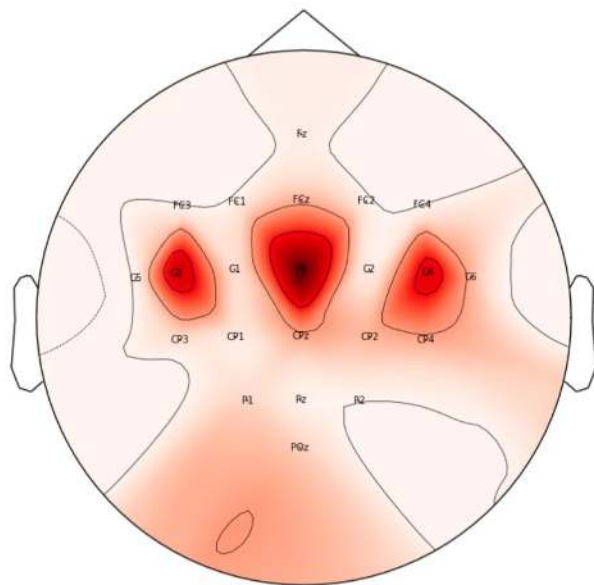
Table 2.7 – Reconstruction R^2 of half of the electrodes for various methods and dataset

	Shin [129]		Zhou [108]	
Config	Accuracy	N	Accuracy	N
Intersection	53.2±2.8	2	61.2±2.0	9
Dataset	63.2±2.3	22	56.2±4.8	14
Union	62.3±2.1	76	46.4±2.8	76

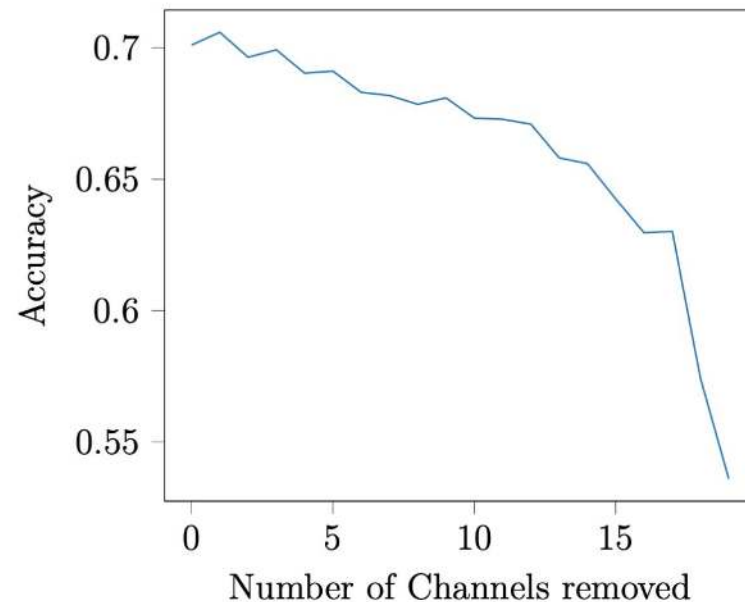
Table 2.8 – Generalization accuracy on the two target datasets. N stands for the number of electrodes. “Intersection” refers to the electrodes common to the 4 datasets. “Dataset” corresponds to the electrodes of the dataset in the column. “Union” represents the union of electrodes across all 5 datasets.



Pruning techniques to select meaningful EEG channels for BCI applications



(a) Topographic map showing the most commonly selected electrodes ($K=4$), across multiple runs



(b) Classification accuracy in function of the number of removed EEG channels

Figure 2.4 – Selection of EEG channels via pruning on the BNCI MI dataset

EEG-to-speech decoding with a CLIP-based approach

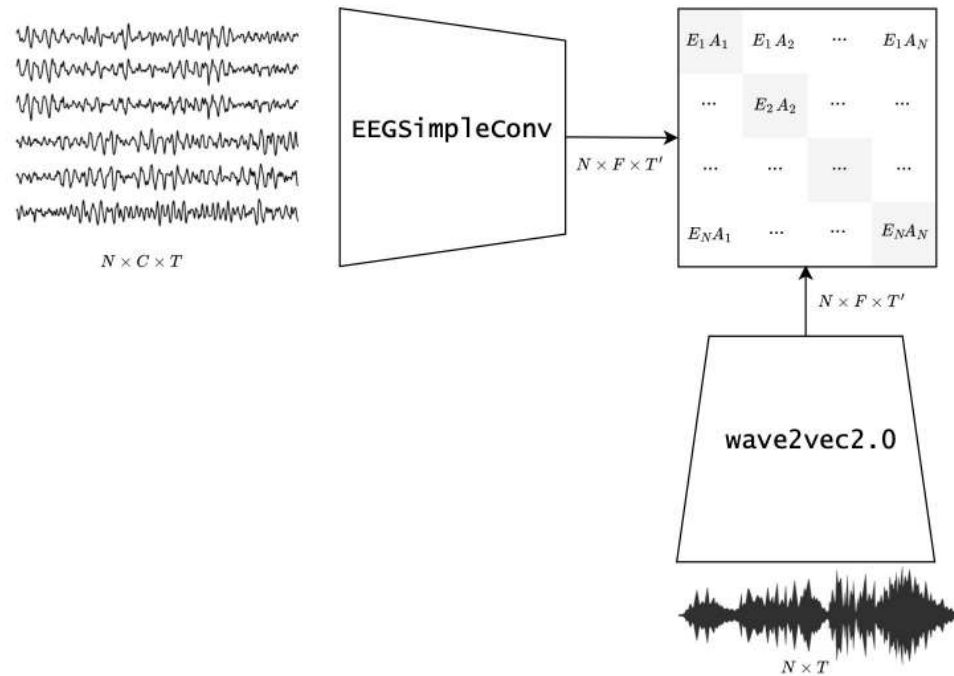


Figure 2.5 – CLIP EEG Audio Framework

EEG-to-speech decoding with a CLIP-based approach

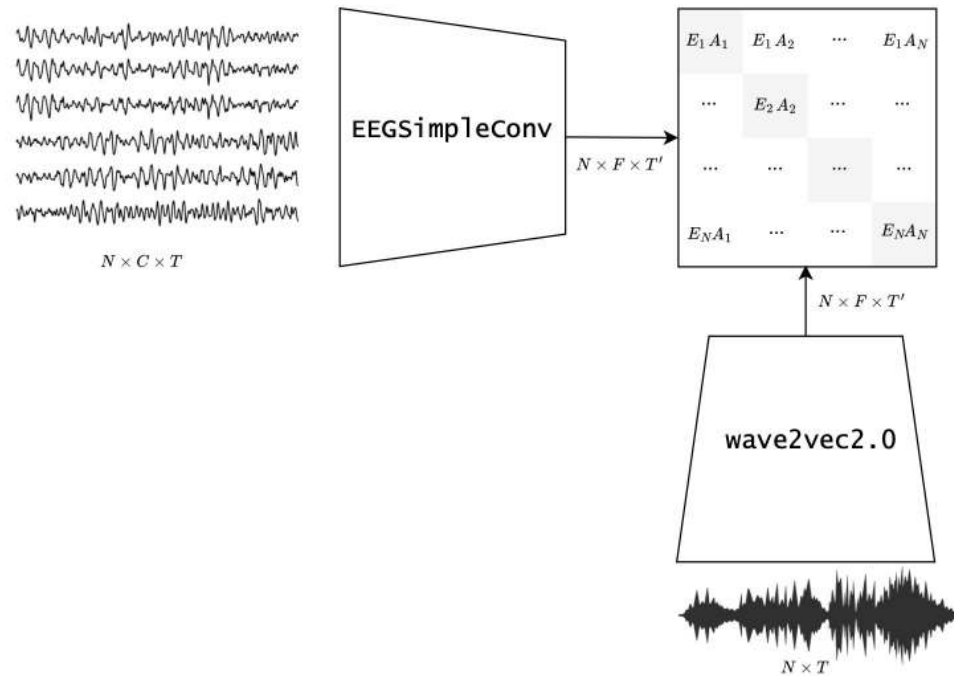


Figure 2.5 – CLIP EEG Audio Framework

Unsupervised adaptive deep learning method for BCI motor imagery decoding

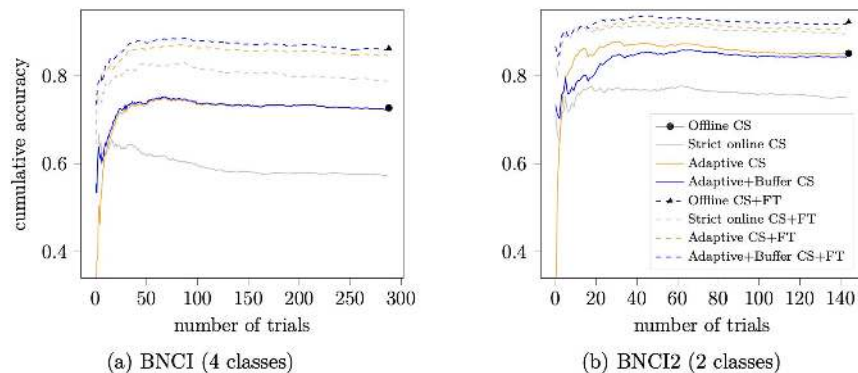


Figure 3.1 – Adaptive classifier performance compared to strict online and offline baseline in Cross-Subject (CS) and Cross-Subject Fine-Tuning (CS+FT) evaluations.

	BNCI	BNCI2	Large
	Cross-Subject		
Strict online	56.9	75.1	77.5
Adaptive	72.1	84.9	86.5
Adaptive + Buffer	72.2	84.2	88.3
Offline	72.7	85.1	89.3
	Cross-Subject + Fine-Tuning		
Strict online	78.8	89.5	91.2
Adaptive	84.6	90.7	91.0

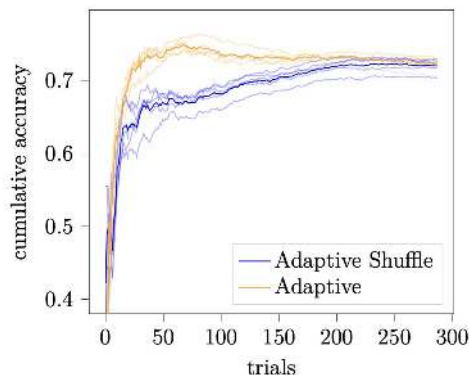


Figure 3.2 – Impact of shuffling the data to test the hypothesis on user fatigue (BNCI dataset, Cross-Subject without buffer). Lighter lines correspond to each run and thicker lines are the average across runs.

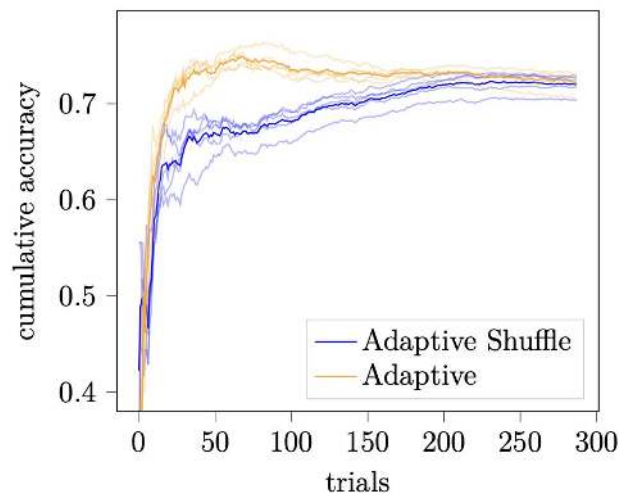


Figure 3.2 – Impact of shuffling the data to test the hypothesis on user fatigue (BNCI dataset, Cross-Subject without buffer). Lighter lines correspond to each run and thicker lines are the average across runs.

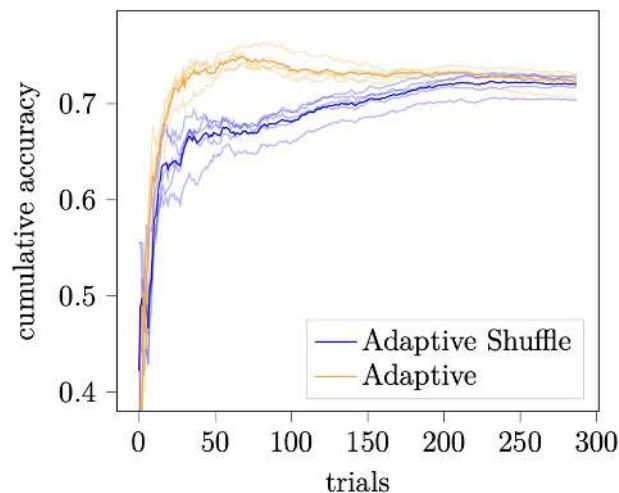
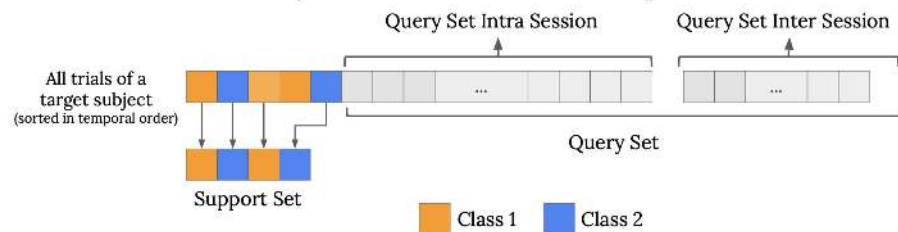
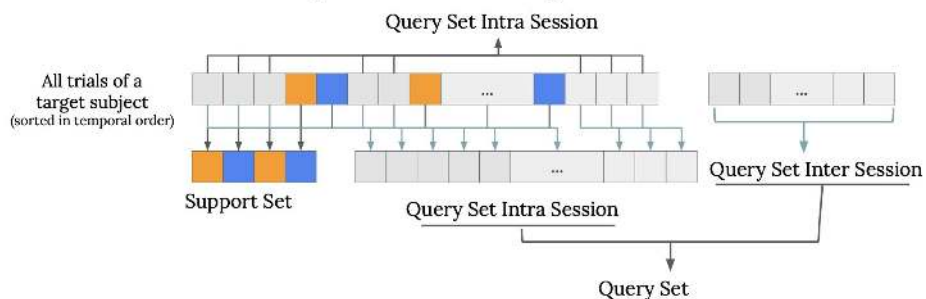


Figure 3.2 – Impact of shuffling the data to test the hypothesis on user fatigue (BNCI dataset, Cross-Subject without buffer). Lighter lines correspond to each run and thicker lines are the average across runs.

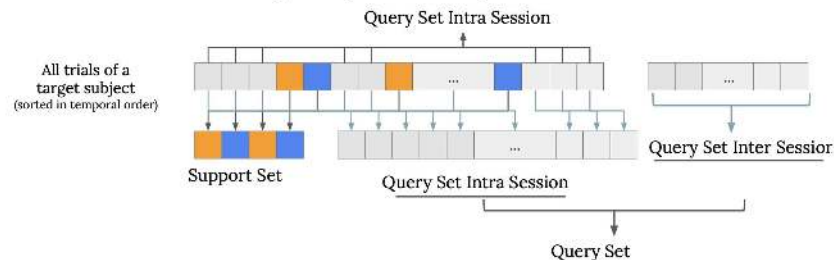
1) Online Inductive Setup



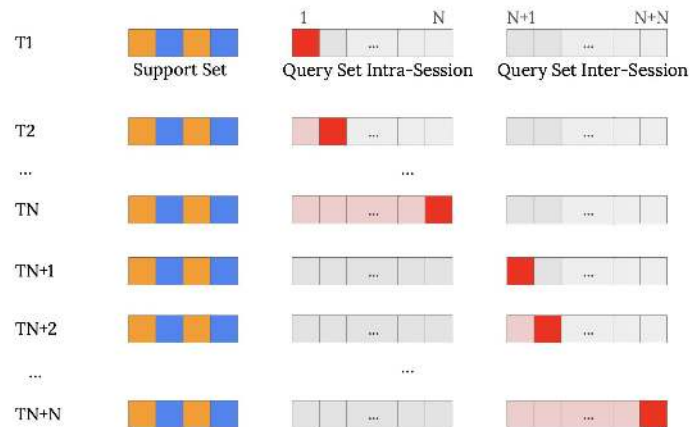
2) Inductive Setup



3) Adaptive Setup



At T_i ■ Query being classified ■ Query available ■ Query unavailable



(Query Set = Query Set Intra-Session + Query Set Inter-Session)

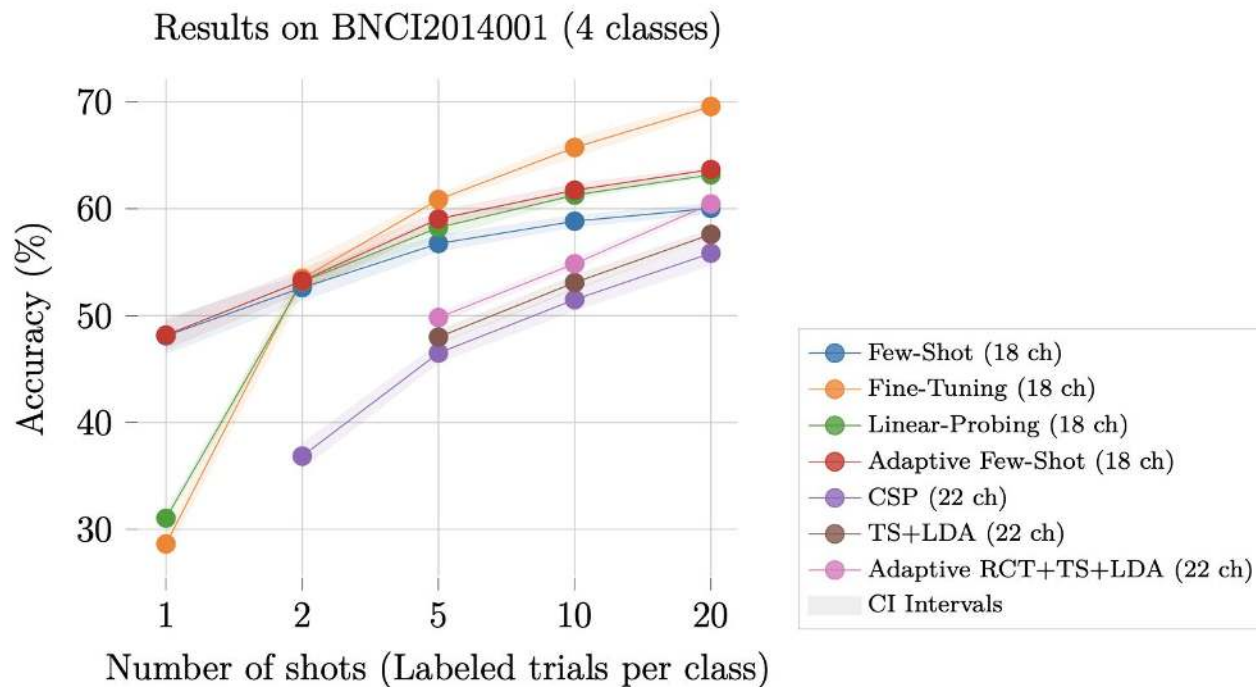
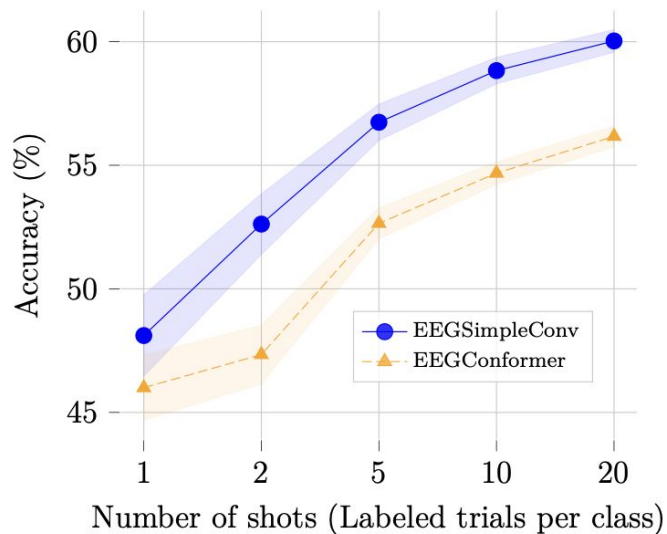
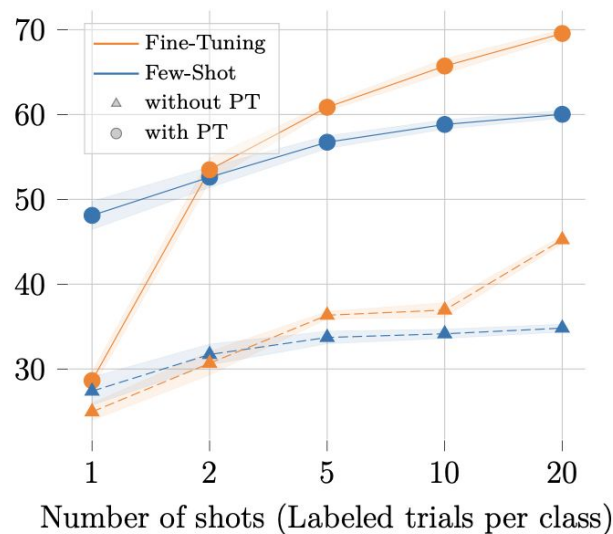


Figure 3.6 – Comparison of deep learning methods using EEGSimpleConv vs machine learning methods



(a) Few-Shot method per backbones.



(b) Impact of Pretraining (PT).

Figure 3.7 – Performance analysis on BNCI2014001 (4 classes). The left panel compares the Few-Shot method with different backbone architectures, while the right panel evaluates the impact of pretraining on performance.

Shots	Adaptive	Average		Intra-session		Inter-session	
Few-Shot							
1	✓/ ✗/ Δ	48.11 ± 1.65 / 48.18 ± 1.31 / +0.07	48.86 ± 1.79 / 47.68 ± 1.27/ -1.18	47.36 ± 1.67 / 48.68 ± 1.68/ +1.32			
2	✓/ ✗/ Δ	52.62 ± 1.23 / 53.25 ± 1.05 / +0.63	53.61 ± 1.33 / 53.35 ± 0.99/ -0.26	51.63 ± 1.30 / 53.16 ± 1.18/ +1.53			
5	✓/ ✗/ Δ	56.74 ± 0.74 / 59.04 ± 0.74 / +2.30	57.91 ± 0.85 / 59.69 ± 0.81/ +1.78	55.58 ± 0.79 / 58.38 ± 0.79/ +2.80			
10	✓/ ✗/ Δ	58.83 ± 0.54 / 61.73 ± 0.60 / +2.90	60.04 ± 0.60 / 62.54 ± 0.87/ +2.50	57.61 ± 0.66 / 60.91 ± 0.47/ +3.30			
20	✓/ ✗/ Δ	60.03 ± 0.46 / 63.66 ± 0.22 / +3.63	61.38 ± 0.59 / 64.65 ± 0.37/ +3.27	58.68 ± 0.57 / 62.67 ± 0.39/ +3.99			
TS+LDA							
5	✓/ ✗/ Δ	49.82 ± 0.49 / 47.99 ± 0.61 / +1.83	50.35 ± 0.64 / 50.21 ± 0.84 / +0.14	49.29 ± 0.44 / 45.77 ± 0.72 / +3.52			
10	✓/ ✗/ Δ	54.85 ± 0.50 / 53.09 ± 0.67 / +1.76	56.04 ± 0.59 / 55.78 ± 0.61 / +0.26	53.66 ± 0.82 / 50.40 ± 1.18 / +3.26			
20	✓/ ✗/ Δ	60.45 ± 0.33 / 57.60 ± 0.48 / +2.85	61.64 ± 0.51 / 60.77 ± 0.72 / +0.87	59.26 ± 0.37 / 54.43 ± 0.68 / +4.83			

Table 3.5 – Performance comparison between intra-session and inter-session setups on BNCI2014001 for the FSL and Riemannian methods, with and without adaptive strategies. Δ indicates the performance gain from using the adaptive setup.

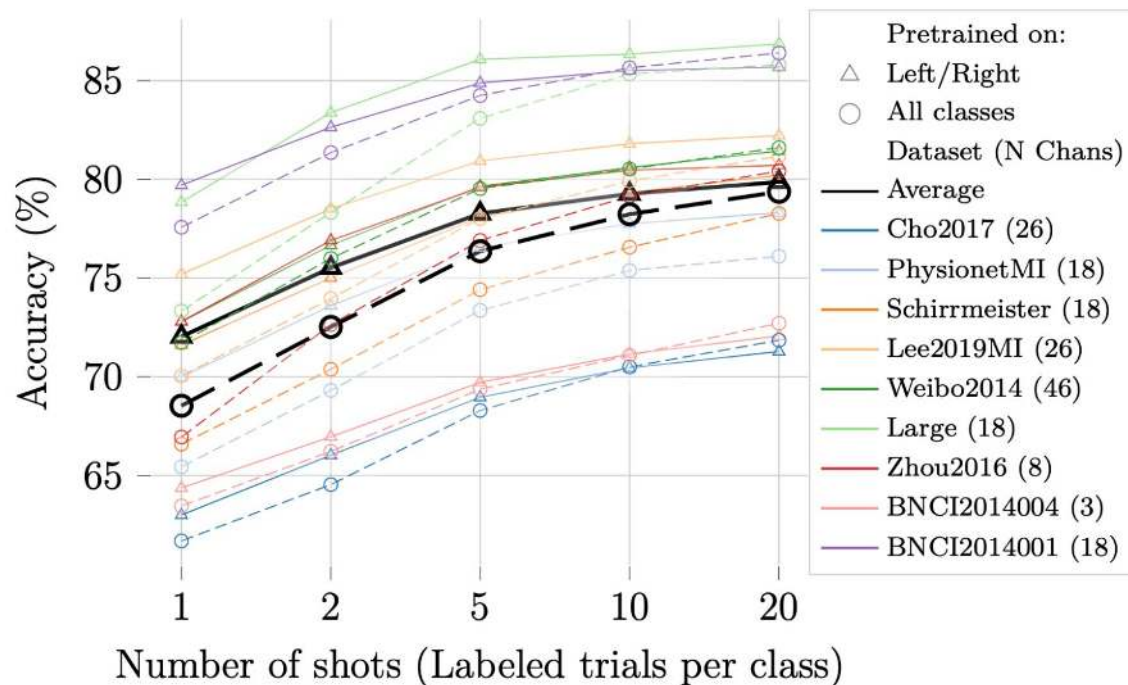


Figure 3.8 – Left vs Right MI Classification, Pretraining classes analysis for the Few-Shot method

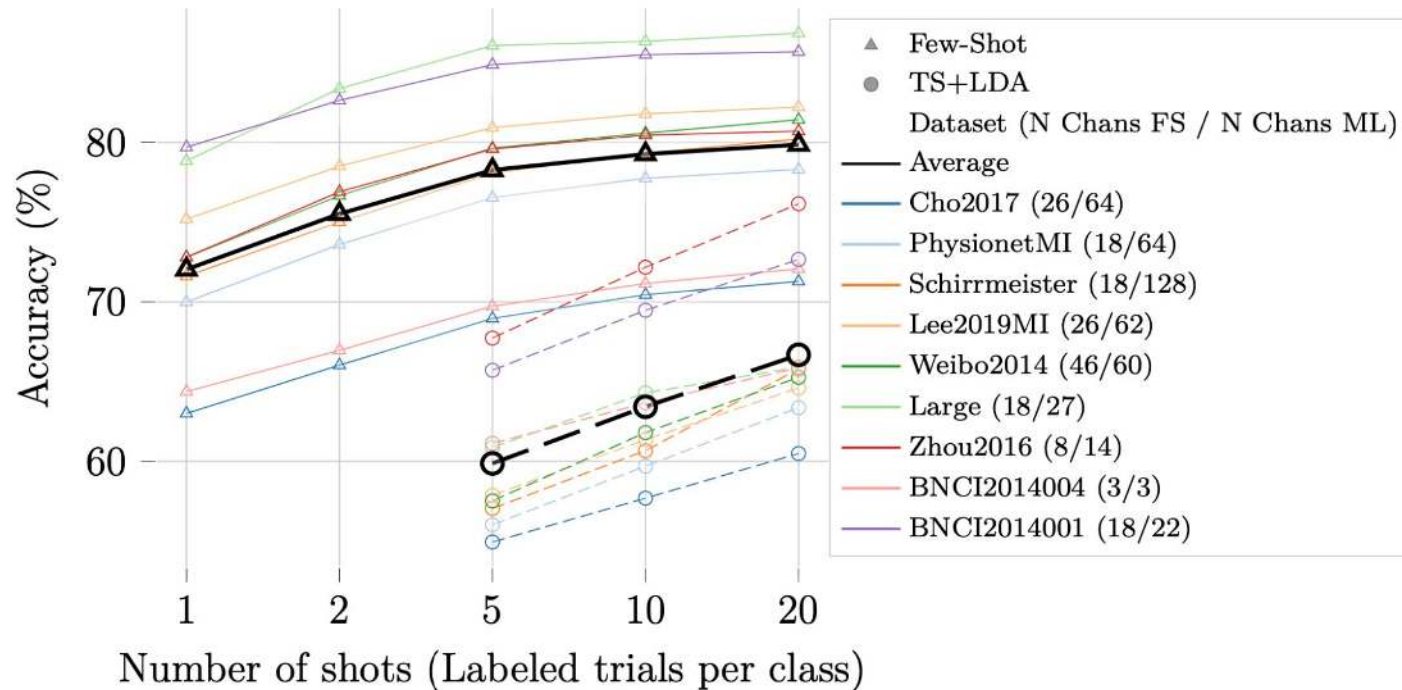


Figure 3.9 – Left vs Right MI Classification - Few-Shot vs TS+LDA

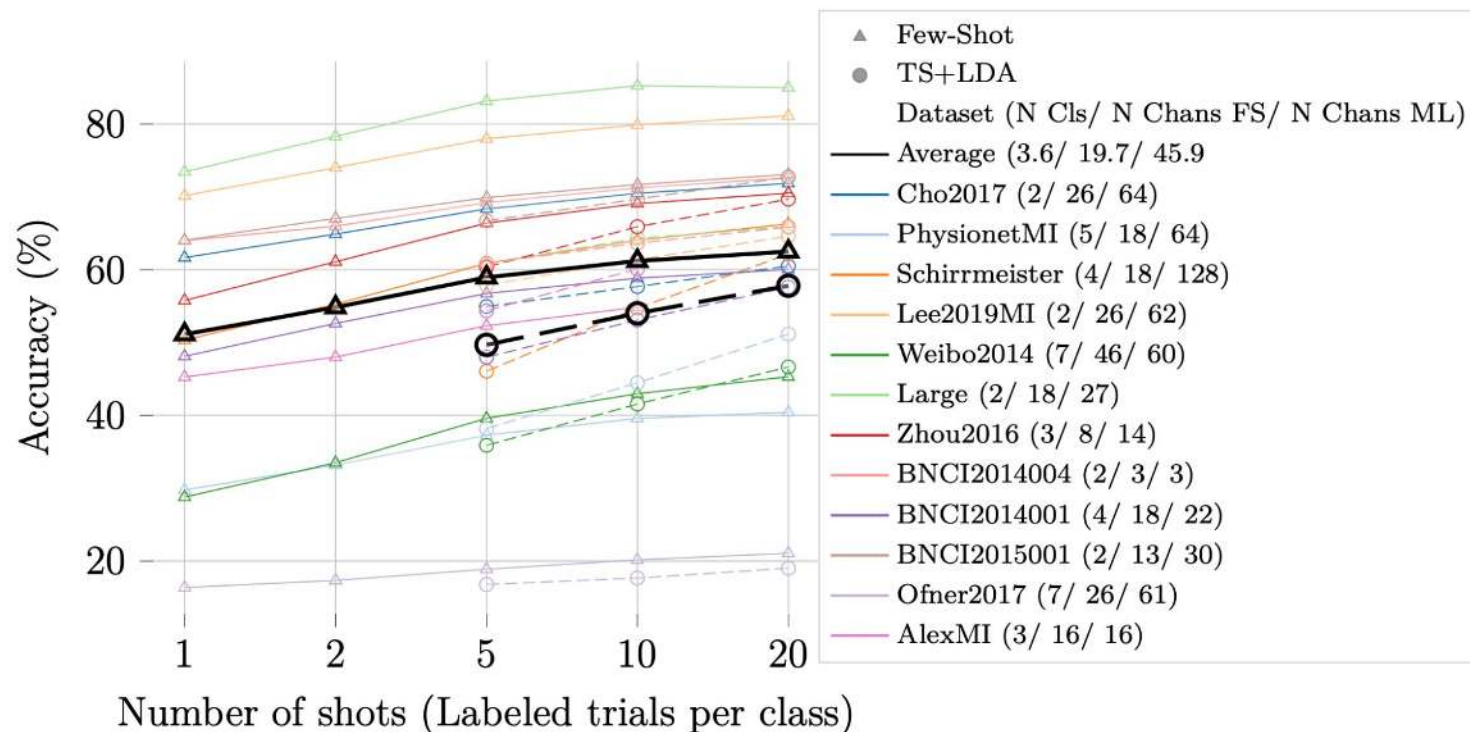
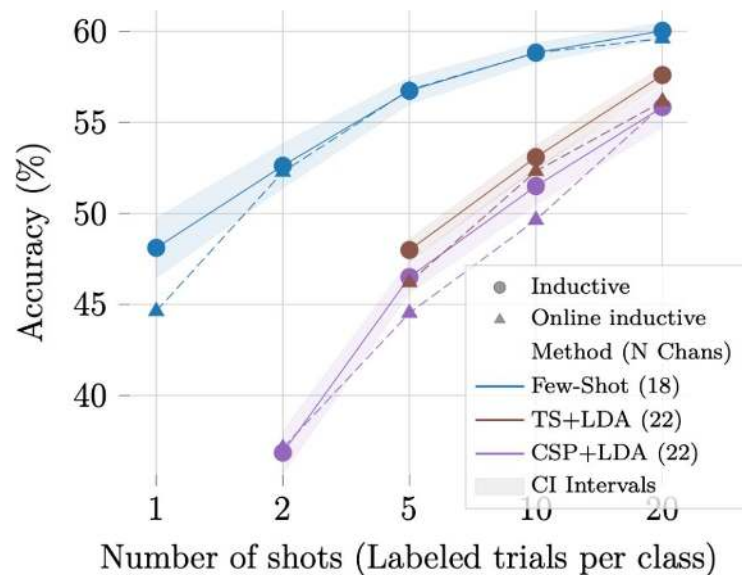
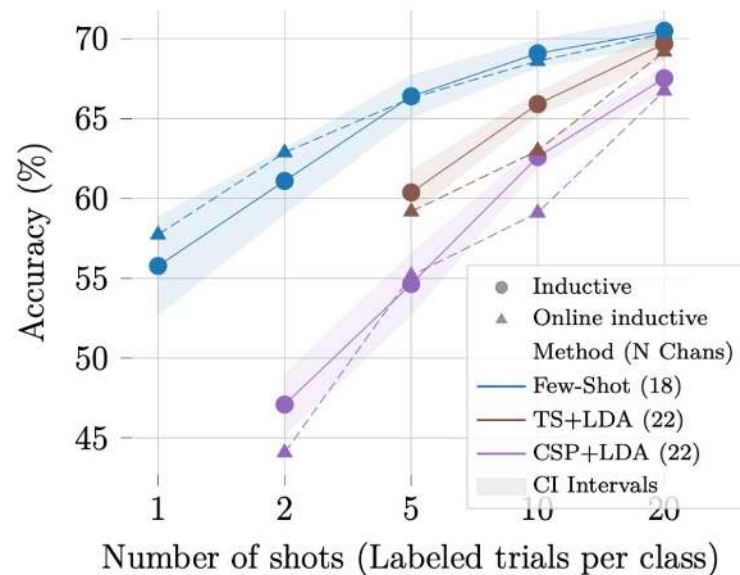


Figure 3.10 – All classes MI Classification - machine learning (TS+LDA) vs Few-Shot



(a) Analysis on BNCI2014001 (4 classes)



(b) Analysis on Zhou2016 (3 classes)

Figure 3.11 – Inductive vs Online Inductive setup

Method	Task 2 Score
Winning Methods	
Cogitat [67, 169]	76.33
wduong [67]	71.33
ms01 [67]	59.87
Our Methods	
FSL	72.00
FSL + Ensembling	75.60
FT	75.40
FT + Ensembling	78.00

Table 3.6 – Performance Comparison on task 2 of the BEETL Challenge: Winning vs. Our methods.

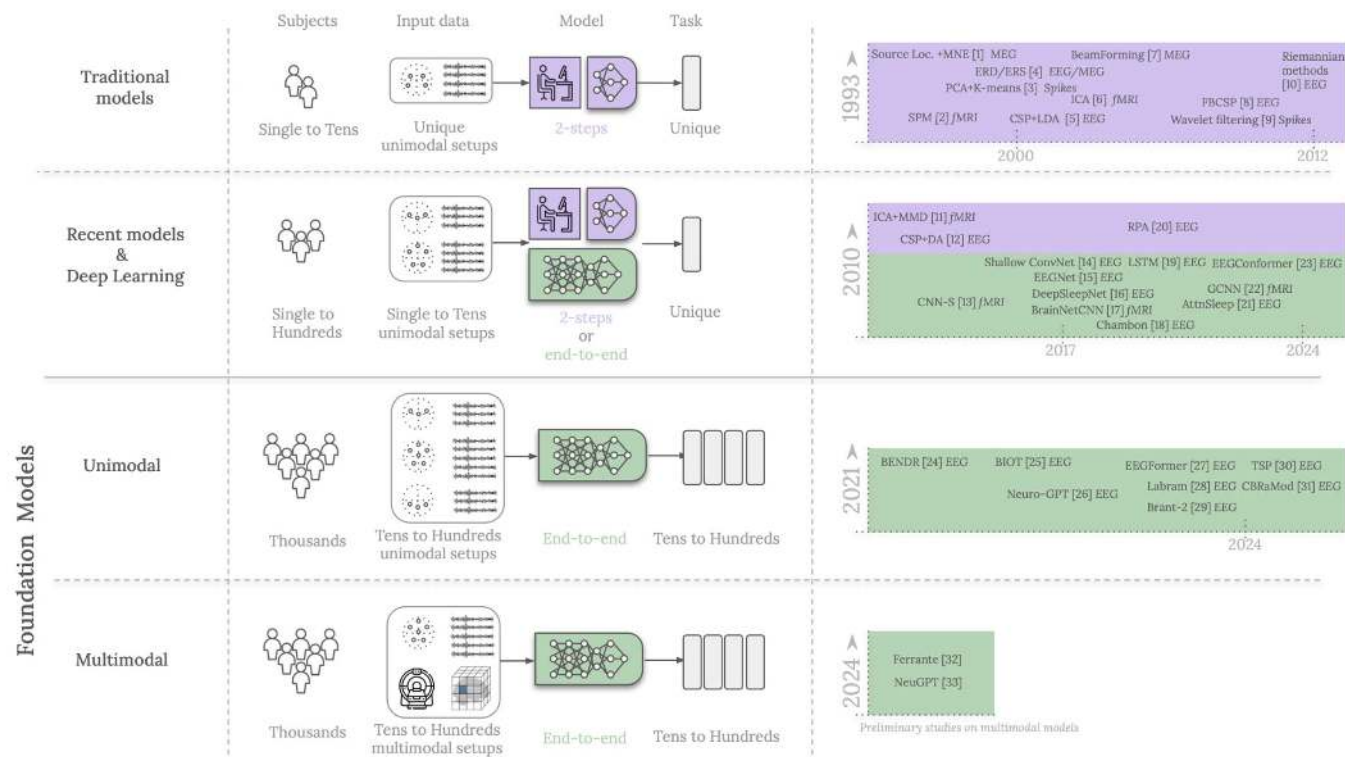


Figure 4.1 – Evolution of models for neuroscience. We display on the left the generalization dimensions per category of model, with examples of influential studies on the right.

	CHARACTERISTICS						USABILITY		
Name Year	Input type	Input Signal duration 1-token 2-overlap	Preprocessing Steps	Normalization	SSL method / reconstruction space	Pretraing Datasets	Electrode config: 1- Trained on multiple 2- Can infer any	Model Size (params)	Open: 1-code 2-weights
BENDR 2021 [24]	raw EEG	1- 0.375s. 2- 0s.	RS: 256 Hz LP: 120 Hz	Linear scale + shift + add a channel with the relative amplitude	CPC compare distorted (masked) segments / latent space	TUEG	1- No 2- No	>1 B.	1- Yes 2- Yes
BIOT 2023 [25]	frequency amplitude	1- 1s. 2- 0.5s.	RS: 200 Hz	Channel-wise normalization	CL predict signal from a perturbed version / latent space	PREST SHHS	1- No 2- No	3.3 M.	1- Yes 2- Yes
Neuro-GPT [26] 2023	raw EEG	1- 2s. 2- 0.2s.	RS: 250 Hz HP: 0.5 Hz LP: 100 Hz N: 60 Hz Avg re-ref DC correct	Standard scaling across the time	SP predict next token / latent space	TUEG	1- No 2- No	tofill	1- Yes 2- Yes
EEGFormer 2024 [27]	frequency amplitude	—	RS: 250Hz	Instance Normalization	SP Auto-Encoder / input data	TUEG	1- No 2- Yes	—	1- No 2- No
Labram 2024 [28]	raw EEG	1- 1s 2- 0s.	RS: 200 Hz HP: 0.1 Hz LP: 75 Hz N: 50Hz	Normalize with unit to 0.1 mV	SP mask tokens prediction / latent space (previously learned codebook)	TUH subsets + 19 sources	1- Yes 2- Yes	5.8 M. / 46 M. / 369 M.	1- Yes 2- Yes
TSP 2024 [30]	frequency amplitude	1- 1s. 2- 0.5s.	RS: 200 Hz HP: 75 Hz	Channel-wise normalization	SP + CPC mask channels prediction + forecast / latent space (BIOT embeddings)	SEEDs EP TUAB HBN TDBRAIN PRED+CT	1- Yes 2- Yes	—	—
Brant-2 2024 [29]	raw EEG + PSD	—	Denosing Filtering	—	SP mask tokens prediction + forecasting / input data	TUEG	1- No 2- Yes	—	1- No 2- No
CBraMod 2025 [31]	raw EEG + frequency amplitude	1- 1s. 2- 0s.	RS: 200 Hz HP: 0.3 Hz LP: 75 Hz N: 60 Hz Bad sample removal	Normalize with unit to 0.1 mV	SP Masked token prediction	TUEG	1- No 2- Yes	4 M.	1- Will be 2- Will be

Notation: RS: Resampling, LP: Low-Pass, HP: High-Pass, N: Notch, SP: Self-Predictive, CL: Contrastive Learning, CPC: Contrastive Predictive Coding, "—": Unknown, M.: Millions, B.: Billions

T

Methods	TUAB, 2-class			TUEV, 6-class		
	Balanced Accuracy	AUC-PR	AUROC	Balanced Accuracy	Cohen's Kappa	Weighted F1
EEGNet	0.7642 ± 0.0036	0.8299 ± 0.0043	0.8412 ± 0.0031	0.3876 ± 0.0143	0.3577 ± 0.0155	0.6539 ± 0.0120
EEGConformer	0.7758 ± 0.0049	0.8427 ± 0.0054	0.8445 ± 0.0038	0.4074 ± 0.0164	0.3967 ± 0.0195	0.6983 ± 0.0152
SPaRCNet	0.7896 ± 0.0018	0.8414 ± 0.0018	0.8676 ± 0.0012	0.4161 ± 0.0262	0.4233 ± 0.0181	0.7024 ± 0.0104
ContraWR	0.7746 ± 0.0041	0.8421 ± 0.0104	0.8456 ± 0.0074	0.4384 ± 0.0349	0.3912 ± 0.0237	0.6893 ± 0.0136
CNN-Transformer	0.7777 ± 0.0022	0.8433 ± 0.0039	0.8461 ± 0.0013	0.4087 ± 0.0161	0.3815 ± 0.0134	0.6854 ± 0.0293
FFCL	0.7848 ± 0.0038	0.8448 ± 0.0065	0.8569 ± 0.0051	0.3979 ± 0.0104	0.3732 ± 0.0188	0.6783 ± 0.0120
ST-Transformer	0.7966 ± 0.0023	0.8521 ± 0.0026	0.8707 ± 0.0019	0.3984 ± 0.0228	0.3765 ± 0.0306	0.6823 ± 0.0190
BIOT	0.7959 ± 0.0057	0.8792 ± 0.0023	0.8815 ± 0.0043	0.5281 ± 0.0225	0.5273 ± 0.0249	0.7492 ± 0.0082
LaBraM-Base	0.8140 ± 0.0019	0.8965 ± 0.0016	0.9022 ± 0.0009	0.6409 ± 0.0065	0.6637 ± 0.0093	0.8312 ± 0.0052
LaBraM-Large	0.8226 ± 0.0015	0.9130 ± 0.0005	0.9127 ± 0.0005	0.6581 ± 0.0156	0.6622 ± 0.0136	0.8315 ± 0.0040
LaBraM-Huge	0.8258 ± 0.0011	0.9204 ± 0.0011	0.9162 ± 0.0016	0.6616 ± 0.0170	0.6745 ± 0.0195	0.8329 ± 0.0086
Cbramod	0.8289 ± 0.0022	0.9258 ± 0.0008	0.9227 ± 0.0011	0.6671 ± 0.0107	0.6772 ± 0.0096	0.8342 ± 0.0064
REVE-Small (ours)	0.8110 ± 0.0046	0.8928 ± 0.0101	0.8964 ± 0.0112	0.6360 ± 0.0095	0.6227 ± 0.0114	0.8249 ± 0.0058
REVE-Base (ours)	0.8315 ± 0.0014	0.9281 ± 0.0009	0.9245 ± 0.0013	0.6759 ± 0.0229	0.6783 ± 0.0199	0.8451 ± 0.0129
Methods	PhysioNet-MI, 4-class			BCIC-IV-2a, 4-class		
	Balanced Accuracy	Cohen's Kappa	Weighted F1	Balanced Accuracy	Cohen's Kappa	Weighted F1
EEGNet	0.5814 ± 0.0125	0.4468 ± 0.0199	0.5796 ± 0.0115	0.4482 ± 0.0094	0.2693 ± 0.0121	0.4226 ± 0.0108
EEGConformer	0.6049 ± 0.0104	0.4736 ± 0.0171	0.6062 ± 0.0095	0.4696 ± 0.0106	0.2924 ± 0.0141	0.4533 ± 0.0128
SPaRCNet	0.5932 ± 0.0152	0.4564 ± 0.0234	0.5937 ± 0.0147	0.4635 ± 0.0117	0.2847 ± 0.0147	0.4432 ± 0.0126
ContraWR	0.5892 ± 0.0133	0.4527 ± 0.0248	0.5918 ± 0.0116	0.4678 ± 0.0125	0.2905 ± 0.0160	0.4413 ± 0.0142
CNN-Transformer	0.6053 ± 0.0118	0.4725 ± 0.0223	0.6041 ± 0.0105	0.4600 ± 0.0108	0.2800 ± 0.0148	0.4460 ± 0.0114
FFCL	0.5726 ± 0.0092	0.4323 ± 0.0182	0.5701 ± 0.0079	0.4470 ± 0.0143	0.2627 ± 0.0176	0.4238 ± 0.0139
ST-Transformer	0.6035 ± 0.0081	0.4712 ± 0.0199	0.6053 ± 0.0075	0.4575 ± 0.0145	0.2733 ± 0.0198	0.4471 ± 0.0142
BIOT	0.6153 ± 0.0154	0.4875 ± 0.0272	0.6158 ± 0.0197	0.4748 ± 0.0093	0.2997 ± 0.0139	0.4607 ± 0.0125
LaBraM-Base	0.6173 ± 0.0122	0.4912 ± 0.0192	0.6177 ± 0.0141	0.4869 ± 0.0085	0.3159 ± 0.0154	0.4758 ± 0.0103
CBramod	0.6417 ± 0.0091	0.5222 ± 0.0169	0.6427 ± 0.0100	0.5138 ± 0.0066	0.3518 ± 0.0094	0.4984 ± 0.0085
REVE-Small (ours)	0.5833 ± 0.0097	0.4443 ± 0.0130	0.5847 ± 0.0095	0.5809 ± 0.0104	0.4412 ± 0.0138	0.5753 ± 0.0137
REVE-Base (ours)	0.6480 ± 0.0140	0.5306 ± 0.0187	0.6484 ± 0.0170	0.6396 ± 0.0095	0.5194 ± 0.0126	0.6339 ± 0.0110

Table 4.3 – Results of different methods on various EEG classification tasks

Settings	PhysioNet-MI, 4-class		
	Balanced Accuracy	Cohen's Kappa	Weighted F1
CBramod	0.6417 $\pm$ 0.0091	0.5222 $\pm$ 0.0169	0.6427 $\pm$ 0.0100
Cbramod (Frozen)	0.3845 $\pm$ 0.0345	0.2983 $\pm$ 0.0498	0.3946 $\pm$ 0.0378
CBramod (w/o PT)	0.6196 $\pm$ 0.0143	0.4994 $\pm$ 0.0289	0.6289 $\pm$ 0.0179
BIOT	0.6153 $\pm$ 0.0154	0.4875 $\pm$ 0.0272	0.6158 $\pm$ 0.0197
BIOT (Frozen)	0.3698 $\pm$ 0.0318	0.2703 $\pm$ 0.04722	0.3723 $\pm$ 0.0364
LaBraM-Base	0.6173 $\pm$ 0.0122	0.4912 $\pm$ 0.0192	0.6177 $\pm$ 0.0141
LaBraM (Frozen)	0.3715 $\pm$ 0.0458	0.2814 $\pm$ 0.0586	0.3796 $\pm$ 0.0472
REVE-Base	0.6480 $\pm$ 0.0140	0.5306 $\pm$ 0.0187	0.6484 $\pm$ 0.0170
REVE-Tiny (Frozen)	0.3625 $\pm$ 0.0347	0.1500 $\pm$ 0.0462	0.3557 $\pm$ 0.0396
REVE-Small (Frozen)	0.5331 $\pm$ 0.0190	0.3774 $\pm$ 0.0254	0.5343 $\pm$ 0.0194
REVE-Base (Frozen)	0.5371 $\pm$ 0.0052	0.3827 $\pm$ 0.0070	0.5376 $\pm$ 0.0033
REVE-Large (Frozen)	0.5510 $\pm$ 0.0014	0.4013 $\pm$ 0.0019	0.5515 $\pm$ 0.0006
REVE-Tiny (w/o PT)	0.5212 $\pm$ 0.0097	0.3615 $\pm$ 0.0129	0.5202 $\pm$ 0.0092
REVE-Small (w/o PT)	0.5240 $\pm$ 0.0122	0.3652 $\pm$ 0.0162	0.5251 $\pm$ 0.0124
REVE-Base (w/o PT)	0.5409 $\pm$ 0.0094	0.3879 $\pm$ 0.0125	0.5421 $\pm$ 0.0101
REVE-Large (w/o PT)	0.5391 $\pm$ 0.0111	0.3856 $\pm$ 0.0147	0.5395 $\pm$ 0.0098

Table 4.4 – Impact of pretraining (PT) and freezing weights on REVE and baseline models for PhysioNet-MI

Variable	Meaning	Value
Data preprocessing		
w	Window size	1s
o	Overlap	0.1s
σ_{noise}	Position noise std	0.25cm
Masking parameters		
M_r	Total masking ratio	55%
R_s	Spatial masking radius	3 cm
R_t	Temporal masking radius	3 seconds
D_r	Dropout ratio	10%
R_d	Dropout spatial radius	4 cm
Training dynamics		
	Optimizer	StableAdamW
	Scheulder	Warmup Stable Decay
η	Peak learning rate	$\eta = 2.4 \cdot 10^{-4}$
β_1, β_2	Momentum constants	0.9, 0.95
ε	Numerical stability bias	10^{-9}
σ_{init}	Initialization std	0.02
	Batch size	4,096
λ	Secondary loss multiplier	0.1

Table C.1 – Exhaustive list of all hyperparameter values

Size	depth	n_heads	dim	params (M)	n_{freq}
Tiny	4	4	200	2	3
Small	4	8	512	12	4
Base	22	12	512	69	4
Large	22	19	1216	460	5

Table C.2 – Summary of encoder configurations for different sizes

PhysioNet-MI, 4-class				
Second Loss	Balanced Accuracy		Cohen's Kappa	Weighted F1
REVE-small	0.5833 $\pm$ 0.0097		0.4443 $\pm$ 0.0130	0.5847 $\pm$ 0.0095
REVE-small (without)	0.5625 $\pm$ 0.0109		0.4166 $\pm$ 0.0145	0.5629 $\pm$ 0.0113
REVE-Small-Frozen	0.5331 $\pm$ 0.0190		0.3774 $\pm$ 0.0254	0.5343 $\pm$ 0.0194
REVE-Small-Frozen (without)	0.3791 $\pm$ 0.0584		0.1721 $\pm$ 0.0779	0.3777 $\pm$ 0.0597

Masking Ratio	Frozen		Full Fine-Tuning	
	Random	Block	Random	Block
0.25	0.4510 $\pm$ 0.0388	0.3721 $\pm$ 0.0177	0.5718 $\pm$ 0.0107	0.5856 $\pm$ 0.0133
0.55	0.4438 $\pm$ 0.0647	0.5331 $\pm$ 0.0190	0.5721 $\pm$ 0.0086	0.5833 $\pm$ 0.0097
0.75	0.3833 $\pm$ 0.0178	0.3833 $\pm$ 0.0178	0.5714 $\pm$ 0.0053	0.5714 $\pm$ 0.0053

Table C.5 – Performance comparison across different masking ratios (0.25, 0.55, 0.75) between block masking strategy and random masking, evaluated for full fine-tuning versus frozen embeddings. We display balanced accuracy on the small model on PhysionetMI.

Datasets

Dataset	BNCI	Zhou	Physionet	Cho
Considered Subjects	9	4	103	49
Sessions per subject	2	3	1	1
Trials per session	288	150	90	200
Total Trials	5184	1800	9270	9880
Classes	L/R/F/T	L/R/F	L/R/F/BH	L/R
EEG electrodes	22	14	64	64
EOG electrodes	3	2	0	0
Sampling frequency (Hz)	250	250	160	512
Trial duration (s)	4	5	3.5	3

Table 2.1 – MI datasets considered. L = Left hand, R = Right hand, F = Feet, BH = Both hands, T = Tongue.

Dataset	BNCI	BNCI2	Large
Considered subjects	9	9	85
Sessions per subject	2	2	5
Trials per session	288	144	40
Total trials	5184	2592	17000
Classes	L/R/F/T	L/R	L/R
EEG electrodes	22	22	27
Sampling frequency (Hz)	250	250	512
Trial duration (s)	4	4	4.5

Table 3.1 – MI datasets considered. L = Left hand, R = Right hand, F = Feet, T = Tongue.

MI dataset	Subjects	Electrodes	Samples	Sampling frequency	Dur.	Tasks
Schirrmeister [14]	14	76	13484	500 Hz	4 s.	L/R/F/R
BNCI2014 [107]	9	22	5184	250 Hz	4 s.	L/R/F/T
Cho2017 [109]	52	64	9880	512 Hz	3 s.	L/R
PhysionetMI [110]	109	62	9838	160 Hz	3 s.	L/R/F/BH
Zhou2016 [108]	4	14	1800	250 Hz	3 s.	L/R/F
Shin2017 [129]	29	22	1740	200 Hz	3 s.	L/R

Table 2.6 – MI datasets considered. L = Left hand, R = Right hand, F = Feet, R = Rest, BH = Both hands

MI dataset	Subjects	Electrodes	Samples	Sampling frequency	Duration	Tasks
BNCI2014001 [107]	9	22	5184	250 Hz	4 s.	L/R/F/T
PhysionetMI [160]	103	64	18540	160 Hz	3 s.	L/R/F/BH/Re
AlexMI [161]	8	16	480	512 Hz	3 s.	R/F/Re
Weibo2014 [162]	10	60	5540	200 Hz	4 s.	L/R/F/BH/Re/2× C
Ofner2017 [163]	15	61	6300	512 Hz	3 s.	Re / 6× R
Cho2017 [109]	49	64	9880	512 Hz	3 s.	L/R
Lee2019MI [164]	54	62	10800	1000 Hz	4 s.	L/R
Schirrmeister [14]	14	128	13484	500 Hz	4 s.	L/R/F/Re
Zhou2016 [165]	4	14	1800	250 Hz	5 s.	L/R/F
BNCI2015001 [166]	12	13	5600	512 Hz	5 s.	R/F
BNCI2014004 [167]	9	3	6520	250 Hz	4.5 s.	L/R
Large [145]	85	27	20400	512 Hz	4 s.	L/R

Table 3.4 – MI datasets considered in our study. L = Left hand, R = Right hand, F = Feet, Re = Rest, BH = Both hands, C = Compound

Datasets

Group	Subjects	Duration (hours)	Datasets
Category			
BCI	791	457	28
Cognition	4,193	10,376	56
Clinic	19,290	50,581	8
Platform			
TUH	14,987	26,847	1
Physionet	607	22,707	2
OpenNeuro	4153	10,194	56
MOABB	711	384	27
Other	3,802	1,250	6
Channels			
[3 – 30[	19,871	50,870	31
[30 – 80[	1,781	1,516	48
[80 – 129]	2,622	9,027	13
Total	24,274	61,415	92

Table 4.2 – Detailed overview of the pretraining datasets.

BCI Tasks	Datasets	Rate	# Channels	Duration	# Samples	Label
I. Motor Imagery Classification	PhysioNet-MI [160]	160Hz	64	4s	9,837	4-class
	BCIC-IV-2a [107]	250Hz	22	4s	5,184	4-class
II. Event Type Classification	TUEV [214]	256Hz	16	5s	112,491	6-class
III. Abnormal Detection	TUAB [214]	256Hz	16	10s	409,455	2-class

Table C.3 – Overview of downstream BCI tasks and datasets.

MOABB [106]: AlexMI [161], BNCI2014004 [167], BNCI2015001 [166], BNCI2015004 [268], Cho2017 [269], Lee2019MI [164], Liu2024 [270], Ofner2017 [163], Shin2017A [271], Weibo2014 [162], Zhou2016 [165], Schirrmeister2017 [14], Kalunga2016 [272], Lee2019SSVEP [273], Nakan- ishi2015 [274], BI2014a [275], BI2014b [276], BNCI2014008 [277], BNCI2014009 [278], BNCI2015003 [279], EPFLP300 [280], BI2015a [281], BI2015b [276], Sosulski2019 [282], Lee2019ERP [283]

Physionet [110]: Siena [284, 285], ICARE [286],

OpenNeuro: ds004706 [287], ds004582 [288], ds004356 [289], ds004817 [290], ds005189 [291], ds003887 [292], ds004043 [293], ds003885 [294], ds004357 [295], ds003825 [296], ds004816 [297], ds004840 [298], ds005262 [299], ds004477 [300], ds005273 [301], ds004561 [302], ds004951 [303], ds004324 [304], ds005095 [305], ds005509 [306], ds005505, ds005506, ds005507, ds005510, ds005511, ds005512, ds005514 [307, 308] ds001787 [309], ds003690 [310], ds004603 [311], ds003969 [312], ds004147 [313], ds003004 [314], ds002721 [315], ds004152 [316], ds005089 [317], ds004264 [318], ds004315 [319], ds004408 [320], ds005121 [321], ds003775 [322], ds004572 [323], ds002778 [324], ds003846 [325], ds004279 [326], ds004148 [327], ds004902 [328], ds002680 [329], ds004284 [330], ds004395 [331], ds005508 [332], ds005697 [333], ds005620 [334], ds005594 [335], ds005586 [336]

Other sources: NMT [337], HMS [338], SparrKULee [339], Inria Large [145], THINGS2 [340], TDBRAIN [341], TUH [214]

BrainWave: A Brain Signal Foundation Model for Clinical Applications

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Gefei Gu¹, Fei Wu¹, Chenhao Tan⁶, Yang Yang^{1*}

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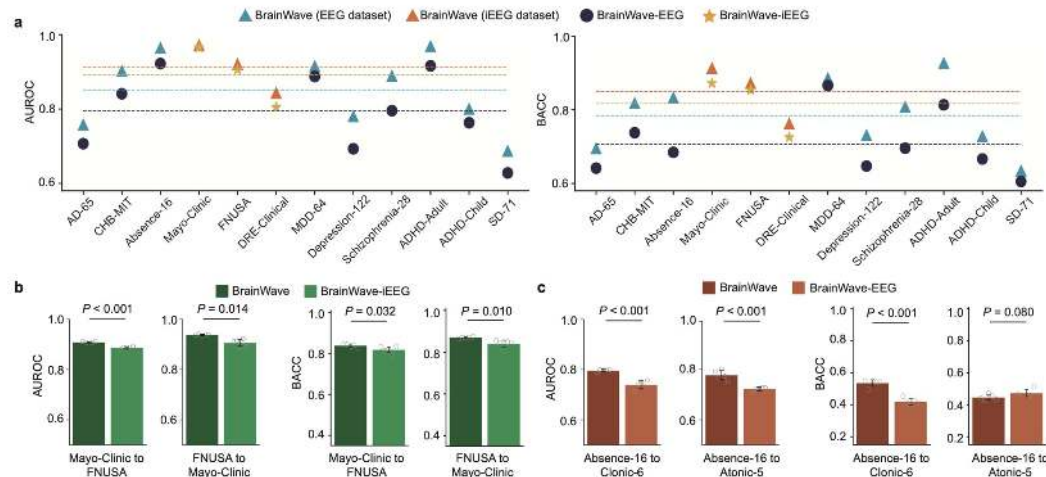
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CBRAMod: A CRISS-CROSS BRAIN FOUNDATION MODEL FOR EEG DECODING

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godot.lzl@alibaba-inc.com

Table 4: The results of ablation study on pre-training.

Settings	FACED, 9-class			SEED-V, 5-class		
	Balanced Accuracy	Cohen's Kappa	Weighted F1	Balanced Accuracy	Cohen's Kappa	Weighted F1
clean pre-training	0.5509 $\pm$ 0.0089	0.5041 $\pm$ 0.0122	0.5618 $\pm$ 0.0093	0.4091 $\pm$ 0.0097	0.2569 $\pm$ 0.0143	0.4101 $\pm$ 0.0108
dirty pre-training	0.5319 $\pm$ 0.0245	0.4768 $\pm$ 0.0348	0.5397 $\pm$ 0.0219	0.3914 $\pm$ 0.0217	0.2224 $\pm$ 0.0318	0.3894 $\pm$ 0.0243
w/o pre-training	0.5232 $\pm$ 0.0216	0.4615 $\pm$ 0.0289	0.5304 $\pm$ 0.0187	0.3902 $\pm$ 0.0241	0.2211 $\pm$ 0.0384	0.3879 $\pm$ 0.0225

Settings	PhysioNet-MI, 4-class			SHU-MI, 2-class		
	Balanced Accuracy	Cohen's Kappa	Weighted F1	Balanced Accuracy	AUC-PR	AUROC
clean pre-training	0.6417 $\pm$ 0.0091	0.5222 $\pm$ 0.0169	0.6427 $\pm$ 0.0100	0.6370 $\pm$ 0.0151	0.7139 $\pm$ 0.0088	0.6988 $\pm$ 0.0068
dirty pre-training	0.6245 $\pm$ 0.0153	0.5063 $\pm$ 0.0256	0.6223 $\pm$ 0.0167	0.6301 $\pm$ 0.0189	0.7026 $\pm$ 0.0176	0.6895 $\pm$ 0.0131
w/o pre-training	0.6196 $\pm$ 0.0143	0.4994 $\pm$ 0.0289	0.6157 $\pm$ 0.0145	0.6289 $\pm$ 0.0179	0.7032 $\pm$ 0.0145	0.6878 $\pm$ 0.0166

Decoding speech perception from non-invasive brain recordings

Alexandre Défossez ✉, Charlotte Caucheteux, Jérémy Rapin, Ori Kabeli & Jean-Rémi King ✉

Nature Machine Intelligence **5**, 1097–1107 (2023) | [Cite this article](#)

Model	Broderick (EEG)	Brennan (EEG)	Schoffelen (MEG)	Gwilliams (MEG)	delta	p-val
Our model	17.7 ± 0.6	25.7 ± 2.9	67.5 ± 0.4	70.7 ± 0.1		
- Spatial attention dropout	16.0 ± 1.7 *	26.8 ± 0.7	67.5 ± 0.2	69.0 ± 0.2 *	0.4	0.009
- GELU + ReLU	16.4 ± 0.1	24.6 ± 2.1	65.8 ± 0.6 *	68.8 ± 1.3 *	1.6	< 10 ⁻¹⁸
- Final convs	14.2 ± 1.1 *	19.0 ± 4.4 *	67.5 ± 0.3	68.9 ± 0.9 *	1.1	< 10 ⁻¹⁰
- Non-residual GLU conv	8.4 ± 6.8 *	6.0 ± 0.2 *	67.0 ± 0.2	70.2 ± 0.2	1.6	< 10 ⁻¹⁰
- Skip connections	13.9 ± 2.0 *	24.2 ± 2.7	65.4 ± 0.4 *	66.2 ± 0.3 *	2.4	< 10 ⁻²¹
- Initial 1x1 conv	15.4 ± 0.6	22.1 ± 1.9 *	62.9 ± 0.9 *	67.7 ± 0.7 *	3.4	< 10 ⁻²⁶
- Spatial attention	15.4 ± 0.6 *	20.6 ± 2.2 *	65.9 ± 0.3 *	65.5 ± 0.4 *	2.5	< 10 ⁻²²
- Subj layer	8.1 ± 1.9 *	20.2 ± 1.3 *	42.4 ± 0.1 *	47.0 ± 1.3 *	14.4	< 10 ⁻²⁸
- Clamping	0.5 ± 0.0 *	14.1 ± 1.0 *	1.5 ± 0.3 *	23.6 ± 24.6 *	26.2	< 10 ⁻²⁹

Extended Data Fig. 2 | Ablations of the brain module. Segment-level top-10 accuracy (%) for our model and its ablated versions. Stars indicate significant gain ($p < 0.001$) across participants (dataset dependent, see Table 1). Confidence intervals are SEM over 3 runs.

Decoding speech perception from non-invasive brain recordings

[Alexandre Défossez](#) ✉, [Charlotte Caucheteux](#), [Jérémy Rapin](#), [Ori Kabeli](#) & [Jean-Rémi King](#) ✉

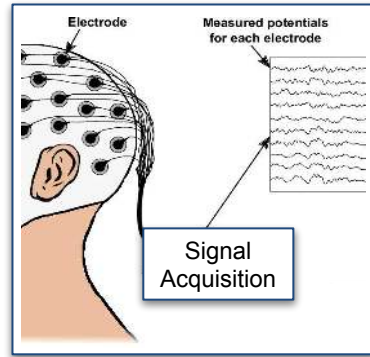
[Nature Machine Intelligence](#) 5, 1097–1107 (2023) | [Cite this article](#)

Model	Brennan (EEG)	Broderick (EEG)	Gwilliams (MEG)	Schoffelen (MEG)
Random model	5.3±0.1	0.5±0.1	0.7±0.1	0.8±0.1
Base model	6.0±0.9	1.0±0.3	12.4±1.2	20.6±1.8
+ Contrastive	8.0±4.8	9.7±1.0	55.1±0.7	55.1±0.9
+ Deep Mel	24.7±3.2	15.4±1.6	64.4±0.8	61.2±0.6
+ wav2vec 2.0	25.7±2.9	17.7±0.6	70.7±0.1	67.5±0.4

31.7%

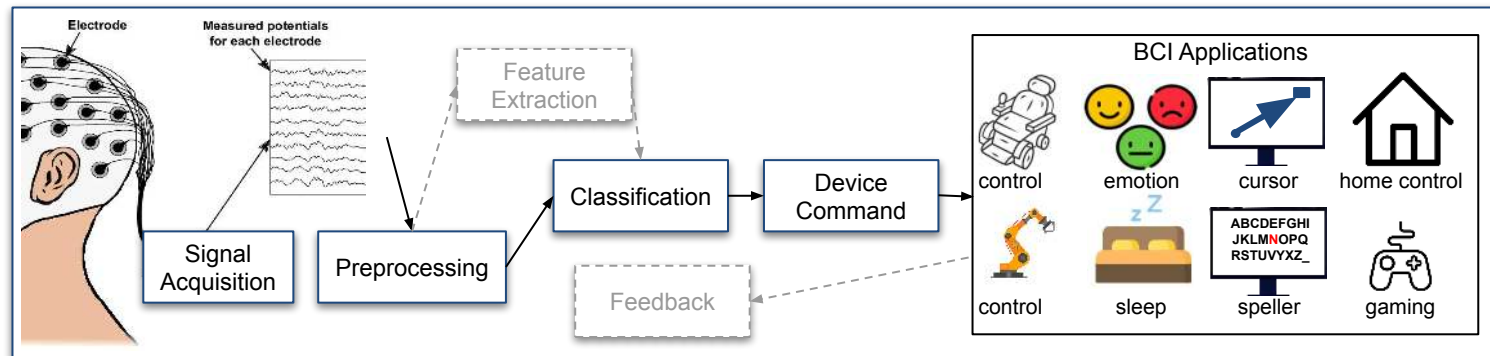
30.3%

Context - Brain Computer Interface (BCI)

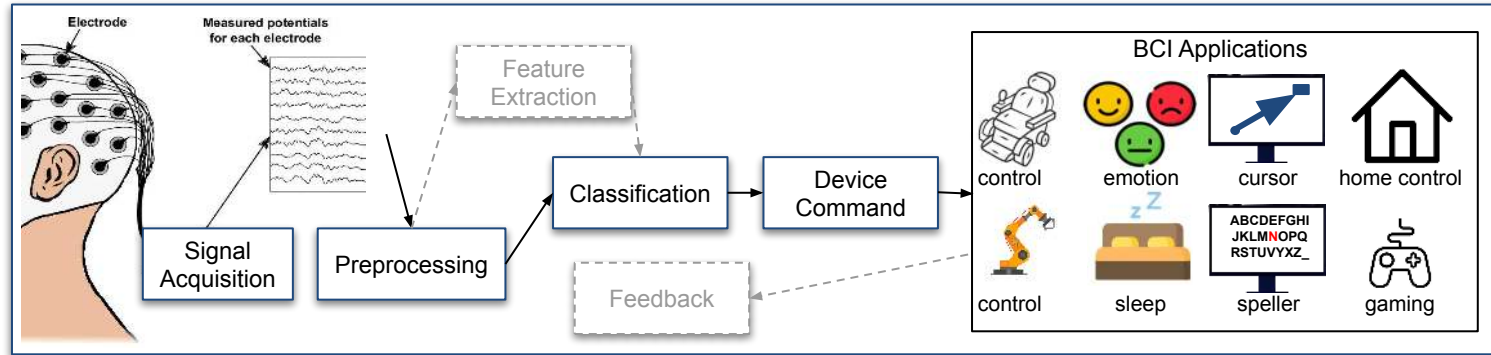


ElectroEncephaloGraphy
(EEG)

Context - Brain Computer Interface (BCI)



Context - Brain Computer Interface (BCI)



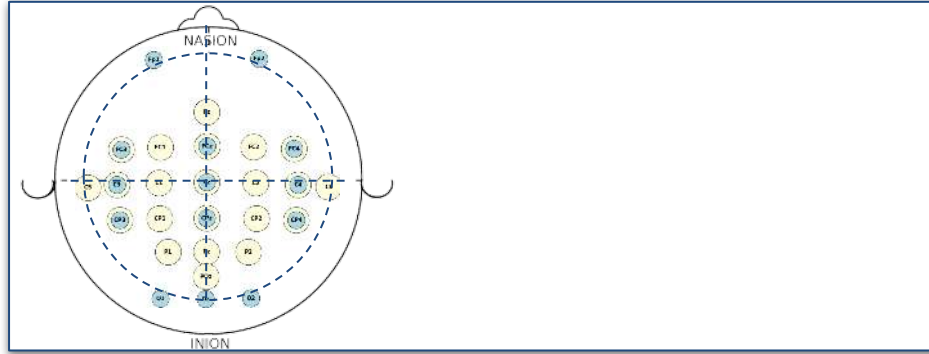
Challenges:

- Lack of big datasets.
 - Small dataset with different recording setups.
- Electrodes layout, sampling frequency, filters, ...



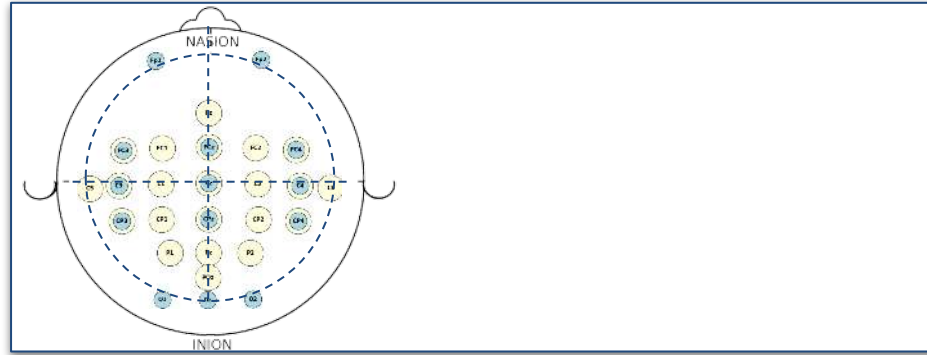
Context - Homogenize BCI datasets

However unifying the spatial aspect of the EEG setups remains a challenge.



Context - Homogenize BCI datasets

However unifying the spatial aspect of the EEG setups remains a challenge.

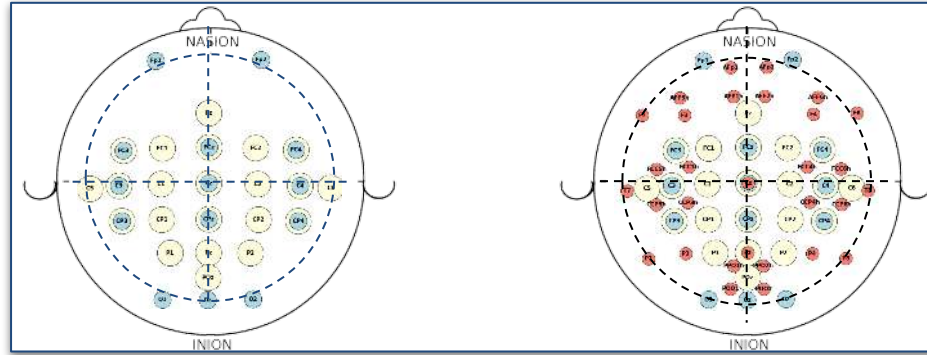


Few approaches have been proposed to unify the spatial aspect of the EEG setups.

- Reduce the spatial dimension:
 - Keeping the intersection
 - Dimension reduction methods (ex : PCA)

Context - Homogenize BCI datasets

However unifying the spatial aspect of the EEG setups remains a challenge.

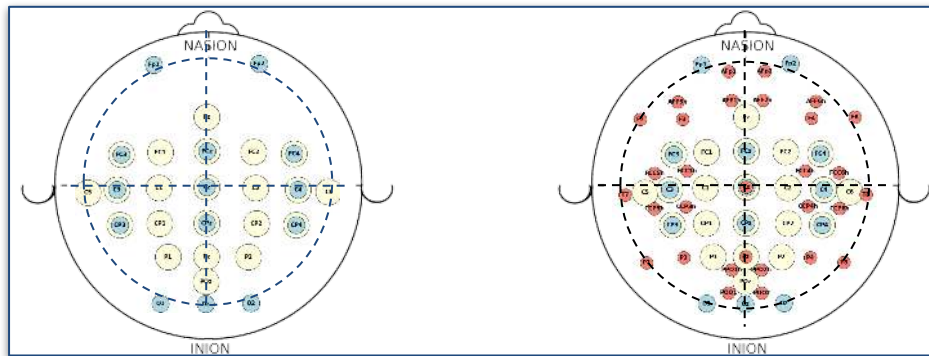


Few approaches have been proposed to unify the spatial aspect of the EEG setups.

- Reduce the spatial dimension:
 - Keeping the intersection
 - Dimension reduction methods (ex : PCA)
- Increase the spatial dimension
 - Riemannian geometry

Context - Homogenize BCI datasets

However unifying the spatial aspect of the EEG setups remains a challenge.



Few approaches have been proposed to unify the spatial aspect of the EEG setups.

- Reduce the spatial dimension:
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 - Dimension reduction methods (ex : PCA)

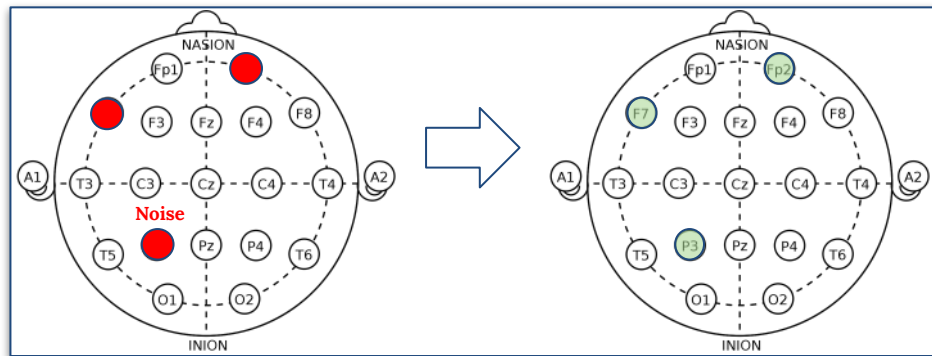
- Increase the spatial dimension
 - Riemannian geometry

No information loss

Context - Electrodes / Graph interpolation

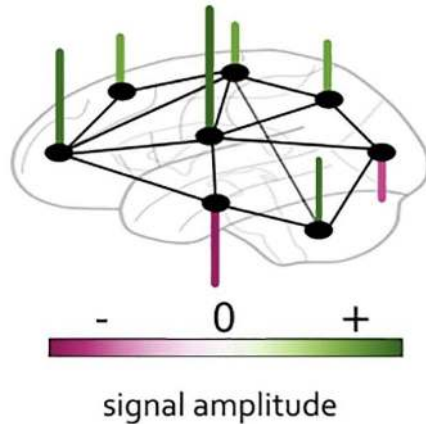
Interpolating electrodes has been mainly used to recover signal from noisy electrodes

Mainly using is Spherical Spline [1]



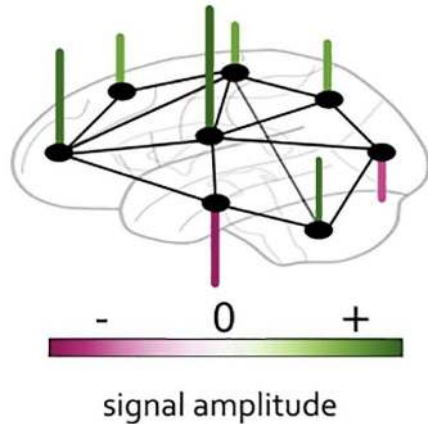
Scientific challenges

Our proposition: Interpolating spatial EEG using Graph Signal Processing



Scientific challenges

Our proposition : Interpolating spatial EEG using Graph Signal Processing

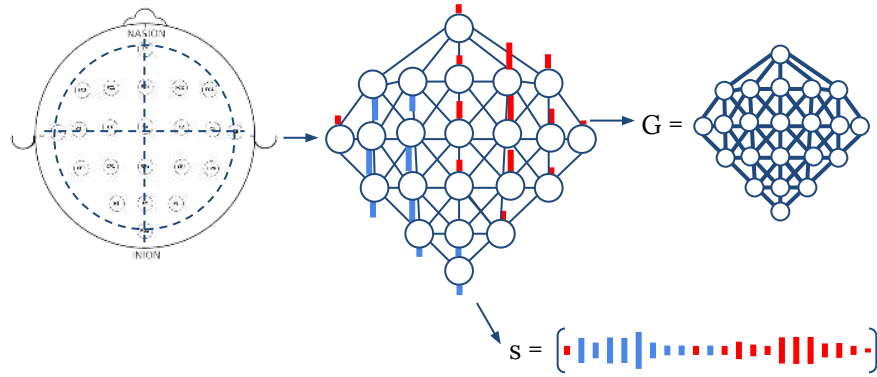


Research questions:

- 1) How to use GSP to interpolate electrodes?
 - Which graph?
 - Which interpolation criterion
- 2) Does unifying multiple EEG datasets with interpolation improve brain decoding?

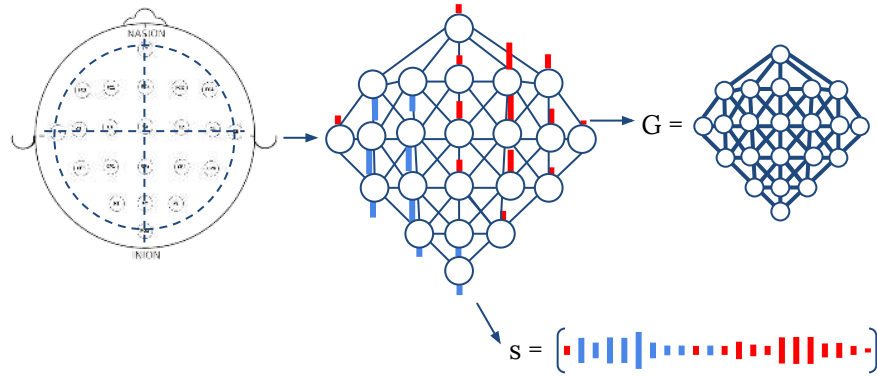
Methods

Few definitions:



Methods

Few definitions:



With D and W , the Degree and weights matrix of G , we have

$L = D - W$, with L the Laplacian

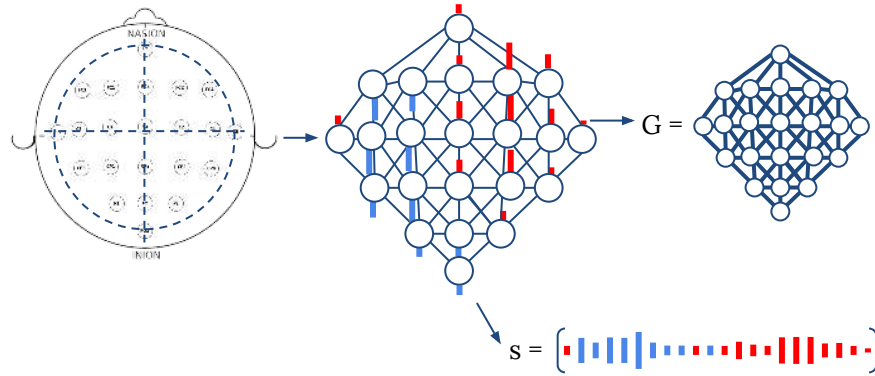
Smoothness of the signal over the graph is defined by:

$$\sigma(s) = s^T L s = \sum_{i=1}^{|V|} \sum_{j=1}^{|V|} W_{ij} (s_i - s_j)^2$$



Methods

Few definitions:

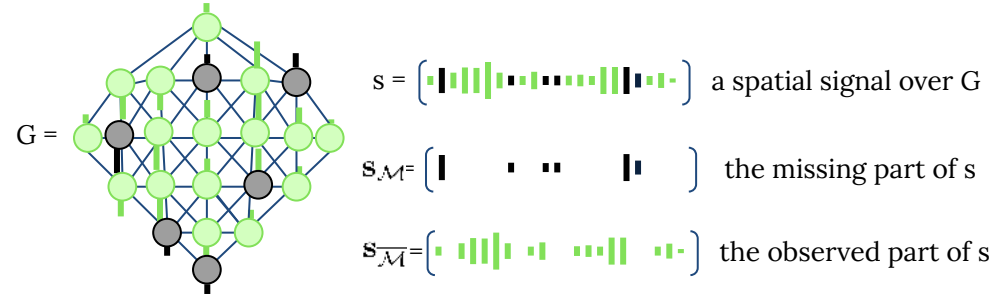


With D and W , the Degree and weights matrix of G , we have
 $\mathbf{L} = D - W$, with $\mathbf{L}$ the Laplacian

Smoothness of the signal over the graph is defined by :

$$\sigma(\mathbf{s}) = \mathbf{s}^\top \mathbf{L} \mathbf{s} = \sum_{i=1}^{|\mathcal{V}|} \sum_{j=1}^{|\mathcal{V}|} W_{ij} (s_i - s_j)^2$$

Interpolation problem formulation:



- 1) How to find SM? Which interpolation criterion?
- 2) How to build a Graph?

1 How to find $\mathbf{s}_{\mathcal{M}}$?

Interpolation criterion : **smoothness**

$$\text{Minimizing } \sigma(\mathbf{s}) = \mathbf{s}^\top \mathbf{L} \mathbf{s} = \sum_{i=1}^{|\mathcal{V}|} \sum_{j=1}^{|\mathcal{V}|} W_{ij} (s_i - s_j)^2$$

We found a closed form of $\sigma(\mathbf{s})$ that provides the optimal $\mathbf{s}_{\mathcal{M}}$

$$(A) \quad \mathbf{s}_{\mathcal{M}} = -\mathbf{L}_{\mathcal{M}}^{-1} \mathbf{L}_{\mathcal{M}\overline{\mathcal{M}}} \mathbf{s}_{\overline{\mathcal{M}}}$$

1 How to found $\mathbf{s}_{\mathcal{M}}$?

Interpolation criterion : **smoothness**

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2 How to build G?

We learn G, from real data using gradient descent.

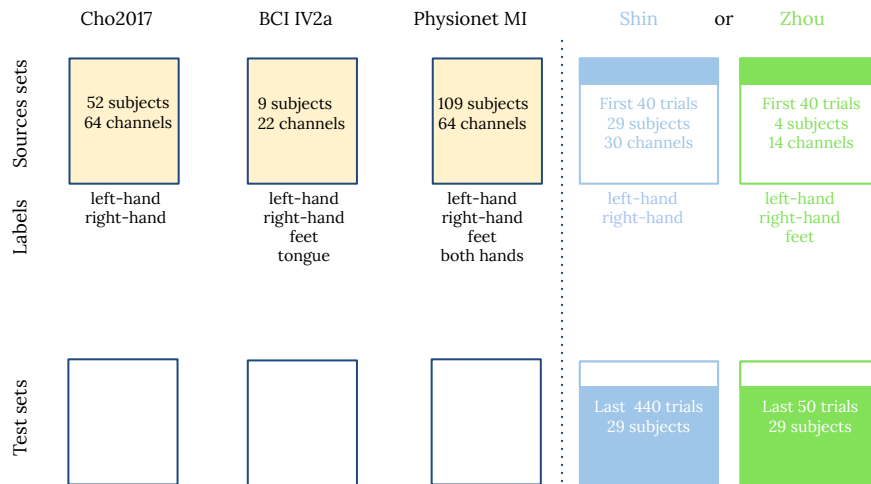
- 1) Initialize a connected graph G
- 2) Create virtual reconstruction problems
- 3) Reconstruct the signal using (A)
- 4) Update G based on the error of reconstruction
- 5) Repeat 2) to 4) until the error is low

Experiments

What is the added value brought by interpolated data?

Does unifying multiple EEG datasets with interpolation improve brain decoding?

We experiment brain decoding on the following realistic setup:



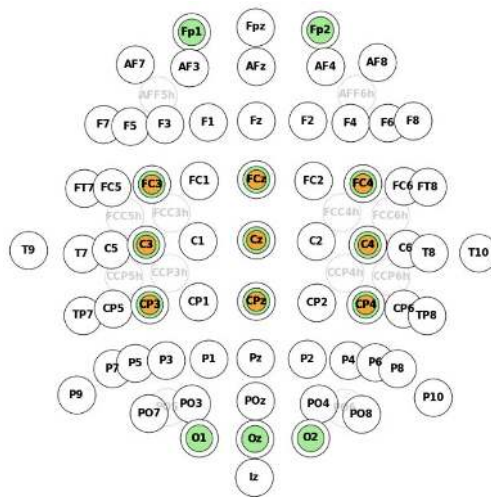
Results

Task : Brain decoding

	Cho2017	BCI IV2a	Physionet MI	Shin	or	Zhou
Sources sets	52 subjects	9 subjects	109 subjects	First 40 trials 29 subjects		First 40 trials 4 subjects
Labels	left-hand right-hand	left-hand right-hand feet tongue	left-hand right-hand feet both hands	left-hand right-hand		left-hand right-hand feet
Test sets				Last 440 trials 29 subjects		Last 50 trials 29 subjects

Classification accuracy

	Zhou	
	Acc	N
$\cap$	<u>61.2 ± 2.0</u>	<u>9</u>
Dataset	56.2 ± 4.8	14
$\cup$	46.4 ± 2.8	66



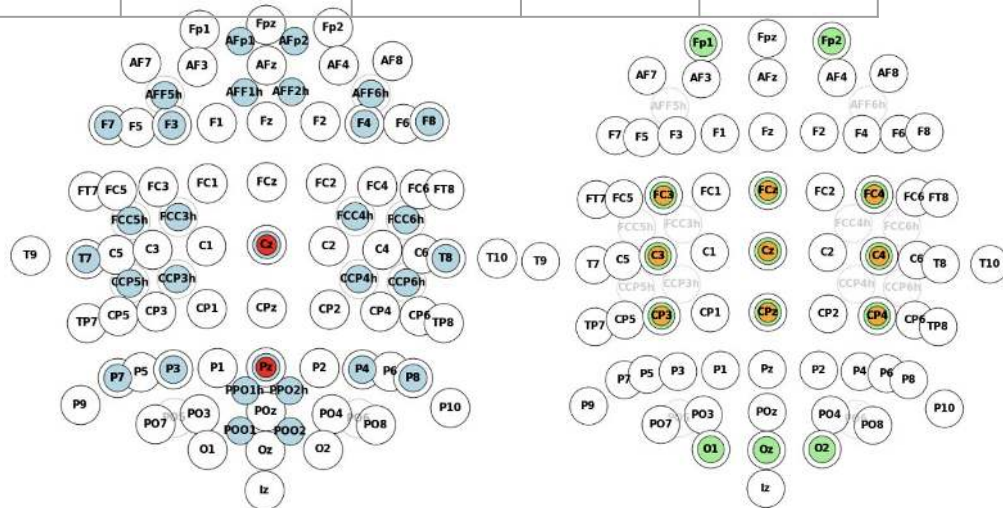
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Test sets				Last 440 trials 29 subjects		Last 50 trials 29 subjects

Classification accuracy

	Shin		Zhou	
	Acc	N	Acc	N
$\cap$	53.2 ± 2.8	2	61.2 ± 2.0	9
Dataset	63.2 ± 2.3	22	56.2 ± 4.8	14
$\cup$	62.3 ± 2.1	76	46.4 ± 2.8	66



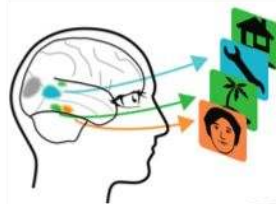
Pruning Graph Convolutional Networks to select meaningful graph frequencies for fMRI decoding

Yassine El
Ouahidi

Hugo Tessier
Giulia Lioi
Nicolas Farrugia
Bastien Padeloup
Vincent Gripon



Context

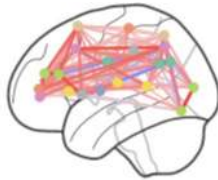
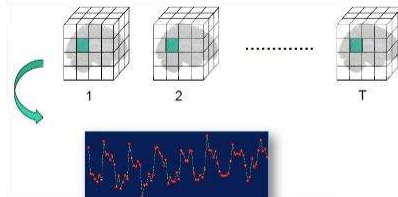


Decoding

Extracting brain activity patterns that discriminate conditions and predict them from unseen neural activity

fMRI

Spatio temporal imagery technic that shows brain activity



Brain & Graphs

The brain can be parcelled in multiple regions that can be connected using graphs

Graph Signal Processing

- Let consider a graph $\mathcal{G} = \langle \mathcal{V}, \mathcal{E}, W \rangle$
 $W \in \mathbb{R}^{N \times N}$ its weight matrix
and $D \in \mathbb{R}^{N \times N}$ its degree matrix
- We can then compute its Laplacian : $L = D - W$
That can be diagonalized : $L = U \Lambda U^T$
- Let $x \in \mathbb{R}^N$ be a signal on $\mathcal{G}$
Its Graph Fourier Transform is given by :
$$\hat{x} = \text{GFT}(x) = U^T x$$

and its inverse by :
$$x = \text{GFT}^{-1}(\hat{x}) = U \hat{x}$$

Challenges

- Which graph frequencies are the most relevant to predict the experimental task from corresponding fMRI activation maps?
- Are these frequencies consistent for different underlying graphs (functional vs. structural) and deep learning architectures?
- Does selecting specific graph frequencies increase brain decoding performance?



Our contribution TL;DR

We propose a new methodology to perform graph frequency selection that we compare to standards methods :

- The core idea is to select by a learning process the appropriate graph frequencies within the signals.
- We consistently show its efficiency, robustness and replicability

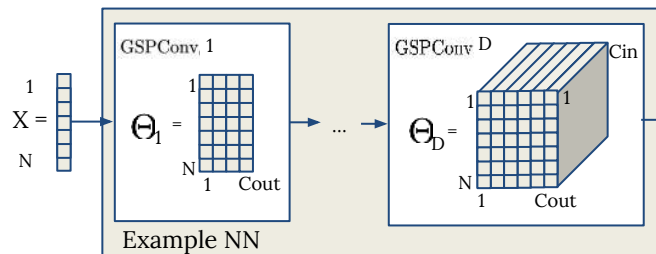


Focus on our pruning / frequency selection methodology

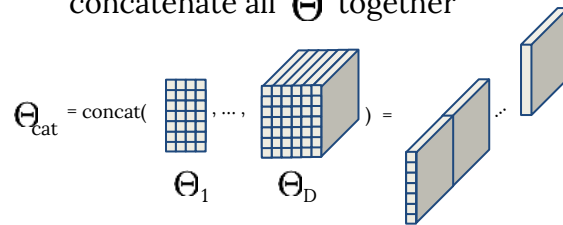
- We introduce a Deep Learning trainable layer that allows to convolute our signal over the graph :

$$\text{GSPConv}_{\text{Cin}, \text{Cout}}(\mathbf{X}) = \sigma \left(\text{GFT}^{-1} \left(\text{GFT}(\mathbf{X}) \otimes \Theta \right) \right) \quad \text{with } \Theta \in \mathbb{R}^{N \times \text{Cin} \times \text{Cout}} \quad (N : \text{number of frequencies})$$

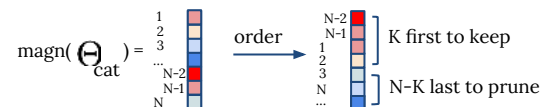
- We use this layer to build Neural Networks



- At each step of the training we concatenate all Θ together

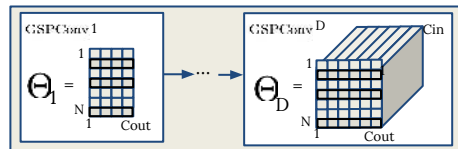


- To identify the frequency to prune

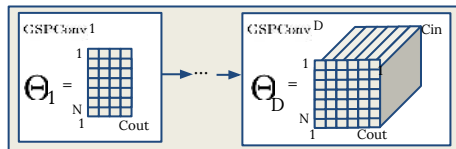


- Pruning phase

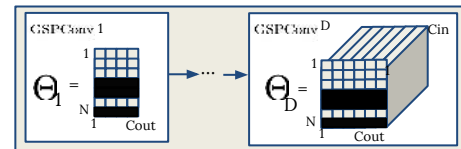
$$\alpha(s) = \alpha_{\min} \left(\frac{\alpha_{\max}}{\alpha_{\min}} \right)^{\frac{s}{s_{\text{final}}}}$$



step 0 : $\alpha = \alpha_{\min}$

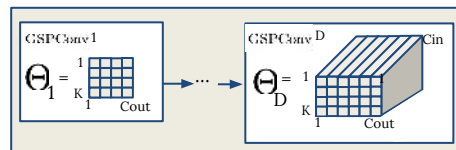


step s : $\alpha(s)$

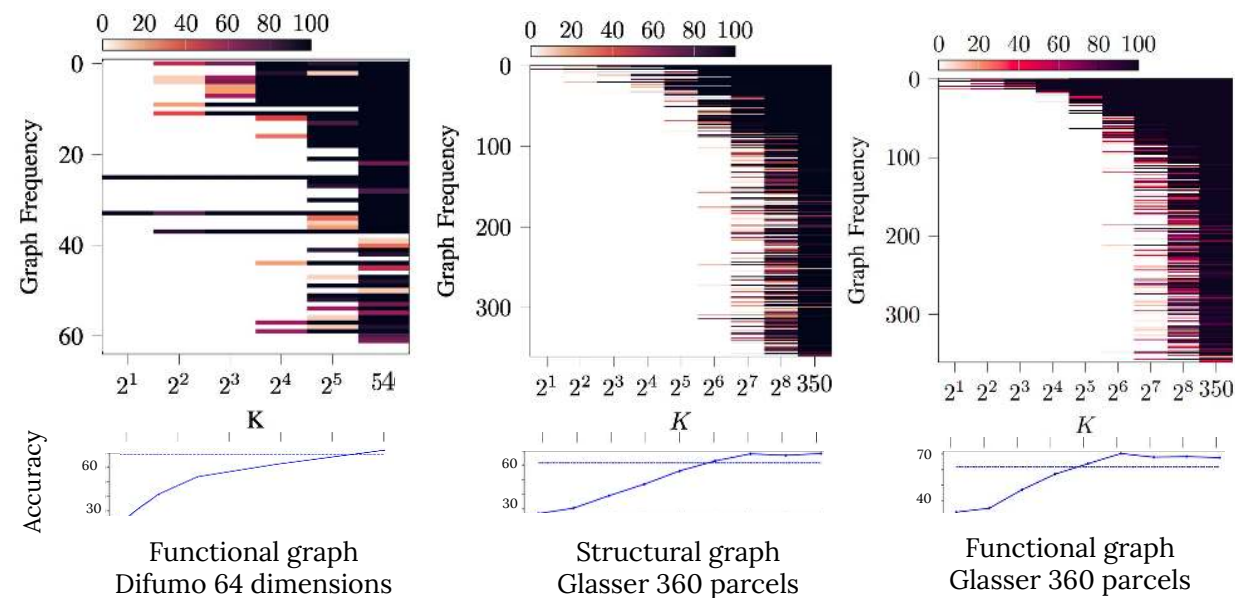


last step : $\alpha = \alpha_{\max}$

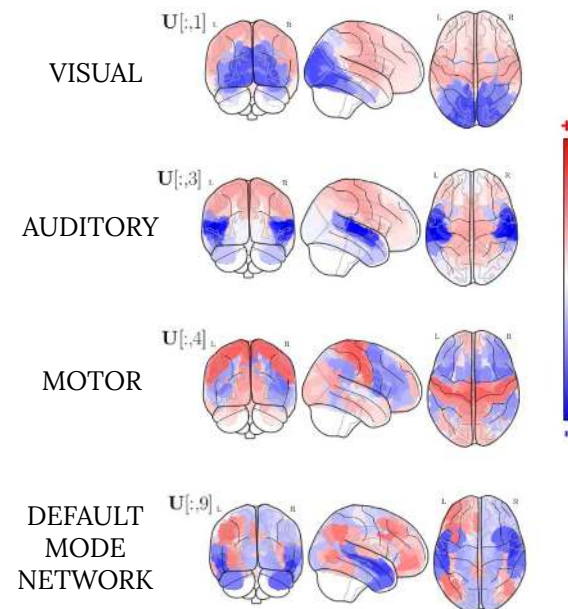
- LR-Rewinding phase



Results



Occurrences of each graph frequency for various fixed K
Corresponding accuracy (23 classes) bellow
(dotted line correspond to baseline without pruning)



Visualization of most frequently selected
graph eigenvectors of the
functional graph (Glasser 360) on the
glass brain.

Conclusions & Perspectives

- Original use of a pruning methodology
 - Validated GNN for fMRI decoding
 - Only few frequencies are required to achieve good performances even among very diverse tasks.
 - Reusable & Adaptable code
-
- Use our methodology to identify spatial patterns characteristics of specific decoded task or of a patient profile.
 - Assess our methodology on other tasks/domains or for other purposes.



Conclusions & Perspectives

- ❖ Original use of a pruning methodology.
 - ❖ A posteriori validation of the use of GNN.
 - ❖ Only few frequencies are required to achieve good performances even among very diverse tasks.
 - ❖ Our code is reusable and easy to adapt.
-
- ❖ Use our methodology to identify spatial patterns characteristics of specific decoded task or of a patient profile.
 - ❖ Assess our methodology on other tasks/domains or for other purposes.



Conclusions & Perspectives

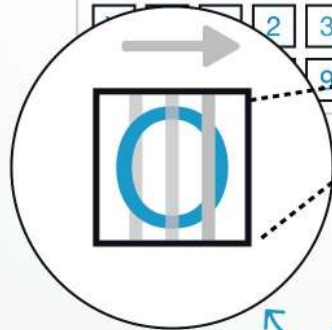


- The generalizability of EEGSimpleConv to other EEG paradigms beyond motor imagery (such as cognitive load, attention states, and sleep stages) is discussed briefly but merits its more detailed examination to strengthen the claim of broad applicability. Further, although motor imagery is a standard task condition in BCI research, it has substantial practical limitations, such as the impossibility of controlling whether subjects correctly perform the task, following instructions. This is briefly mentioned in Section 3.2.3.3, where subject fatigue is evoked as a possible reason for performance degradation. A broader discussion around the limitations of MI would have been of interest.
 - a. Parti sur un modele qui le permet mieux, d'adapter sur des modèles différents via le FM.
- Although EEGSimpleConv demonstrates excellent performance, the impact of hyperparameter choices on model performance could be explored further, particularly in varying conditions or experimental setups. I also found that some of the most technical parameters were missing or not evidently detailed in the relevant Methods sections. I suppose they are all well emphasized in the code provided on GitHub, but those details and related discussions would have strengthened the thesis argumentation.
- The versatility of the proposed architecture for EEGSimpleConv is commendable, yet further discussion of computational trade-offs (e.g., computational complexity versus accuracy in online and adaptive contexts) would be beneficial. A related minor comment concerns the lack of detail on machine learning libraries and computing hardware used to perform the experiments described in the thesis.
- The integration of spatial information in EEGSimpleConv could be more explicitly compared with methods that inherently model spatial relationships, potentially highlighting scenarios where spatial information could play a critical role.
 - a. Graph, conv spatiale,
- One of the promises at the onset of the thesis for using a simpler learning architecture was to make it more transparent and interpretable. However, I do not believe that there is such a discussion around the experiments using EEGSimpleConv; it would have been interesting to get some insight about the EEG features used by the model.
- Chapter 3 effectively addresses calibration challenges using few-shot and adaptive approaches. However, an extended discussion regarding limitations of the simulated online conditions versus actual online environments would strengthen the evaluation.
- For the foundation model (Chapter 4), while the scale of data used is impressive, practical aspects related to computational costs, hardware requirements, and feasibility for widespread adoption could be discussed more explicitly.

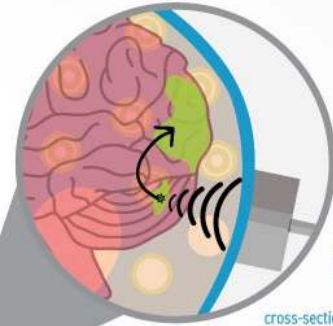
Given that this work is relying heavily on open-source libraries, contributing back to these libraries could have been very appreciated, since learning curves and few-shot learning methodologies could be easily shared. Indeed, I appreciate the fact that an important effort is made to make the code available in open source when possible, but instead publishing code on a separate platform, it could have been useful to help maintain the existing libraries.

H E L L O

A	B	C	D	E	F
G	H	I	J	K	L
M	N	O	P	Q	R
S	T	U	V	W	X
Y	Z	1	2	3	4
5	6	7	8	9	10



letter selection
through eye tracking

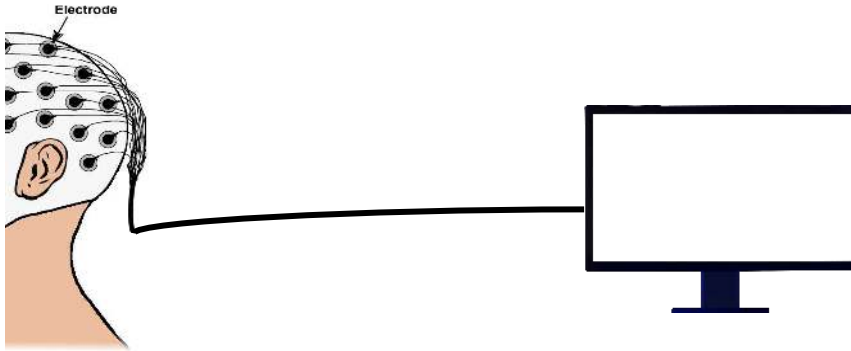


cross-section
showing brain activity

Carnegie Mellon University
College of Engineering

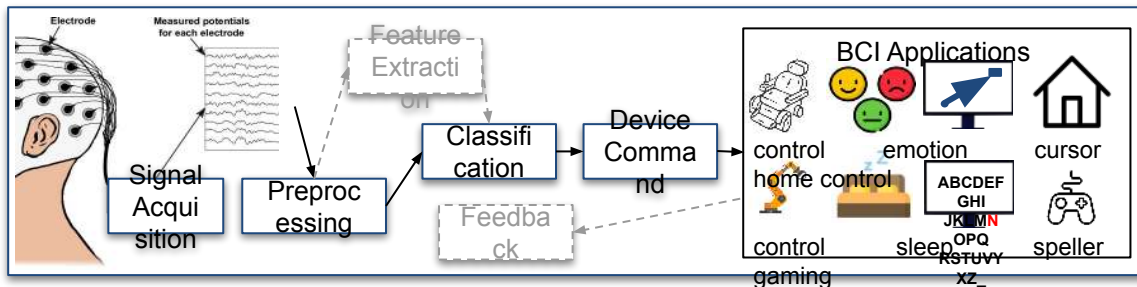
Context: BCI EEG

What is BCI ?



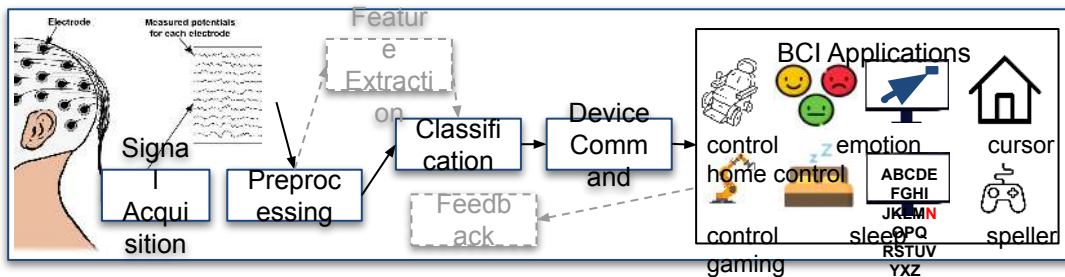
Context: BCI EEG

What is BCI ?

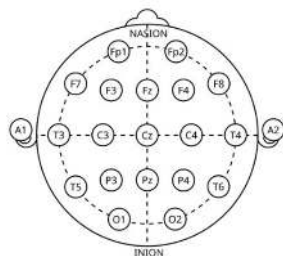


Context: BCI EEG

What is BCI ?



What is EEG ?



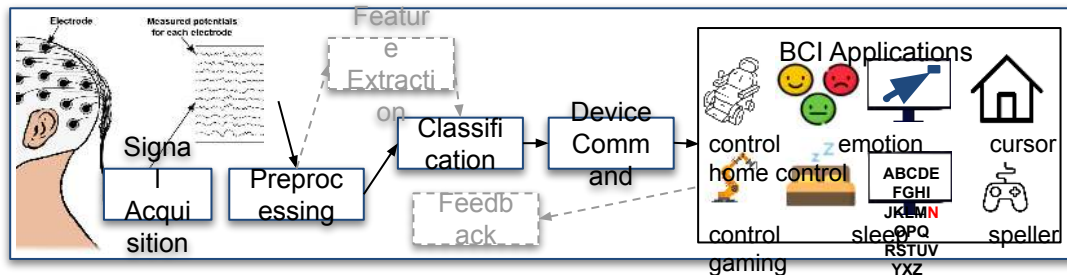
Electrode Layout
(Top View)



Signal Waveforms

Context: BCI EEG

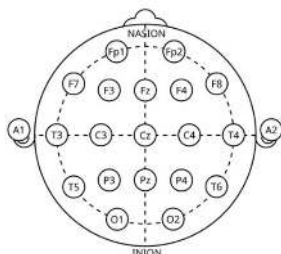
What is BCI ?



How is a BCI dataset characterized



What is EEG ?



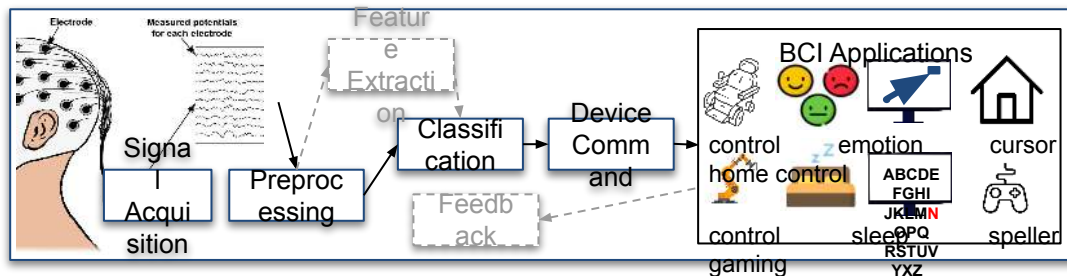
Electrode Layout
(Top View)



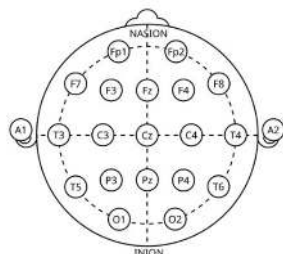
Signal Waveforms

Context: BCI EEG

What is BCI ?



What is EEG ?

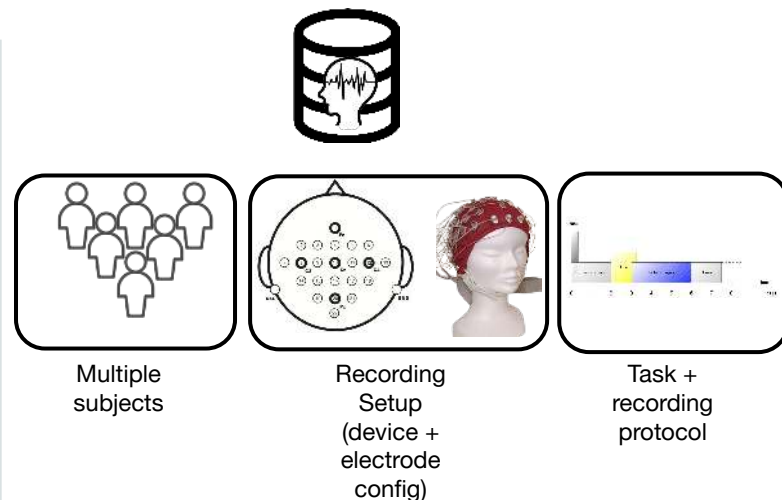


Electrode Layout
(Top View)



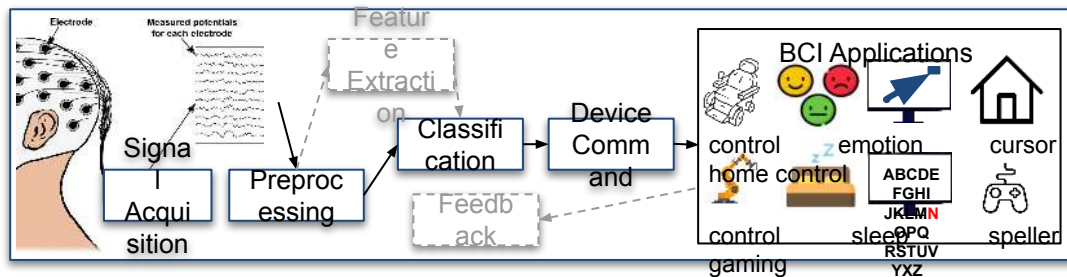
Signal Waveforms

How is a typical BCI dataset characterized

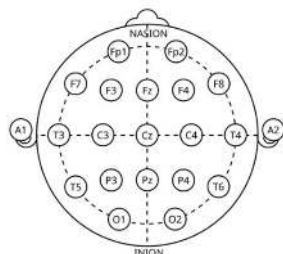


Context: BCI EEG

What is BCI ?



What is EEG ?

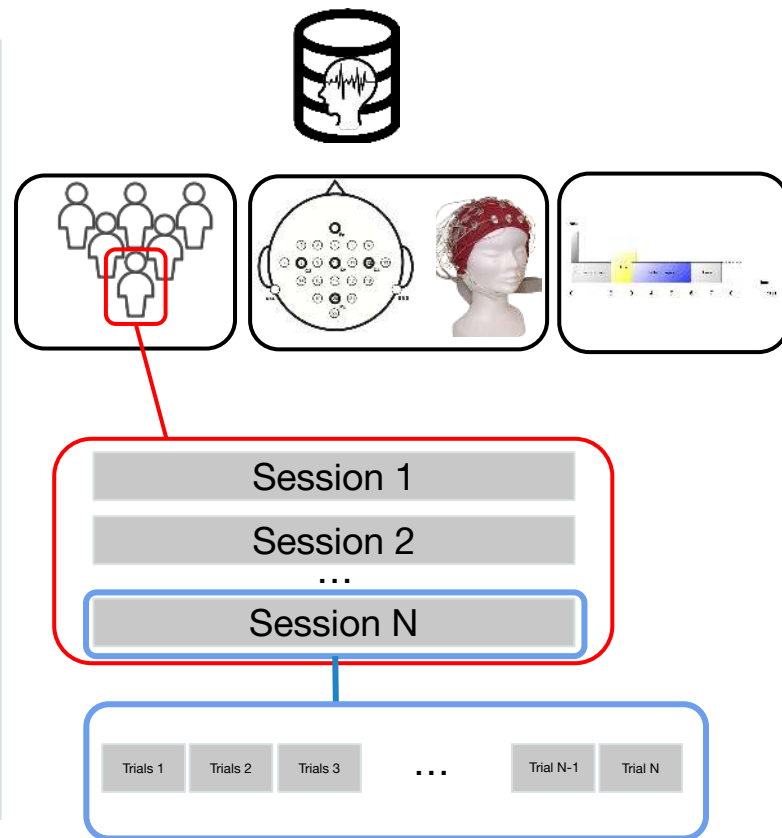


Electrode Layout
(Top View)



Signal Waveforms

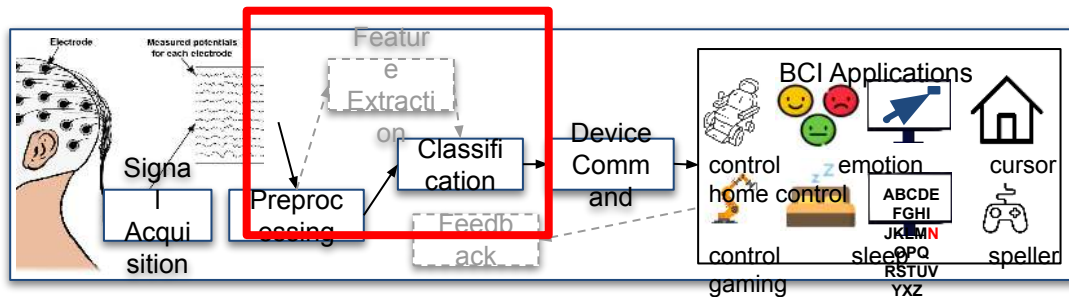
How is a typical BCI dataset characterized



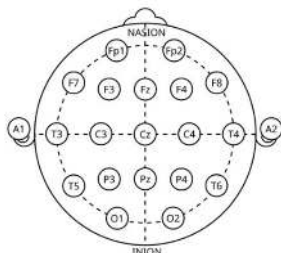
Context: BCI EEG

What is BCI ?

Learning problem : decoding Brain activity



What is EEG ?

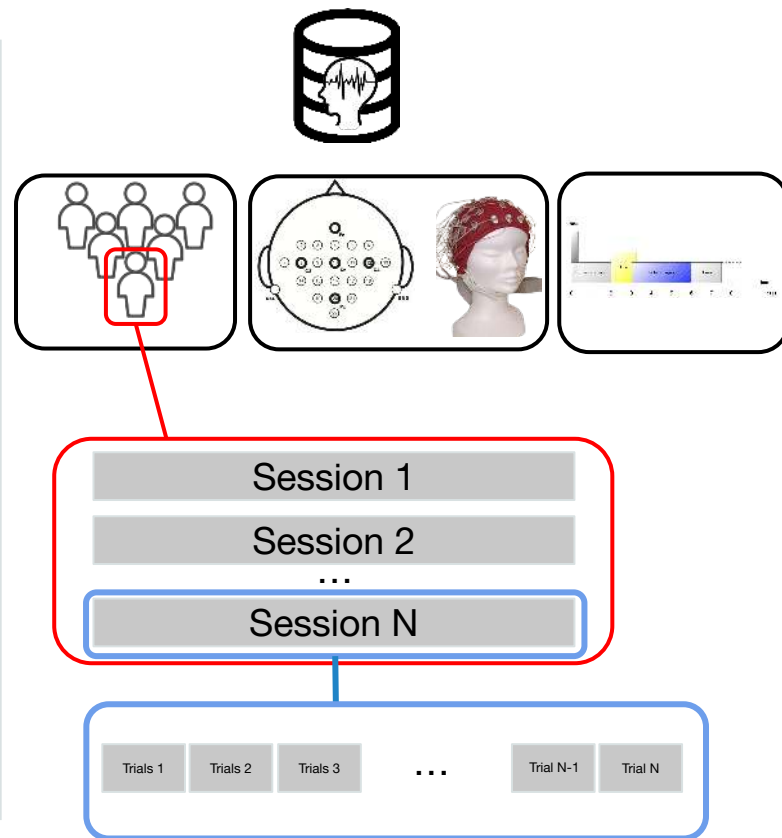


Electrode Layout
(Top View)



Signal Waveforms

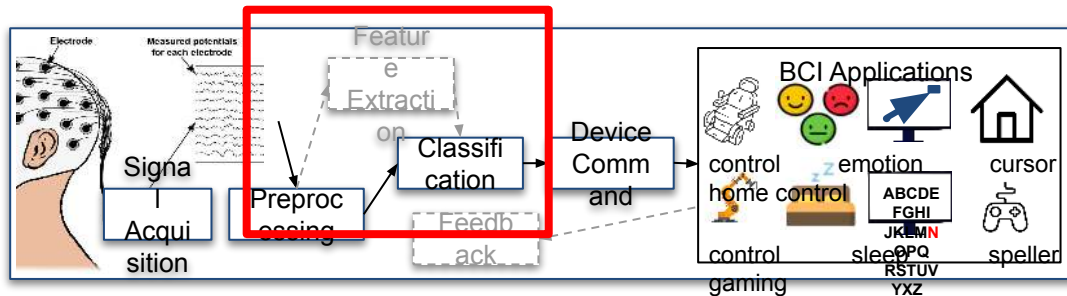
How is a typical BCI dataset characterized



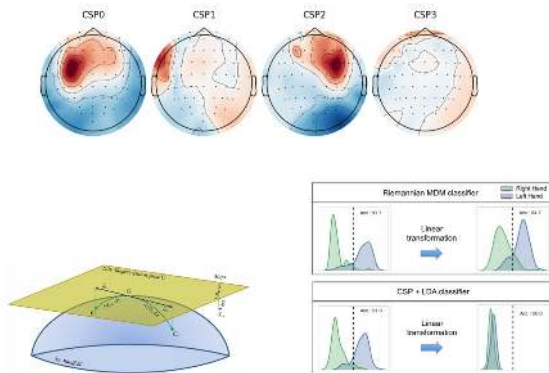
Context: BCI EEG

What is BCI ?

Learning problem : decoding Brain activity

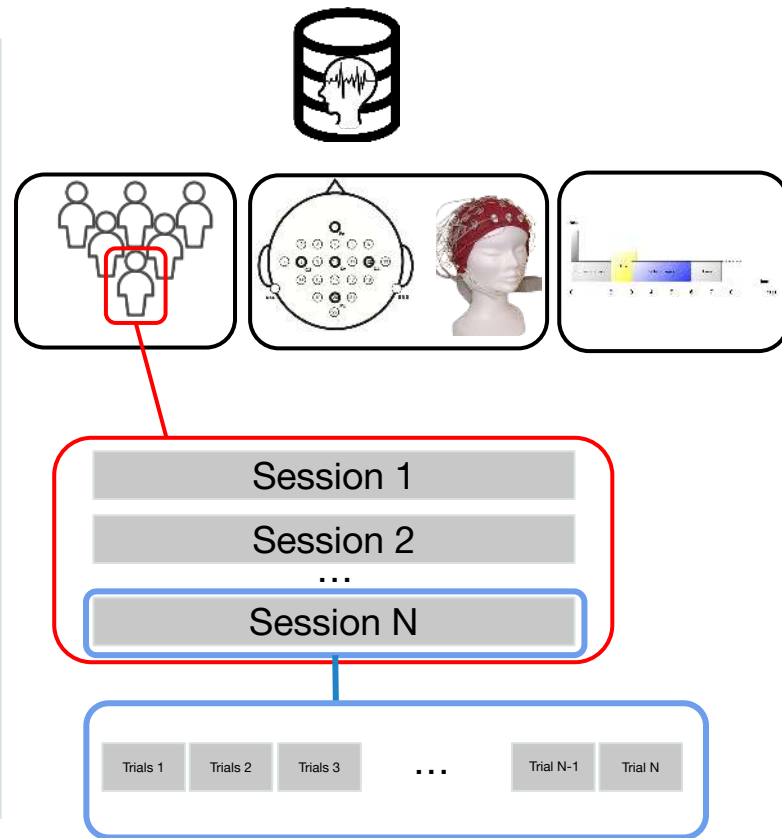


Traditional methods used in BCI systems



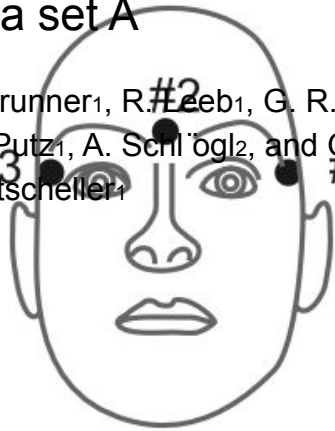
Designed at the session Level

How is a typical BCI dataset characterized

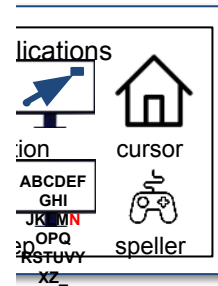


BCI Competition 2008 – Graz data set A

C. Brunner¹, R. Leeb¹, G. R. Müller-
Ler, Putz¹, A. Schlögl², and G.
Pfurtscheller¹



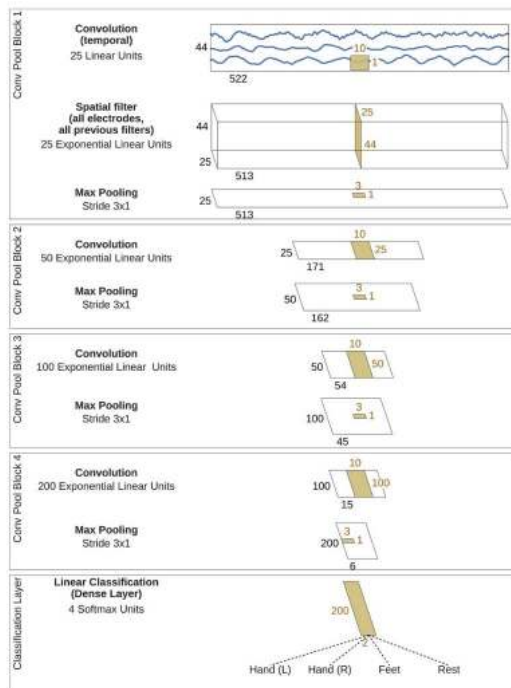
spécificités du domaine



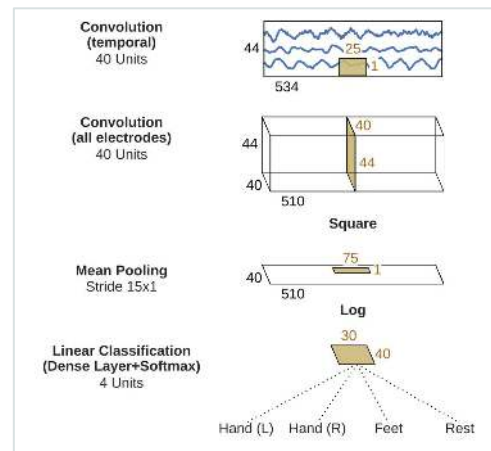
S/S/R/T
S/E C

intro methodes tradi

Deep Learning methods for BCI



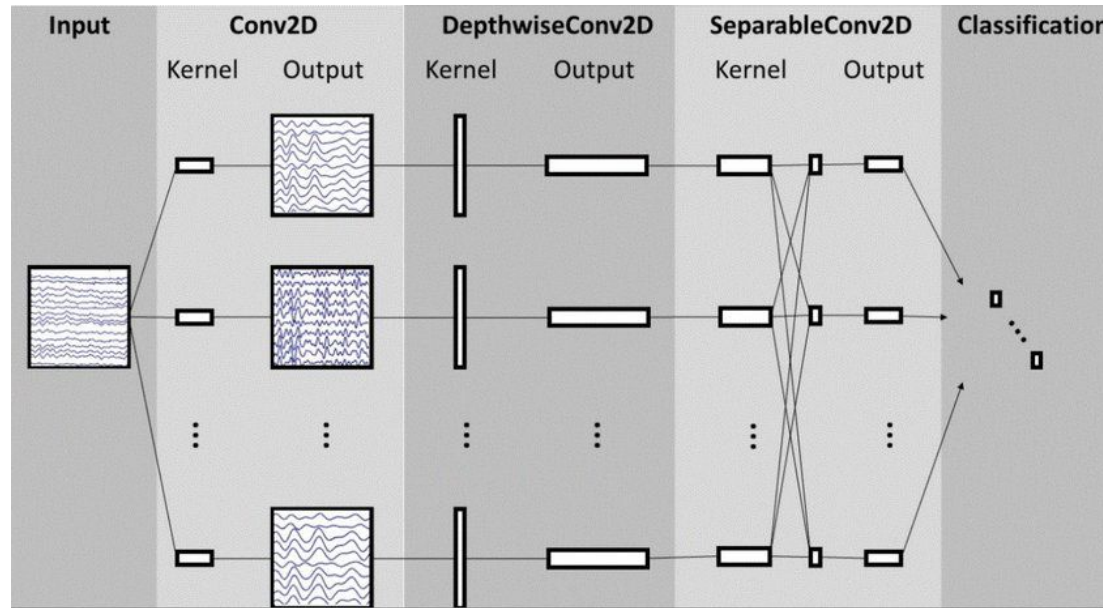
DeepConvNet [2]



Shallow ConvNet [2]

[2] Schirrmeister, Robin Tibor, et al. "Deep learning with convolutional neural networks for EEG decoding and visualization." *Human brain mapping* 38.11 (2017): 5391-5420.

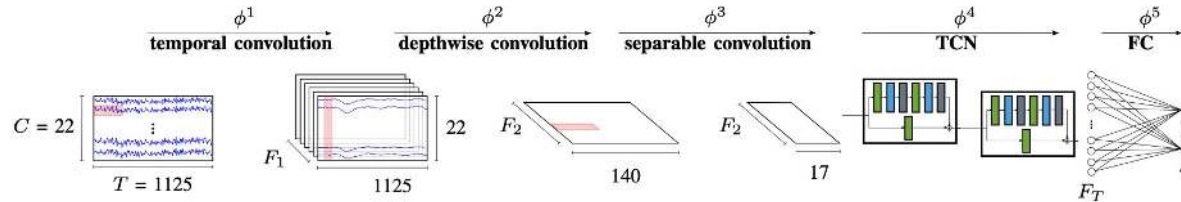
Deep Learning methods for BCI



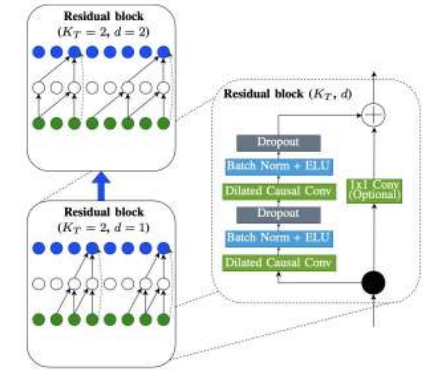
EEGNet [3]

[3] Lawhern, Vernon J., et al. "EEGNet: a compact convolutional neural network for EEG-based brain-computer interfaces." *Journal of neural engineering* 15.5 (2018): 056013.

Deep Learning methods for BCI



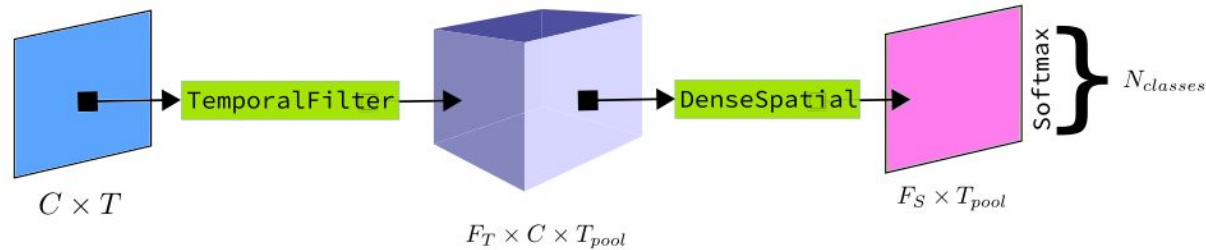
EEG-TCNet [4]



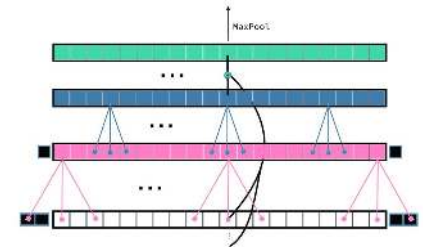
TCN Module

[4] Ingolfsson, Thorir Mar, et al. "EEG-TCNet: An accurate temporal convolutional network for embedded motor-imagery brain-machine interfaces." *2020 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. IEEE, 2020.

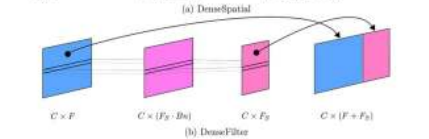
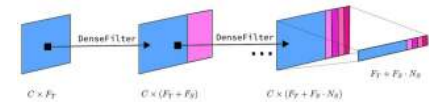
Deep Learning methods for BCI



TIDNet [5]



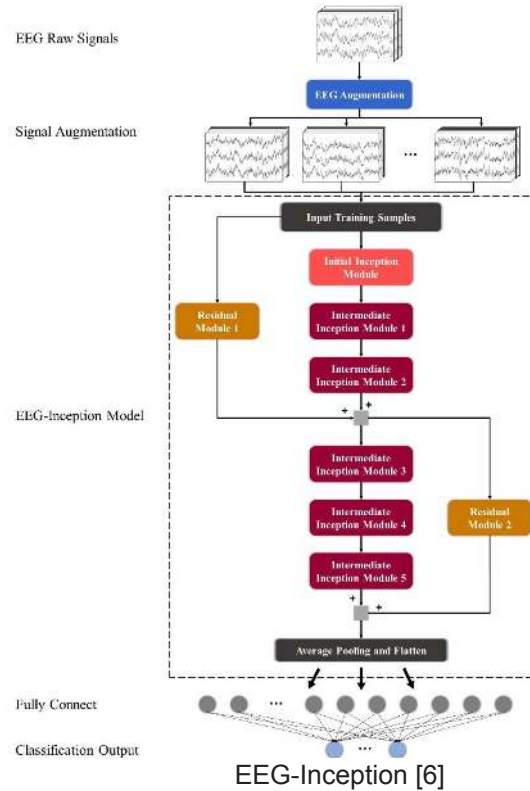
Temporal Filter



Dense Spatial

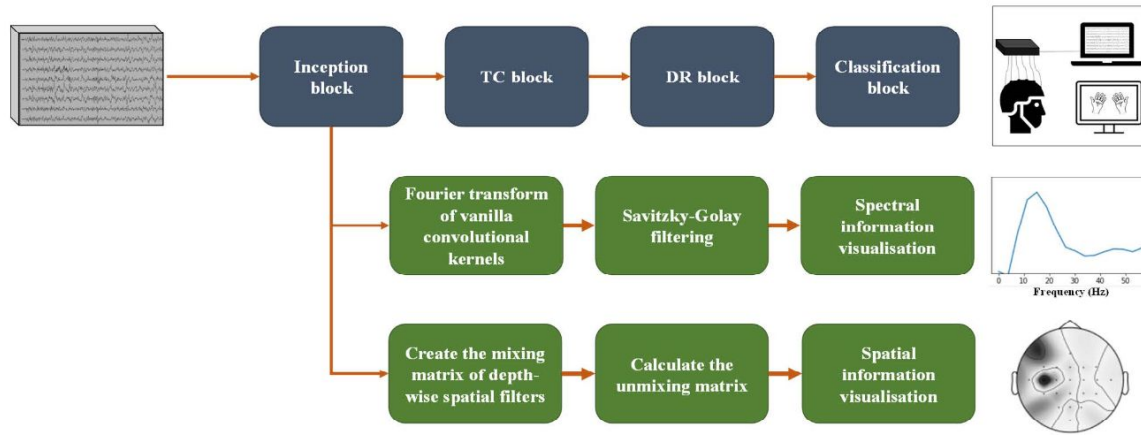
[5] Kostas, Demetres, and Frank Rudzicz. "Thinker invariance: enabling deep neural networks for BCI across more people." *Journal of Neural Engineering* 17.5 (2020): 056008.

Deep Learning methods for BCI

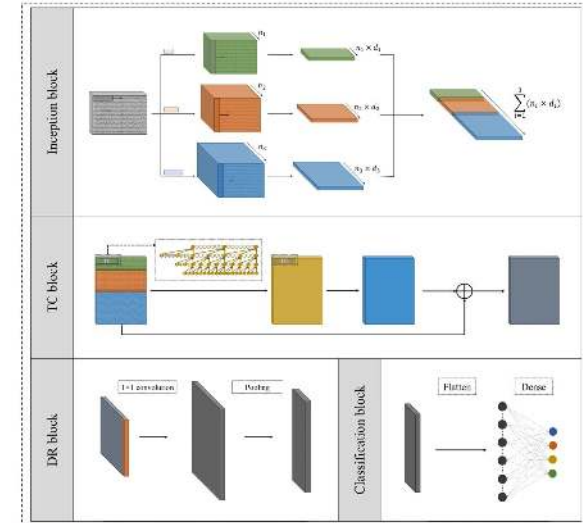


[6] Zhang, Ce, Young-Keun Kim, and Azim Eskandarian. "EEG-inception: an accurate and robust end-to-end neural network for EEG-based motor imagery classification." *Journal of Neural Engineering* 18.4 (2021): 046014.

Deep Learning methods for BCI



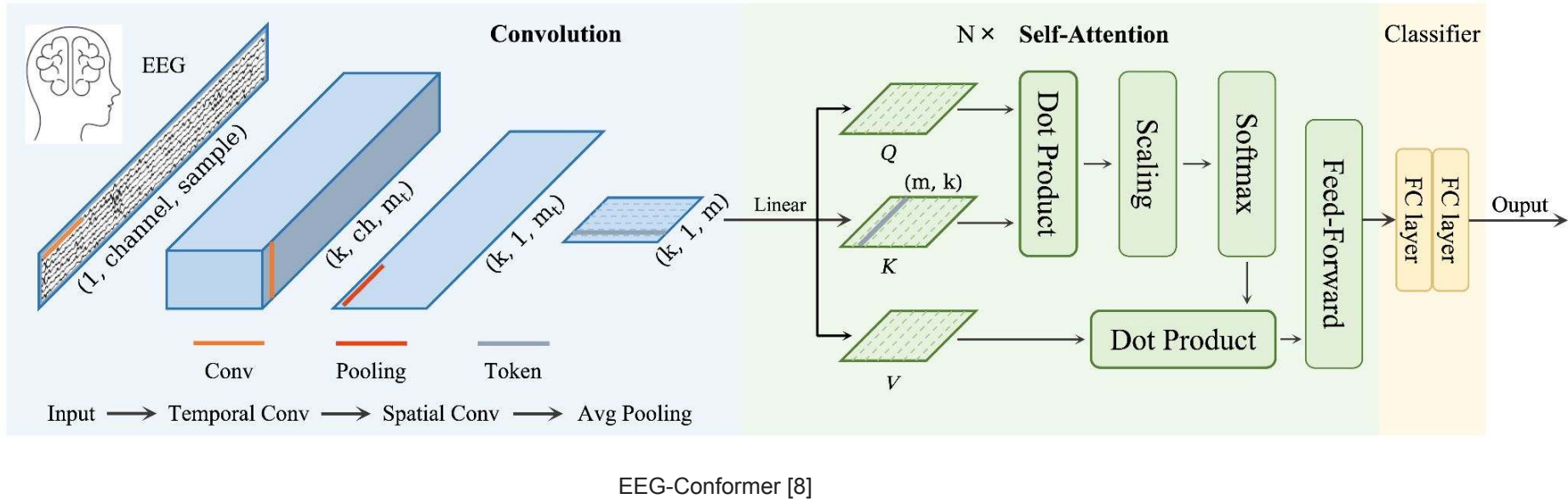
EEG-ITNet [7]



Different blocks details

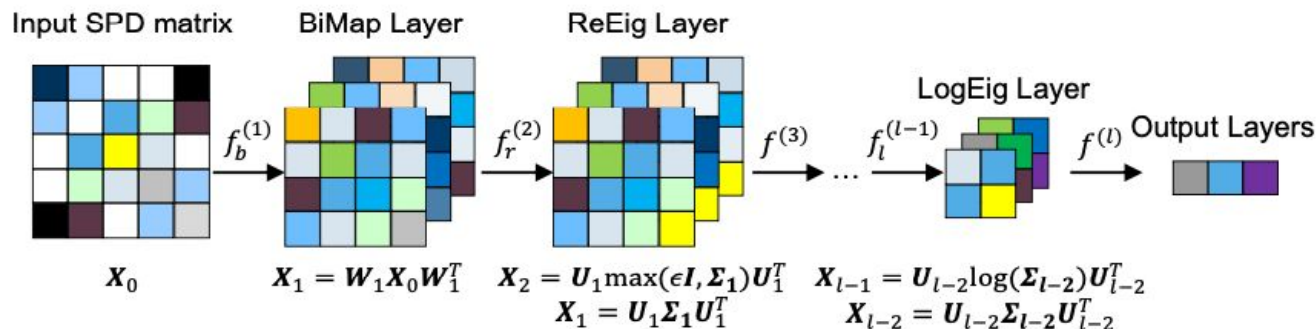
[7] Salami, Abbas, Javier Andreu-Perez, and Helge Gillmeister. "EEG-ITNet: An explainable inception temporal convolutional network for motor imagery classification." *Ieee Access* 10 (2022): 36672-36685.

Deep Learning methods for BCI



[8] Song, Yonghao, et al. "EEG conformer: Convolutional transformer for EEG decoding and visualization." *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 31 (2022): 710-719.

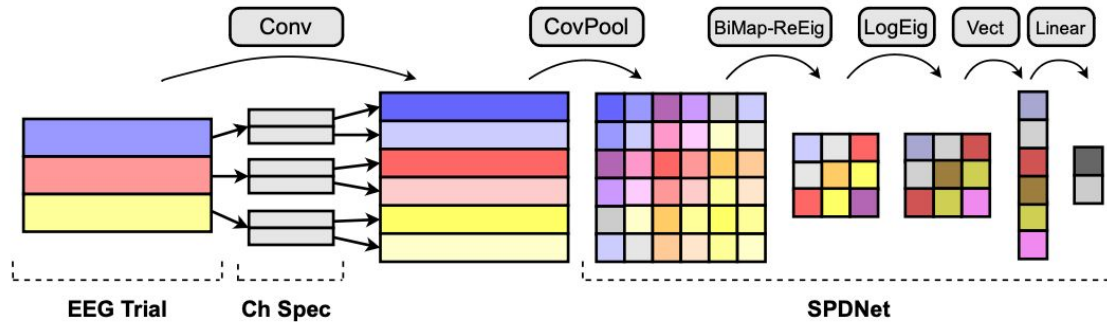
Deep Learning methods for BCI



SPDNet [9]

[9] Huang, Zhiwu, and Luc Van Gool. "A riemannian network for spd matrix learning." *Proceedings of the AAAI conference on artificial intelligence*. Vol. 31. No. 1. 2017.

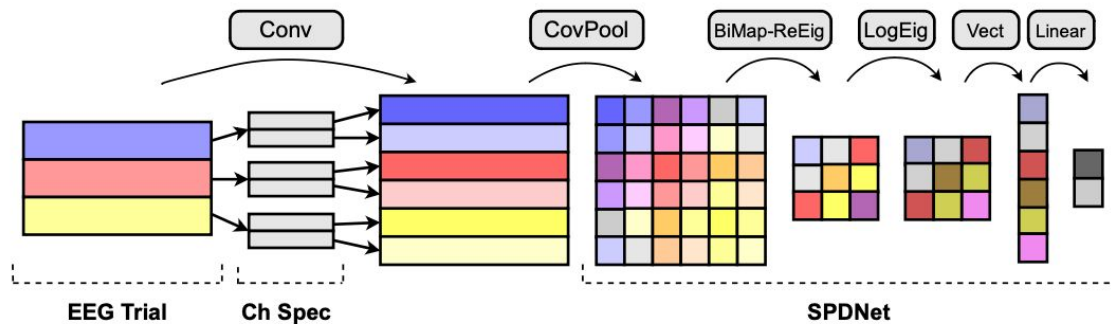
Deep Learning methods for BCI



CNN-SPDNet [10]

[10] Wilson, Daniel, et al. "Deep riemannian networks for eeg decoding." *arXiv preprint arXiv:2212.10426* (2022).

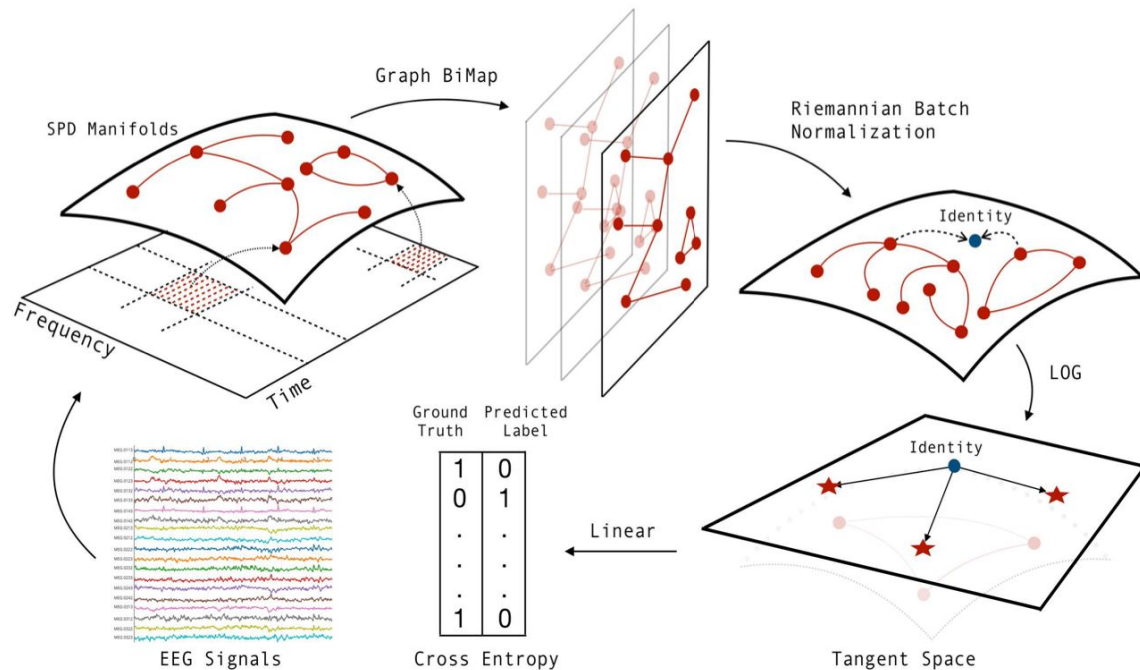
Deep Learning methods for BCI



CNN-SPDNet [10]

[10] Wilson, Daniel, et al. "Deep riemannian networks for eeg decoding." *arXiv preprint arXiv:2212.10426* (2022).

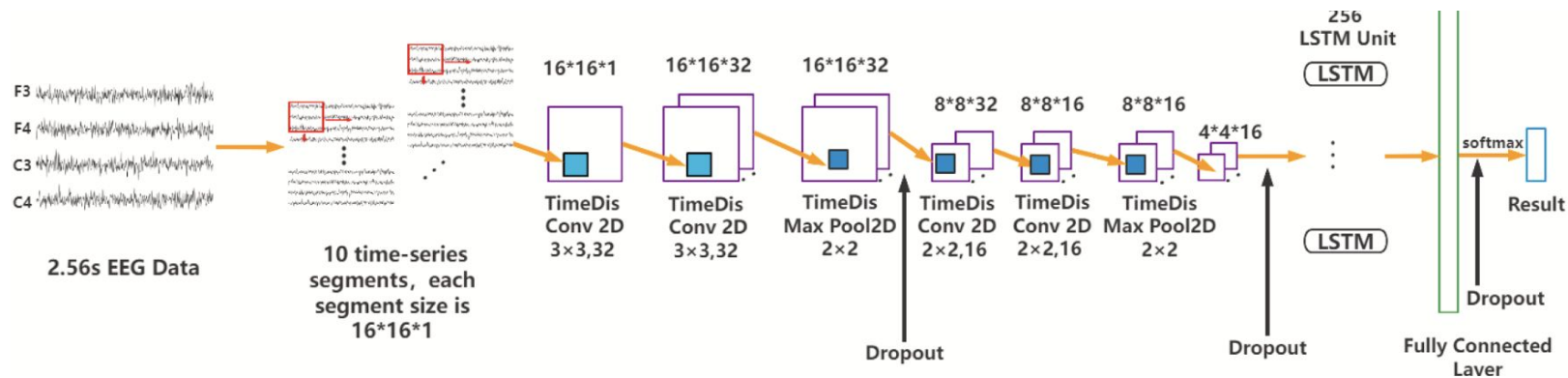
Deep Learning methods for BCI



Graph-CSPNet [11]

[11] Ju, Ce, and Cuntai Guan. "Graph neural networks on spd manifolds for motor imagery classification: A perspective from the time–frequency analysis." *IEEE Transactions on Neural Networks and Learning Systems* (2023).

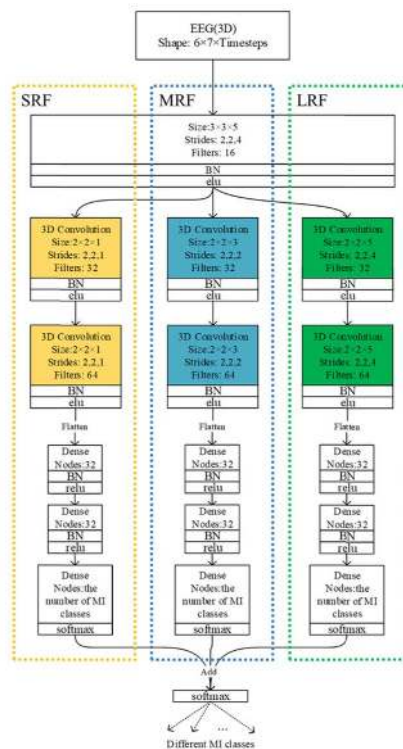
Deep Learning methods for BCI



2DCNN-LSTM [12]

[12] Wang, Jialing, et al. "A 2D CNN-LSTM hybrid algorithm using time series segments of EEG data for motor imagery classification." *Biomedical Signal Processing and Control* 83 (2023): 104627.

Deep Learning methods for BCI



Multi-Branch 3D CNN [13]

[13] Zhao, Xinqiao, et al. "A multi-branch 3D convolutional neural network for EEG-based motor imagery classification." *IEEE transactions on neural systems and rehabilitation engineering* 27.10 (2019): 2164-2177.

Generalization / evolution in leading DL fields

main axes : 1) Domain 2) Task 3) Architecture + Training methods

Generalization / evolution in leading DL fields

Domain

main axes : 1) Domain 2) Task 3) Architecture + Training methods

Vision

Text

EEG

train

test

Foundation
Model

Generalization / evolution in leading DL fields

Domain

main axes : 1) Domain 2) Task 3) Architecture + Training methods

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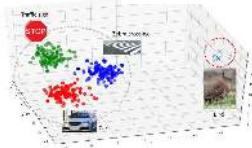
test



ImageNet Dataset



Medical Images



CNN:
AlexNet, VGG,
Resnet
Inception,
DensNet, U-Nets

CNN:
AlexNet, VGG,
Resnet
Inception,
DensNet, U-Nets

Foundation
Model

Problem Statement

Does applying deep learning to BCI demand specialized methods, or can established techniques from computer vision be successfully adapted for EEG?

Plan;

- 1) Dvpt d'un modèle general extracteur de caratéristique
- 2) On l'utilise pour de la reduction de calibration

Evolution of Deep Learning

main axes : 1) Domain 2) Task 3) Architecture + Training methods

Vision

Text

EEG

train

test



MNIST
IMAGENET
CIFAR

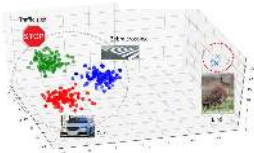
CNN:
AlexNet, VGG,
Resnet
Inception,
DensNet, U-Nets

ImageNet Dataset



Medical Images

CNN:
AlexNet, VGG,
Resnet
Inception,
DensNet, U-Nets



airplane

automobile

bird

cat

deer

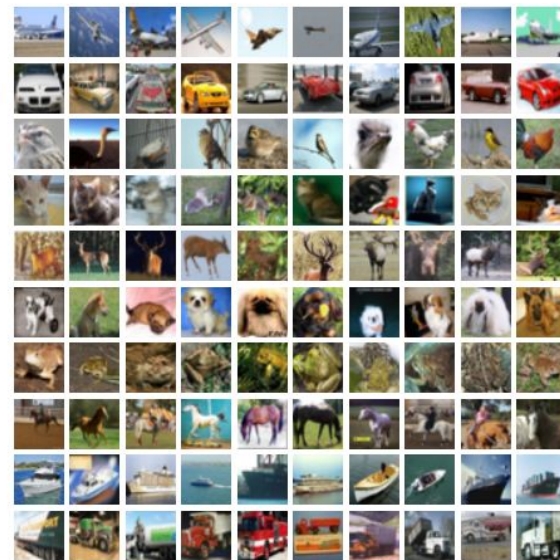
dog

frog

horse

ship

truck



Foundation
Model

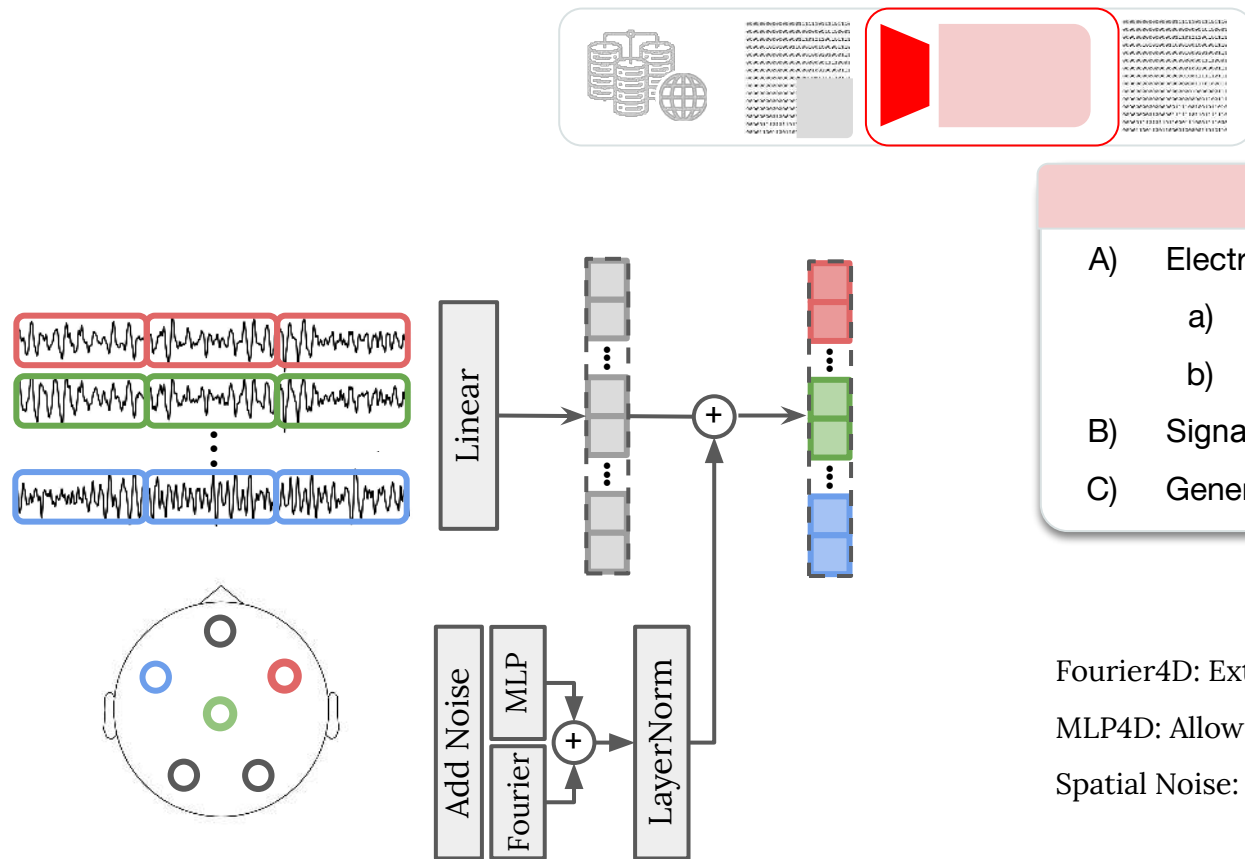
Problem Statement

- Old way: given domain D , and task T , train ad-hoc architecture $f_{\{T,D\}}$
- New way: train a generic f , on a pretext task and an internet scale dataset, then adapt to D and T

Does applying deep learning to BCI demand specialized methods, or can we leverage recent advances and established techniques and adapt them for EEG?

REVE: A foundation model for EEG adapting to any setup with large-scale pretraining

└ 4D Position Encoding Strategy



Challenges

- A) Electrode configurations
 - a) Number of electrodes
 - b) Electrode placement
- B) Signal duration
- C) Generalize to unseen configurations

Fourier4D: Extension of the 2D proposed in [x]

MLP4D: Allow more adaptation

Spatial Noise: Introduce more diversity

Problem Statement

- Old way: given domain D , and task T , train ad-hoc architecture $f_{\{T,D\}}$

$$TD \quad f_{T,D}$$

f

Problem Statement

- Old way: given domain D , and task T , train ad-hoc architecture $f_{\{T,D\}}$
- New way: train a generic f , on a pretext task and an internet scale dataset, then adapt to D and T

Does applying deep learning to BCI demand specialized methods, or can we leverage recent advances and established techniques and adapt them for EEG?

Plan:

- 1) Developing a Foundation model for EEG
- 2) Application study on minimizing Calibration Effort in BCI Systems

Ce qu'on propose

nos setup, adaptive, inductif + le benchmark

MI dataset	Subjects	Electrodes	Samples	Sampling frequency	Duration	Tasks
BNCI2014001 [107]	9	22	5184	250 Hz	4 s.	L/R/F/T
PhysionetMI [160]	103	64	18540	160 Hz	3 s.	L/R/F/BH/Re
AlexMI [161]	8	16	480	512 Hz	3 s.	R/F/Re
Weibo2014 [162]	10	60	5540	200 Hz	4 s.	L/R/F/BH/Re/2× C
Ofner2017 [163]	15	61	6300	512 Hz	3 s.	Re / 6× R
Cho2017 [109]	49	64	9880	512 Hz	3 s.	L/R
Lee2019MI [164]	54	62	10800	1000 Hz	4 s.	L/R
Schirrmeister [14]	14	128	13484	500 Hz	4 s.	L/R/F/Re
Zhou2016 [165]	4	14	1800	250 Hz	5 s.	L/R/F
BNCI2015001 [166]	12	13	5600	512 Hz	5 s.	R/F
BNCI2014004 [167]	9	3	6520	250 Hz	4.5 s.	L/R
Large [145]	85	27	20400	512 Hz	4 s.	L/R

Table 3.4 – MI datasets considered in our study. L = Left hand, R = Right hand, F = Feet, Re = Rest, BH = Both hands, C = Compound

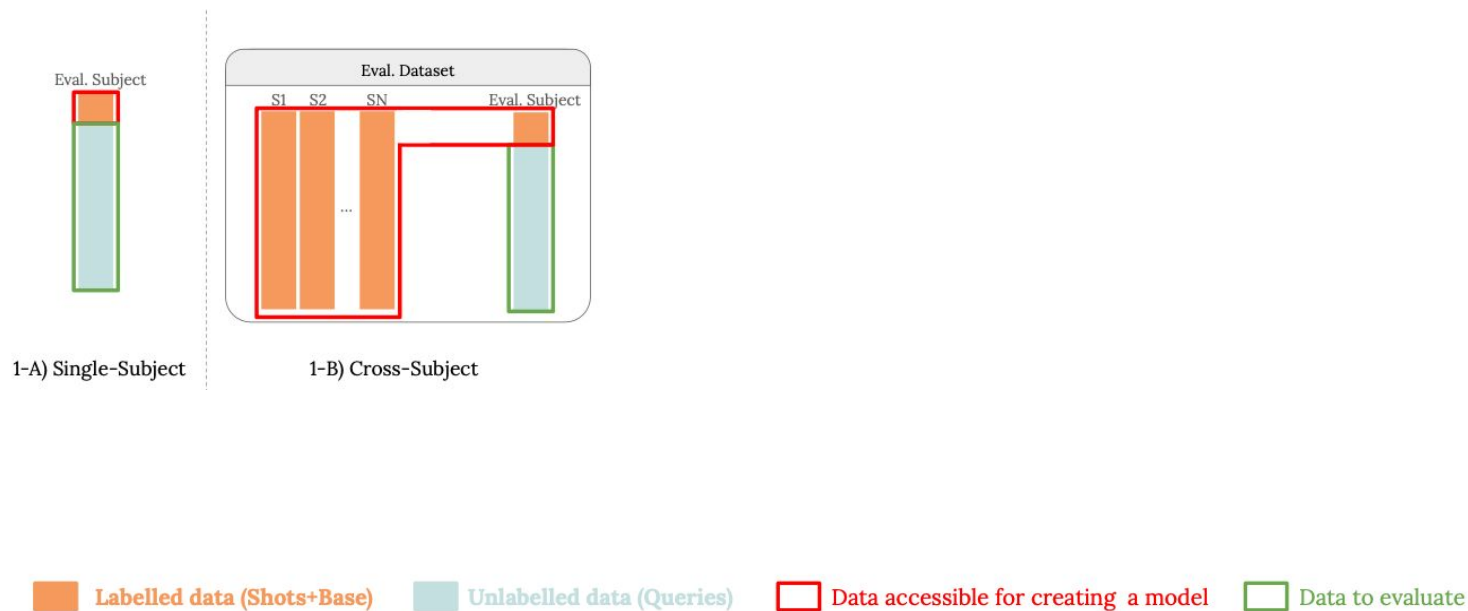
Context: Few-Shot Learning



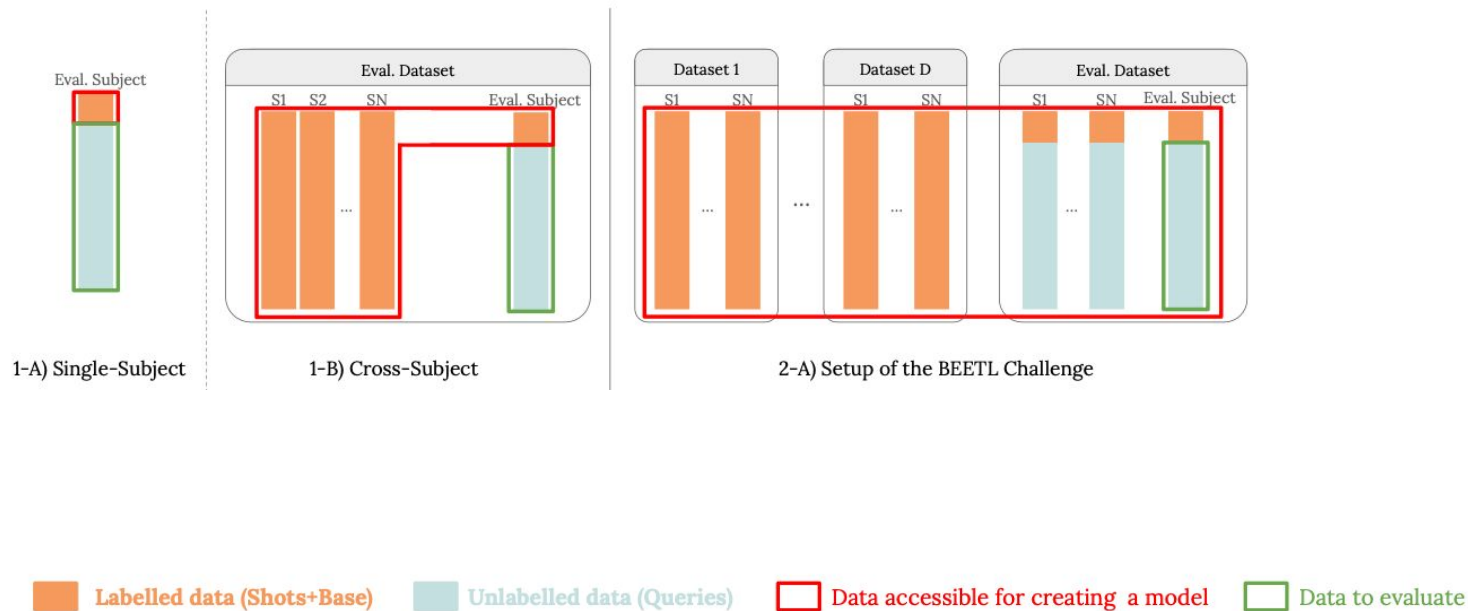
1-A) Single-Subject



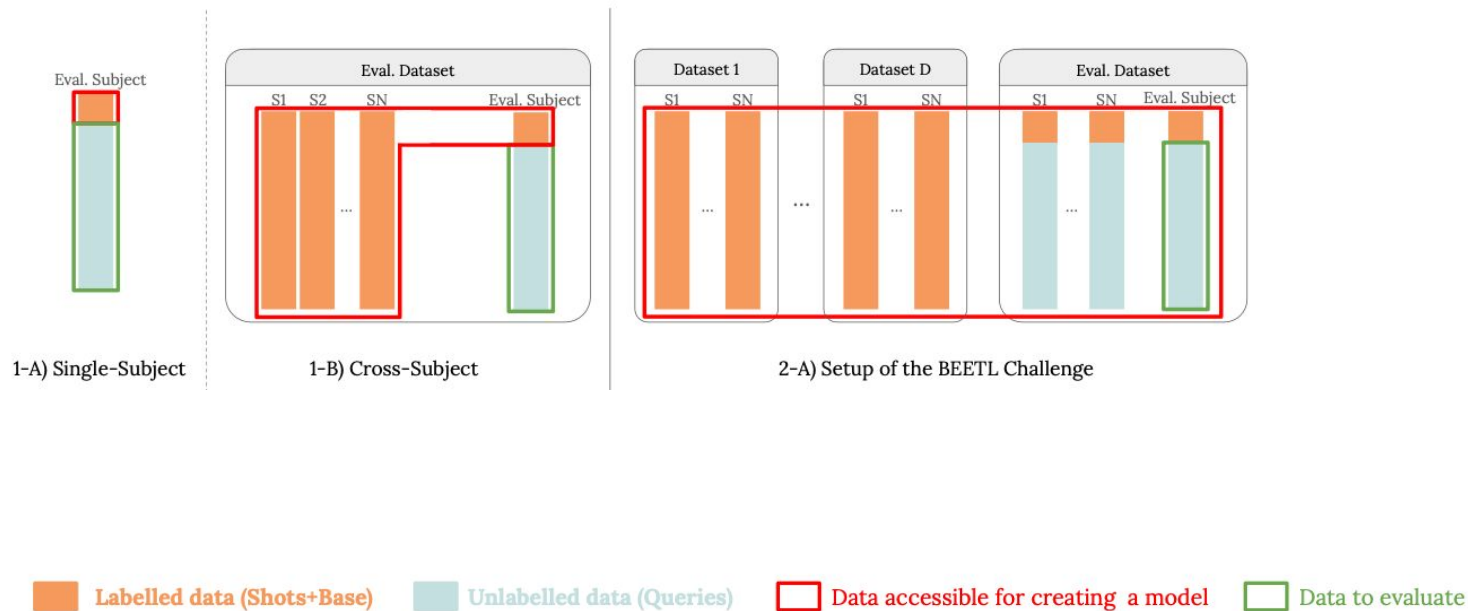
Context: Few-Shot Learning



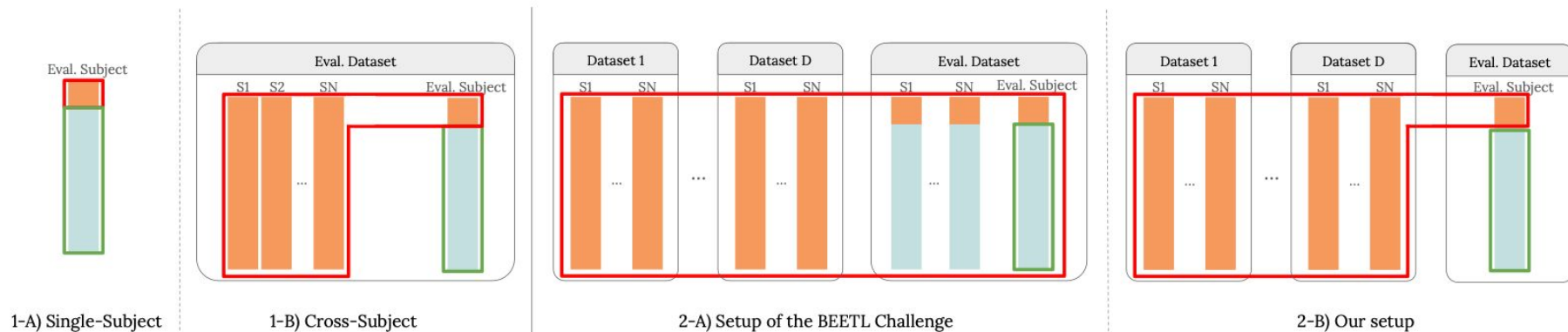
Context: Few-Shot Learning



Context: Few-Shot Learning



Context: Few-Shot Learning



Orange box: Labelled data (Shots+Base) Teal box: Unlabelled data (Queries) Red box: Data accessible for creating a model Green box: Data to evaluate

Other setups aussi on les test, on marche très bien
on fait aussi des comparaison entre les setups

Method	Task 2 Score
Winning Methods	
Cogitat [67, 169]	76.33
wduong [67]	71.33
ms01 [67]	59.87
Our Methods	
FSL	72.00
FSL + Ensembling	75.60
FT	75.40
FT + Ensembling	78.00

Table 3.6 – Performance Comparison on task 2 of the BEETL Challenge: Winning vs. Our methods.

What is a Foundation Model?

A **large, pre-trained deep neural network** serving as the basis for multiple tasks.

Vision: MAE, Dino, Sam, Stable Diffusion

Text: Llama, GPT, Gemini, Bert, Mistral



Shared characteristics:

Large Scale pretraining

Pretrained with SSL

Transformer based



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Self Supervised Learning (SSL)



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Reconstruct



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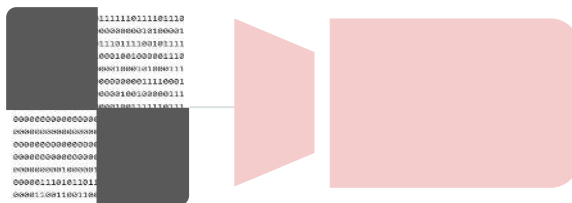


Shared characteristics:

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[illegible]

What's happening in vision and language

└ Foundation Models

Modern paradigm: Train a generic f on a pretext task and an internet scale dataset, then adapt to any D and T

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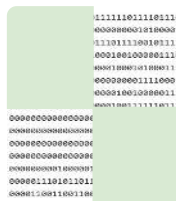
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Large Scale Datasets

Self Supervised Learning (SSL)



Reconstruct



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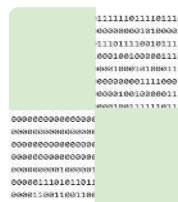
Vision: MAE, Dino, Sam, Stable Diffusion

Text: Llama, GPT, Gemini, Bert, Mistral

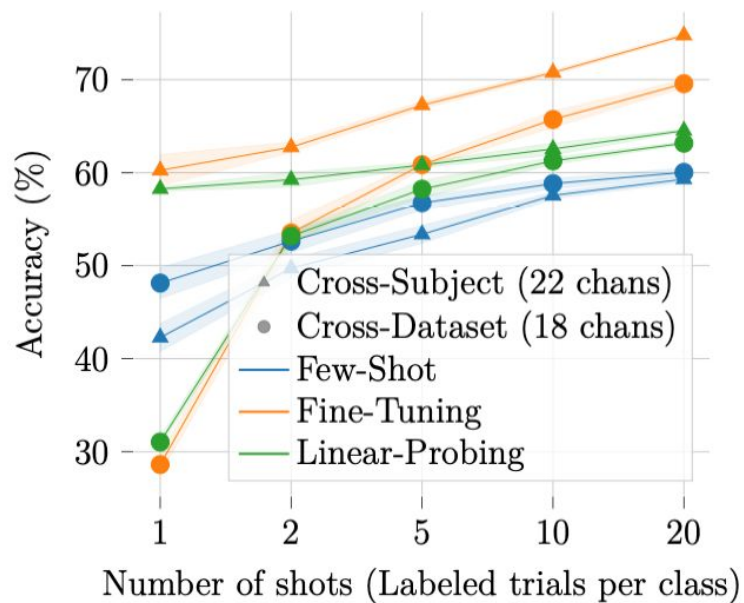


Large Scale Datasets

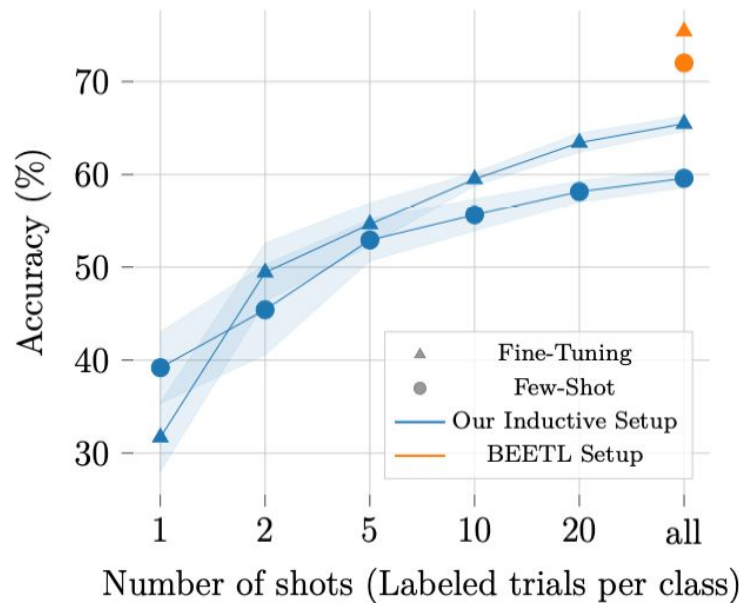
Self Supervised Learning (SSL)



Transformers



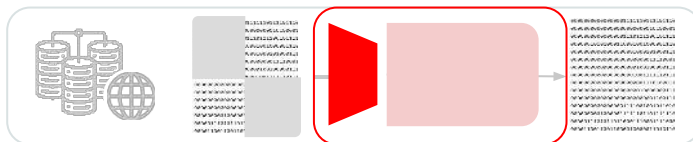
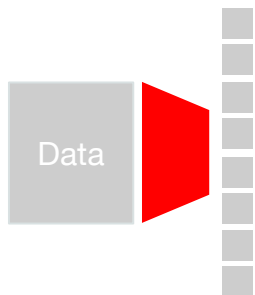
(a) Setup differences on BNCI2014001 (4 classes)



(b) Setup differences on task 2 of the BEETL challenge

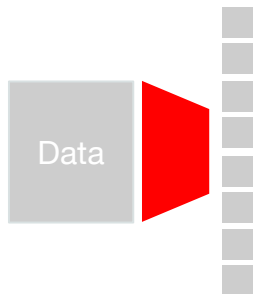
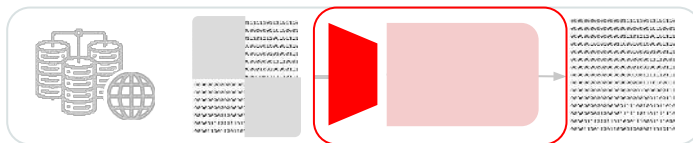
What's happening in vision and language

└ Transformer

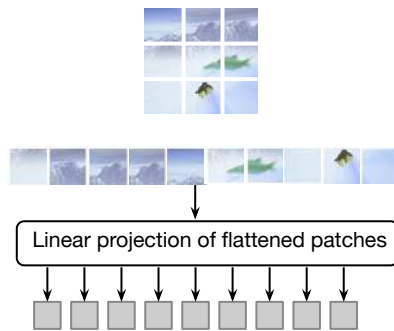
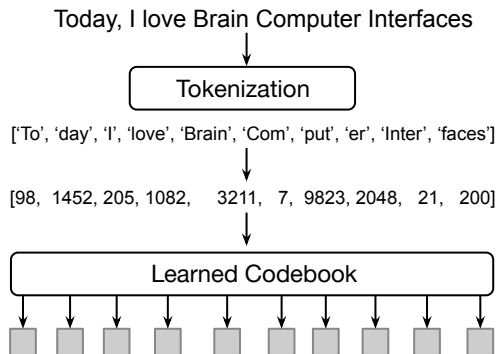


What's happening in vision and language

└ Transformer

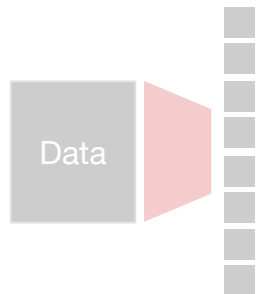


modality dependent

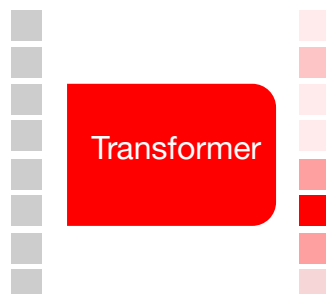
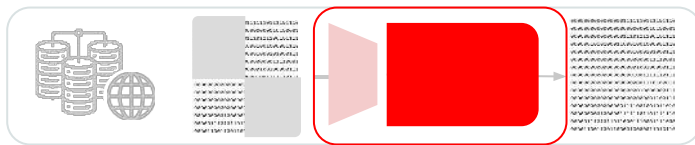


What's happening in vision and language

└ Transformer



modality dependent



modality agnostic

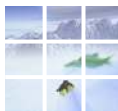
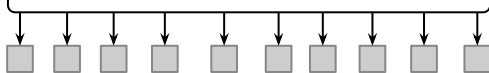
Today, I love Brain Computer Interfaces

Tokenization

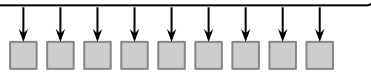
['To', 'day', 'I', 'love', 'Brain', 'Com', 'put', 'er', 'Inter', 'faces']

[98, 1452, 205, 1082, 3211, 7, 9823, 2048, 21, 200]

Learned Codebook



Linear projection of flattened patches



$$Z = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$$

with:

$$X = [x_1, x_2, \dots, x_n] \in \mathbb{R}^{n \times d}$$

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V$$

n : sequence length,

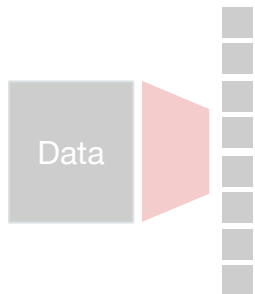
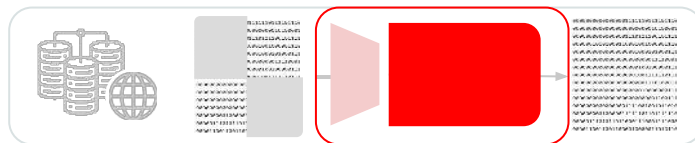
d : embedding dimension,

$W_Q, W_K, W_V \in \mathbb{R}^{d \times d_k}$: learned projection matrices,

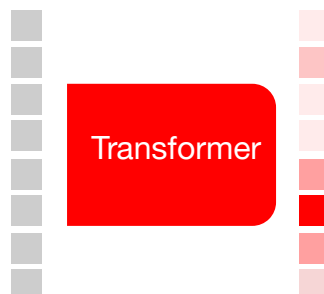
d_k : dimension of each key/query vector.

What's happening in vision and language

└ Transformer



modality dependent
(~1-8% weights)



modality agnostic
(~90-99% weights)

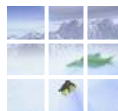
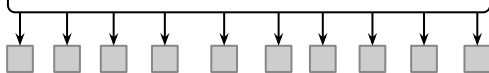
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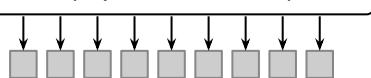
['To', 'day', 'I', 'love', 'Brain', 'Com', 'put', 'er', 'Inter', 'faces']

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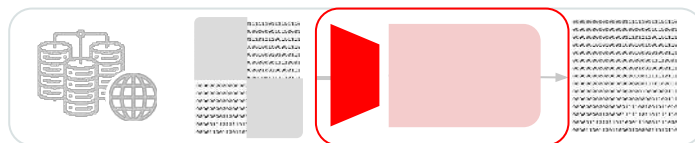
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What's happening in vision and language

└ Positional Encoding



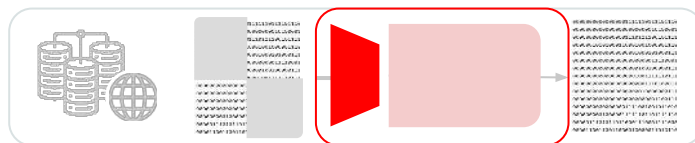
'To', 'day', 'I', 'love', 'Brain', 'Com', 'put', 'er', 'Inter', 'faces'

position

0 1 2 3 4 5 6 7 8 9

What's happening in vision and language

Positional Encoding



'To', 'day', 'I', 'love', 'Brain', 'Com', 'put', 'er', 'Inter', 'faces'

position

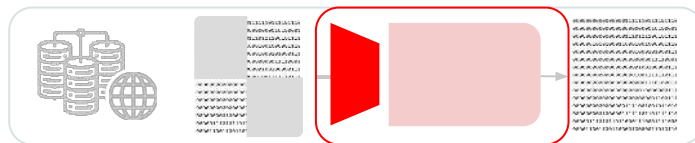
0 1 2 3 4 5 6 7 8 9

Integer encoding

-0.02	5
0.04	5
0.01	5
0.22	5
-0.09	5
0.12	5
-0.02	5
-0.11	5

What's happening in vision and language

└ Positional Encoding



'To', 'day', 'I', 'love', 'Brain', '**Com**', 'put', 'er', 'Inter', 'faces'

position

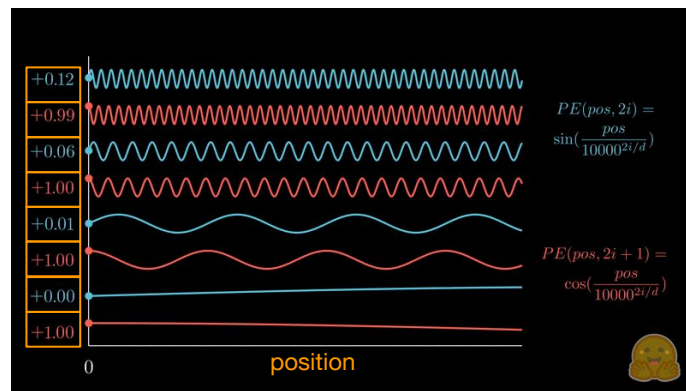
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0.04	5
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0.22	5
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-0.11	5

+

Fourier encoding

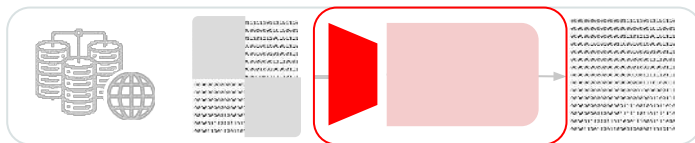


[x]

[x] Blogpost: You could have designed state of the art positional encoding, Christopher Fleetwood <https://huggingface.co/blog/designing-positional-encoding>

What's happening in vision and language

Positional Encoding



'To', 'day', 'I', 'love', 'Brain', 'Com', 'put', 'er', 'Inter', 'faces'

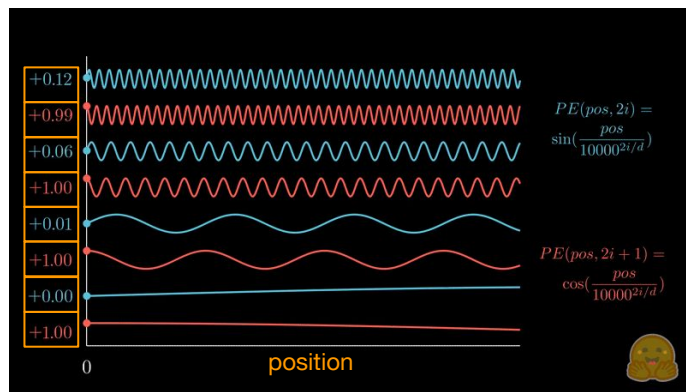
position

0 1 2 3 4 5 6 7 8 9

Integer encoding

-0.02	5
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0.01	5
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-0.09	5
0.12	5
-0.02	5
-0.11	5

Fourier encoding



Learnable embeddings

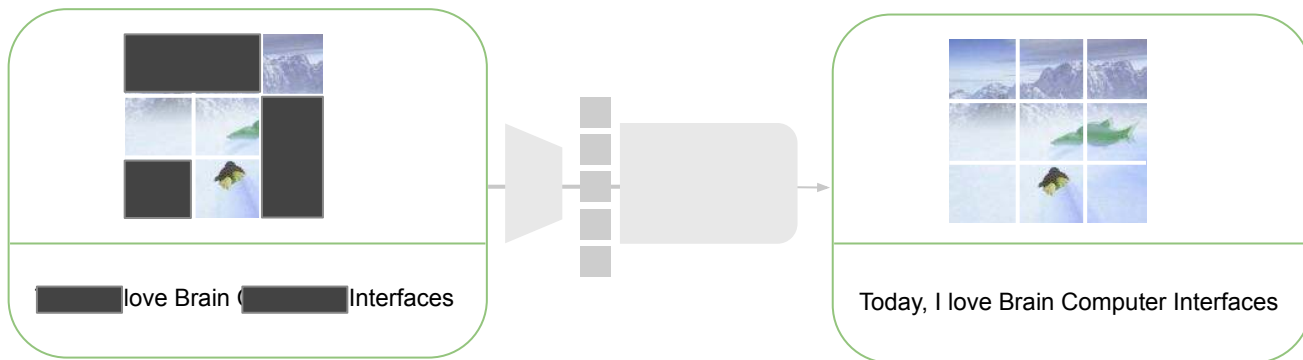
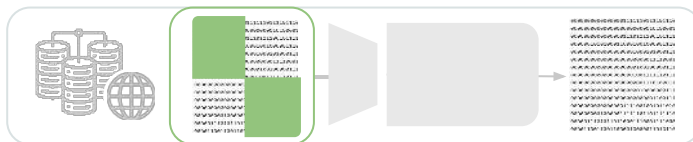
$e_{i,0}$
$e_{i,1}$
$e_{i,2}$
$e_{i,3}$
$e_{i,4}$
$e_{i,5}$
$e_{i,6}$
$e_{i,7}$

[x]

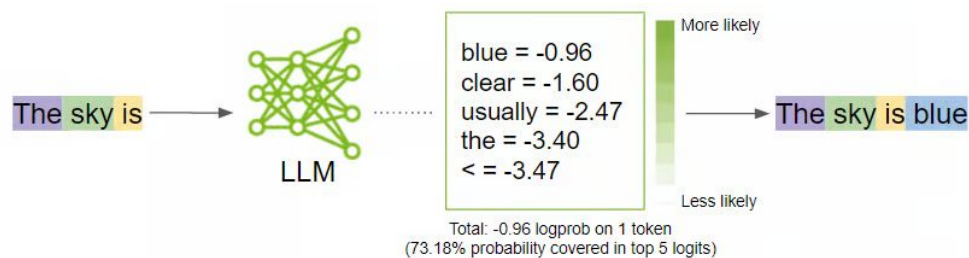
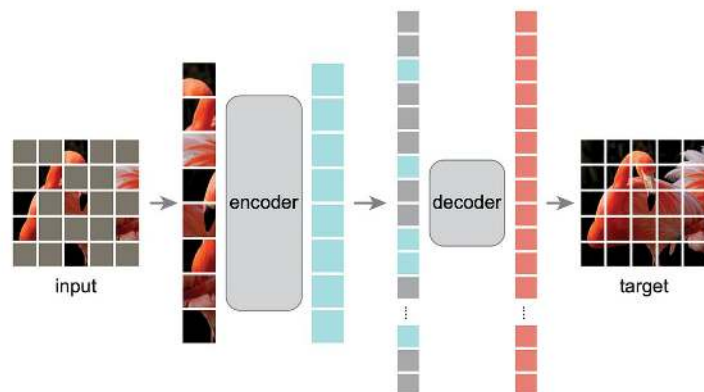
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What's happening in vision and language

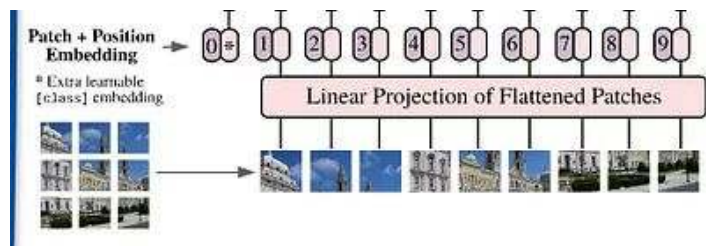
└ Self Supervised Learning



SSL; Masked based SSL, Self-predictif



Patching / Tokenizing and positional encoding



"Today is a beautiful day outside."



["To", "day", "is", "a", "beaut", "iful", "day", "out", "side", "."]

[98, 1]



250 251 252 253 254
 ... The dog chased the cat

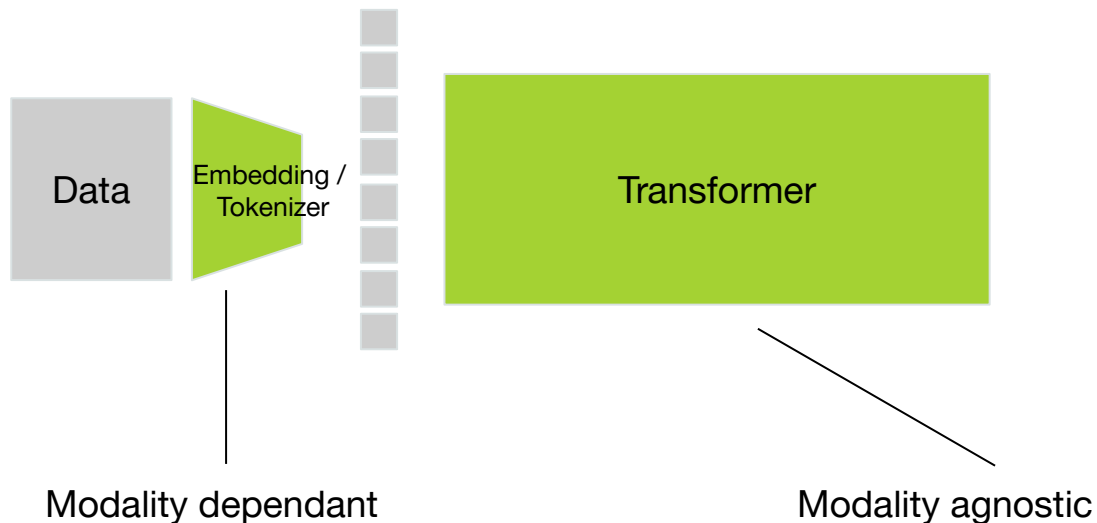
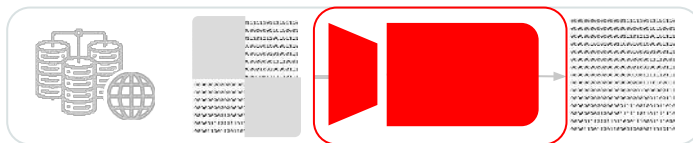
-0.002	252
0.015	252
-0.033	252
-0.015	252
0.016	252
-0.002	252
-0.015	252

+

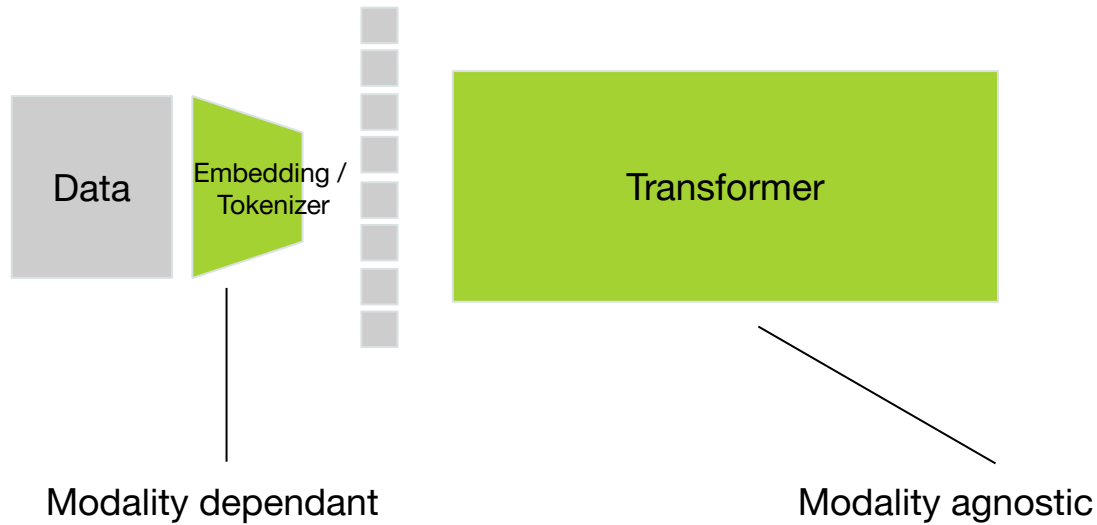
"To", "day", "is",
 "ove"
 Brain"
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What's happening in vision and language

└ Transformer



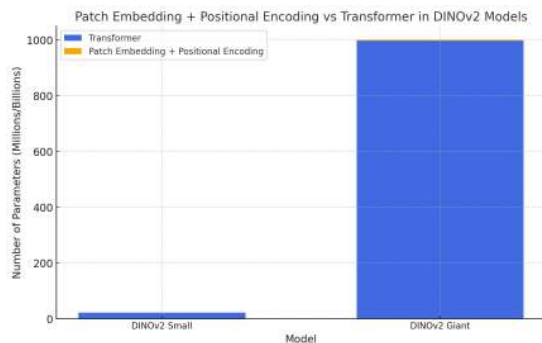
Transformer



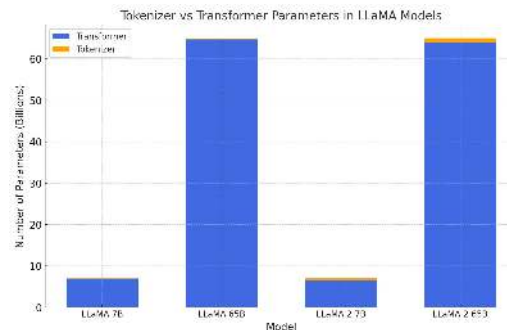
Transformer



The Transformer overwhelmingly dominates parameter count (~90-99%)



Patch Embed + PE = ~1.7% / 0.3%



Tokenizer ~7.5% / 1.6%

Foundation Models for EEG

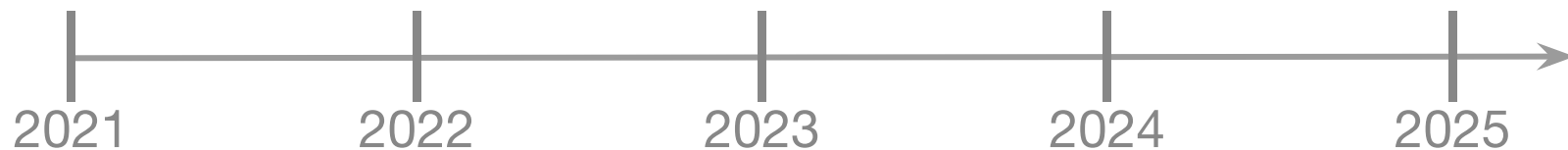
EEGFormer [27] EEG TSP [30] EEG
Labram [28] EEG CBraMod [31] EEG
Neuro-GPT [26] EEG Brant-2 [29] EEG

BIOT [x]
2023 2024

201 ▶

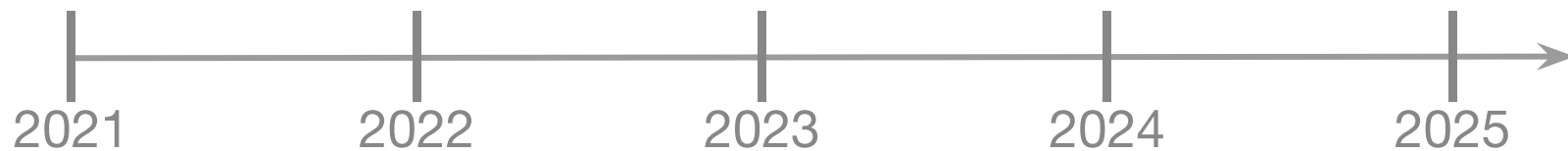
2021 ▶
BENDR [24] EEG BIOT [25] EEG EEGFormer [27] EEG
Neuro-GPT [26] EEG Labram [28] EEG C
Brant-2 [29] EEG
2020

Foundation Models for EEG



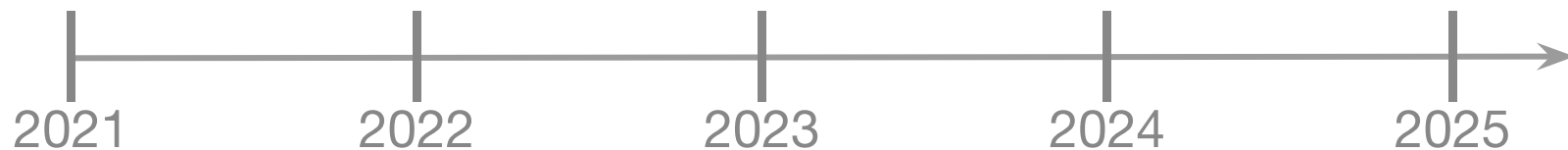
	Infer and generalize to any EEG Configuration	Trained on Multiple config
BIOT	~	
Labram	~	
Cbramod	~	X
REVE (ours)	✓	

Foundation Models for EEG



Infer and generalize to any EEG Configuration	Trained on Multiple config
~	✗
~	✓
~	✗
✓	✓

Foundation Models for EEG



	Trained on Multiple config	Infer and generalize to any EEG Configuration	Can be used without tuning the weights
BIOT	✗	~	~
Labram	~	~	~
Cbramod	✗	~	~
REVE (ours)	✓	✓	✓

REVE: A foundation model for EEG adapting to any setup with large-scale pretraining

└ Dataset



Data Collection



ISIP



Child Mind
Institute

Healthy Brain Network



OpenNEURO

PhysioNet

the research resource for
complex physiologic signals

KU LEUVEN



HARVARD
MEDICAL SCHOOL



Brainclinics
FOUNDATION

Inria

- 19 TB of raw data - 61,415 hours (80% clinic)
- 24,274 subjects - 92 datasets (28 BCI, 8 clinic)
- 150,833 sessions - 396 unique electrode

REVE: A foundation model for EEG adapting to any setup with large-scale pretraining

└ Dataset



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Data Processing

- 1) Resampling 200 Hz, Band-pass [0.5,99.5] Hz

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└ Dataset



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- 1) Resampling 200 Hz, Band-pass [0.5,99.5] Hz
- 2) Z-Normalization (per channel, per session)

Problems addressed

- A) Different amplitudes (2)

REVE: A foundation model for EEG adapting to any setup with large-scale pretraining

└ Dataset



Data Collection



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- 24,274 subjects - 92 datasets (28 BCI, 8 clinic)
- 150,833 sessions - 396 unique electrode

Data Processing

- 1) Resampling 200 Hz, Band-pass [0.5,99.5] Hz
- 2) Z-Normalization (per channel, per session)
- 3) Clipping

Problems addressed

- A) Different amplitudes (2)
- B) Outliers and big spikes (3)

REVE: A foundation model for EEG adapting to any setup with large-scale pretraining

└ Dataset



Data Collection



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- 150,833 sessions - 396 unique electrode

Data Processing

- 1) Resampling 200 Hz, Band-pass [0.5,99.5] Hz
- 2) Z-Normalization (per channel, per session)
- 3) Clipping
- 4) 3D electrode positions alignment
- 5) 10 second windowing

Problems addressed

- A) Different amplitudes (2)
- B) Outliers and big spikes (3)
- C) Different montage with different coordinate systems (4)

REVE: A foundation model for EEG adapting to any setup with large-scale pretraining

└ 4D Position Encoding Strategy



Problems addressed

- A) Electrode configurations (number and placement)
- B) Signal Duration
- C) Generalize to unseen configurations

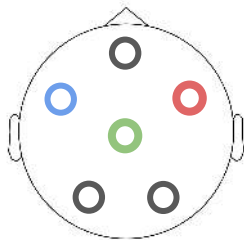
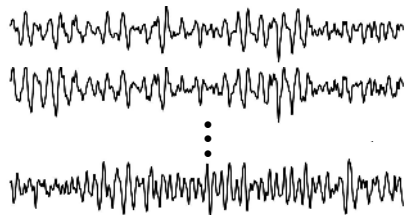
REVE: A foundation model for EEG adapting to any setup with large-scale pretraining

└ 4D Position Encoding Strategy



Problems addressed

- A) Electrode configurations (number and placement) B) Signal Duration C) Generalize to unseen configurations



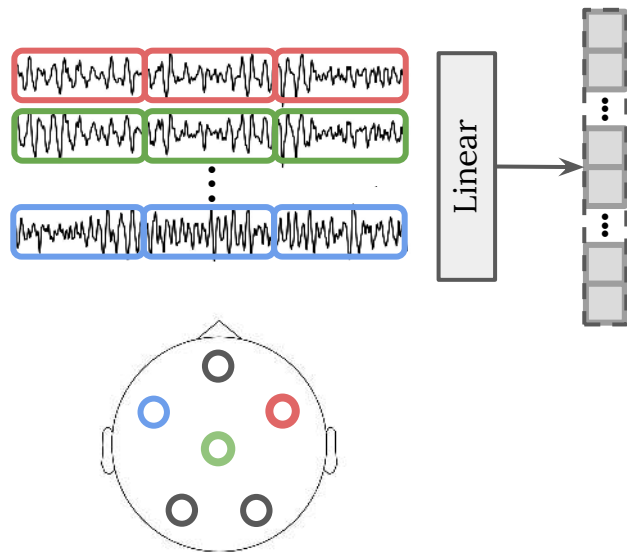
REVE: A foundation model for EEG adapting to any setup with large-scale pretraining

└ 4D Position Encoding Strategy



Problems addressed

- A) Electrode configurations (number and placement) B) Signal Duration C) Generalize to unseen configurations



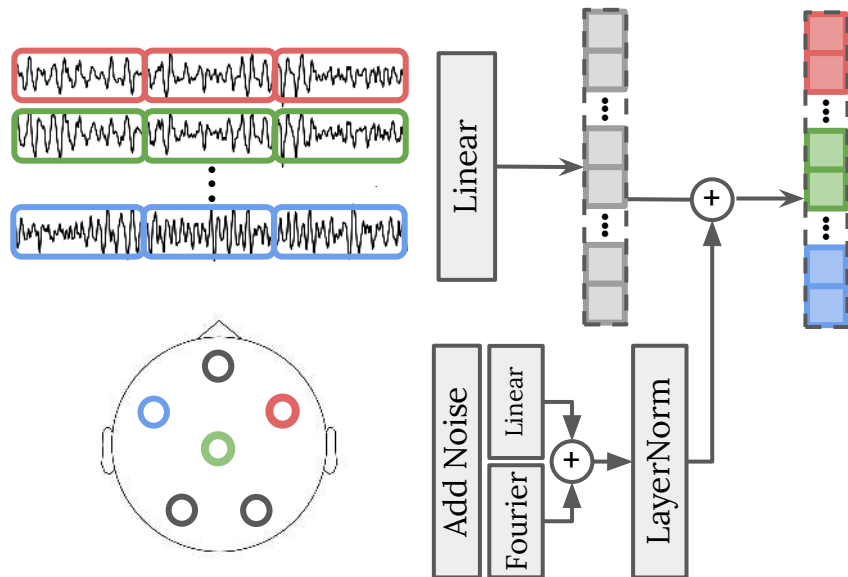
REVE: A foundation model for EEG adapting to any setup with large-scale pretraining

4D Position Encoding Strategy



Problems addressed

- A) Electrode configurations (number and placement) B) Signal Duration C) Generalize to unseen configurations

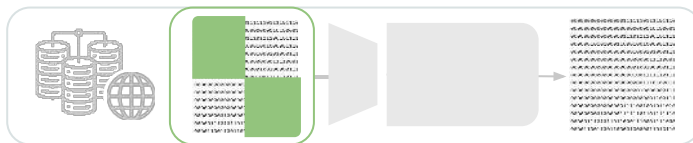


- ▶ Spatial Noise: Introduce more diversity
- ▶ Fourier4D: Extension of the 2D proposed in [x]
- ▶ MLP4D: Allow more adaptation

[x] Défossez, Alexandre, et al. "Decoding speech perception from non-invasive brain recordings." *Nature Machine Intelligence* 5.10 (2023): 1097-1107.

REVE: A foundation model for EEG adapting to any setup with large-scale pretraining

└ Spatio-Temporal Block Masking Strategy



Problems addressed

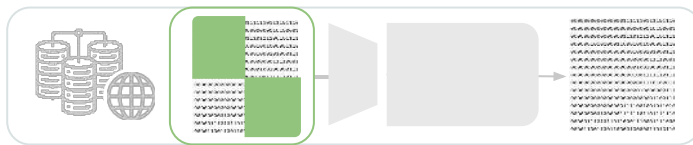
A) Choice of the SSL method

► Same as MAE [x]

[x] He, Kaiming, et al. "Masked autoencoders are scalable vision learners." *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2022.

REVE: A foundation model for EEG adapting to any setup with large-scale pretraining

└ Spatio-Temporal Block Masking Strategy

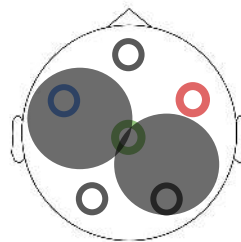
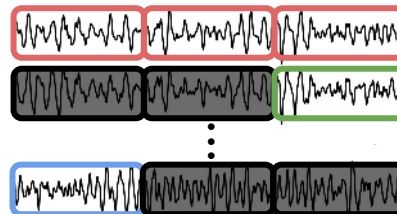


Problems addressed

A) Choice of the SSL method

B) Balancing temporal and spatial characteristics

- ▶ Same as MAE [x]
- ▶ With a novel masking strategy with spatio-temporal blocks
- ▶ Overcome limit of random masking



[x] He, Kaiming, et al. "Masked autoencoders are scalable vision learners." *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 2022.

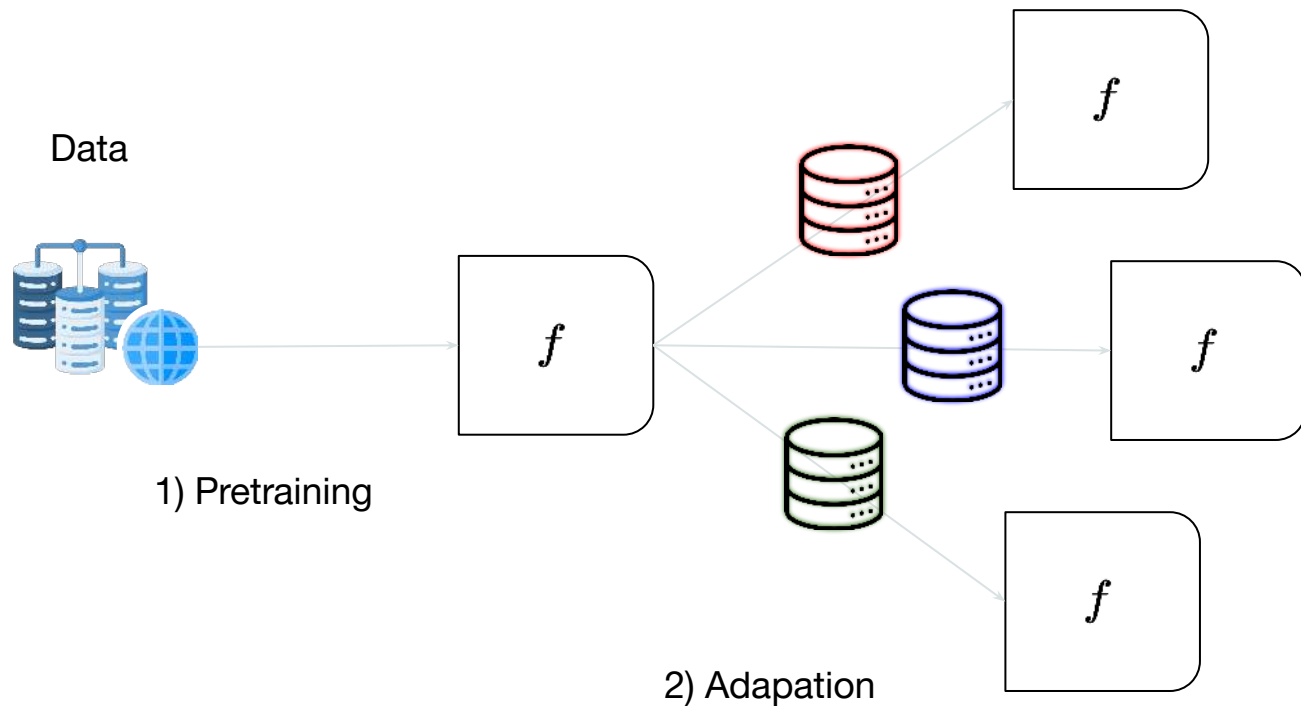
Main results on REVE

1) Pretraining



f

Main results on REVE



- Motor Imagery (PhysionetMI / BNCI2014)
- Event classification (TUEV)
- Abnormal detection (TUAB)
- Sleep staging
- Workload cognition
- Slowing Event Detection
- P300

Context calibration in BCI systems

Why calibration is Needed ?

- variation between users and between sessions
- experimental conditions / mental state / electrode placement

Why it's a problem

- Require collecting new data & retraining
- Time consuming / Impractical

Impact on BCI:

- limit their real world usability
- Barrier to their democratization

Results

Settings	PhysioNet-MI, 4-class		
	Balanced Accuracy	Cohen's Kappa	Weighted F1
CBramod	0.6417 ± 0.0091	0.5222 ± 0.0169	0.6427 ± 0.0100
CBramod (Frozen)	0.3845 ± 0.0345	0.2983 ± 0.0498	0.3946 ± 0.0378
BIOT	0.6153 ± 0.0154	0.4875 ± 0.0272	0.6158 ± 0.0197
BIOT (Frozen)	0.3698 ± 0.0318	0.2703 ± 0.04722	0.3723 ± 0.0364
LaBraM-Base	0.6173 ± 0.0122	0.4912 ± 0.0192	0.6177 ± 0.0141
LaBraM (Frozen)	0.3715 ± 0.0458	0.2814 ± 0.0586	0.3796 ± 0.0472
REVE-Base	0.6480 ± 0.0140	0.5306 ± 0.0187	0.6484 ± 0.0170
REVE-Base (Frozen)	0.5371 ± 0.0052	0.3827 ± 0.0070	0.5376 ± 0.0033

Settings	PhysioNet-MI, 4-class		
	Balanced Accuracy	Cohen's Kappa	Weighted F1
CBramod	0.6417 ± 0.0091	0.5222 ± 0.0169	0.6427 ± 0.0100
CBramod (w/o PT)	0.6196 ± 0.0143	0.4994 ± 0.0289	0.6289 ± 0.0179
REVE-Base	0.6480 ± 0.0140	0.5306 ± 0.0187	0.6484 ± 0.0170
REVE-Base (w/o PT)	0.5409 ± 0.0094	0.3879 ± 0.0125	0.5421 ± 0.0101

C.3 Model interpretability

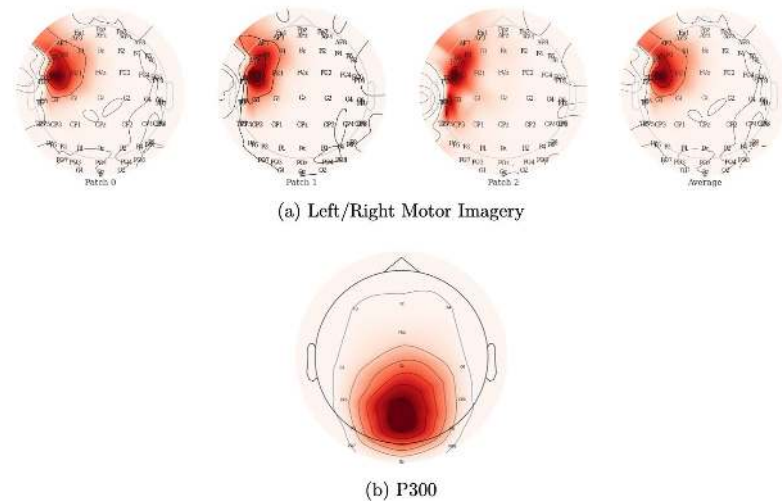


Figure C.1 – Task-Specific Attention Maps for Motor Imagery and P300 Tasks

Additional results

- Provide quality embeddings, with strong generalization even when frozen.
- Important performance gains come from pretraining.

REVE: A foundation model for EEG adapting to any setup with large-scale pretraining

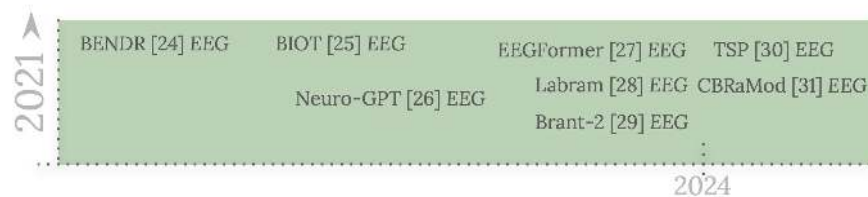
└ Dataset



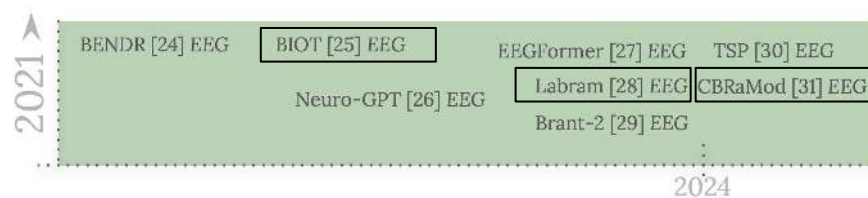
- 19 TB of raw data - 61,415 hours (80% clinic)
- 24,274 subjects - 92 datasets (28 BCI, 8 clinic)
- 150,833 sessions - 396 different electrode

Group	Subjects	Duration (hours)	Datasets
Category			
BCI	791	457	28
Cognition	4,193	10,376	56
Clinic	19,290	50,581	8
Platform			
TUH	14,987	26,847	1
Physionet	607	22,707	2
OpenNeuro	4153	10,194	56
MOABB	711	384	27
Other	3,802	1,250	6
Channels			
[3 – 30[	19,871	50,870	31
[30 – 80[	1,781	1,516	48
[80 – 129]	2,622	9,027	13
Total	24,274	61,415	92

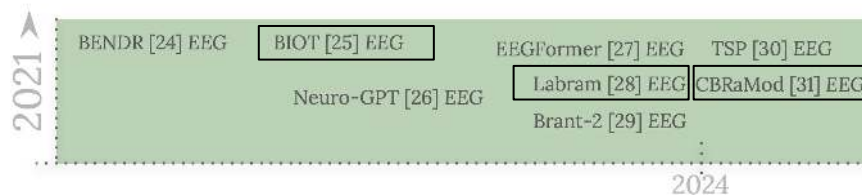
Foundation Models for EEG



Foundation Models for EEG

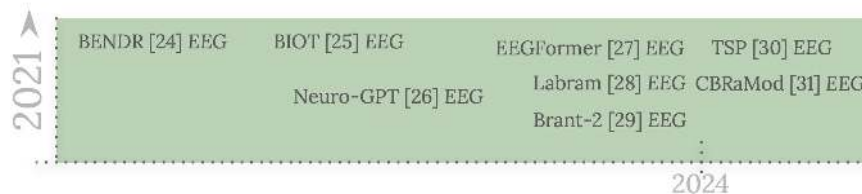


Foundation Models for EEG

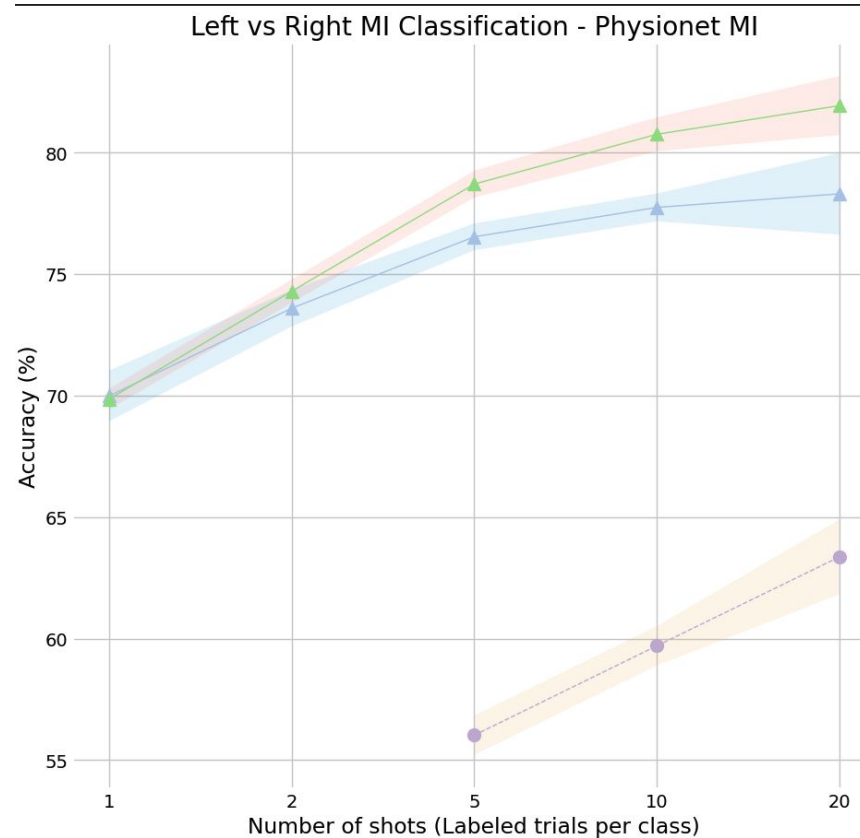


	Large Scale pretraining	Generalize to many DT	Trained on Multiple config	Infer and generalize to any EEG Configuration	Provide ready to use embeddings
BIOT	✓	✓	✗	✗	✗
Labram	✓	✓	✓	✗	✗
CBRaMod	✓	✓	✗	✗	✗

Foundation Models for EEG



	Large Scale pretraining	Generalize to many DT	Trained on Multiple config	Infer and generalize to any EEG Configuration	Provide ready to use embeddings
BIOT	✓	✓	✗	✗	✗
Labram	✓	✓	✓	✗	✗
CBRaMod	✓	✓	✗	✗	✗
Reve (our)	✓	✓	✓	✓	✓



Creation of the novel set pas la même en fonction du modele

Offset ?

REVE vs EEGSimpleConv

- 1) Plus simple de faire le base
- 2) + lent à train et à inférer.
- 3) EEGSimpl sensible au fait d'avoir les mêmes données pour train f (pas que les config mais aussi le windowing), alors que c'est moins le cas pour REVE
- 4)

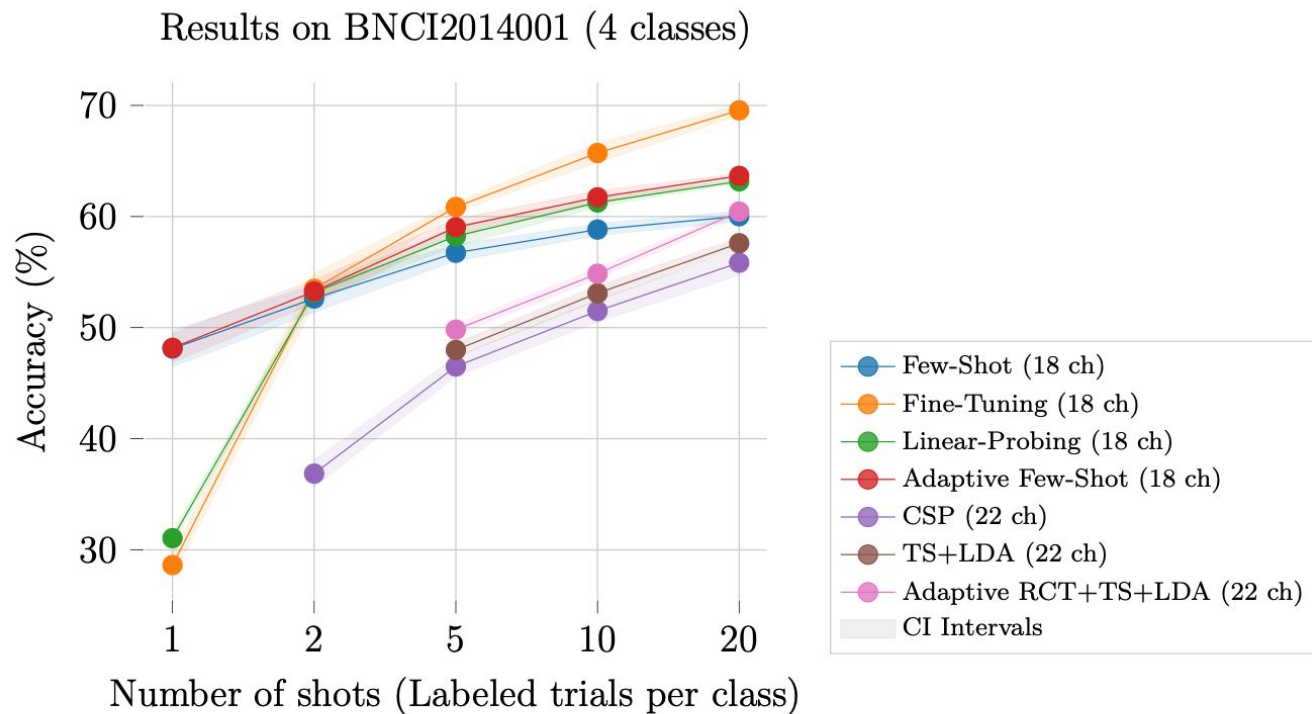


Figure 3.6 – Comparison of deep learning methods using EEGSimpleConv vs machine learning methods

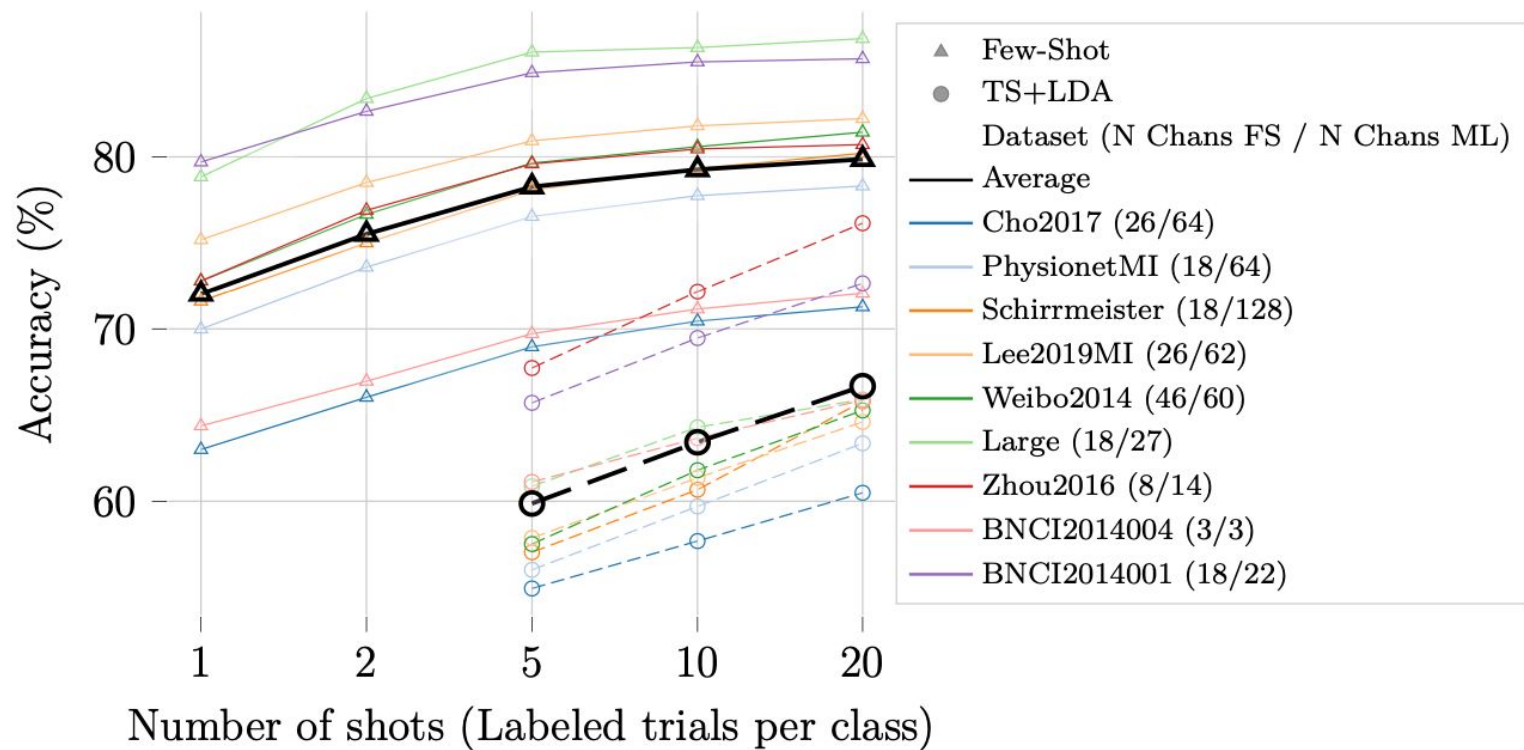


Figure 3.9 – Left vs Right MI Classification - Few-Shot vs TS+LDA